APOLLO DRONES

Group 4 I&E
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Word Count Considering Exclusions: 8214 Words

Executive Summary 03

Problem Outline

In Search and Rescue (*SaR*); As time passes, the search area grows, survival rate decreases, and response personnel are faced with covering large areas in short amounts of time - in a race to save lives.

In US National Parks alone, between 2004 and 2014, 46,609 individuals became lost - requiring search and rescue campaigns. These campaigns costed in excess of $158million. The cost was not solely fiscal, with response personnel taking mental and physical risks.

Current *SaR* programs commonly rely on limited funding, expensive helicopters and ground teams. Existing drones, although helpful for targeted reconnaissance are unable to systematically search large grids due in-part to battery constraints. Cheap drones can become effective once search areas are limited and constrained.

Proposed Solution

We propose a BtB system that uses Machine Learning (ML) to apply probability based targeting to search strategies. The algorithm will initially be trained on data from the International Search & Rescue Incident Database (ISRID), considering a range of inputs. Post-training, the algorithm will take inputs centring around the Initial Planning Point (IPP - Last known location) and Lost Person Types (*LPTs* - Data formed around the demographic of the Lost Person (LP)).

We use a Hardware as a Service (HaaS) model that sells a drone with thermal capabilities to stakeholders, whilst using a subscription model for the use of our ML algorithm, and at an increased cost; API access allowing business' to integrate their use-cases with our protocols.

The drone allows DIY maintenance and focuses design around limiting catastrophic failure - maximising up time, minimising repair costs. Subscription service provides replacement drones during complicated repair however high efficiency, low failure rate, DC motors quell risk of failure.

Value Added

Replace the *ad hoc* model currently used by SaR teams, nullifying the distance ring probability models with intelligent terrain and LP analysis. Thereby reducing search times and saving lives.

Replacing expensive helicopter flights and compounding maintenance with eco-friendly, easy to fix drones.

Sales Channels

Direct Contact

Direct contact to S&R Teams and organisations

Events

INSARAG and other conventions to promote our products

Website

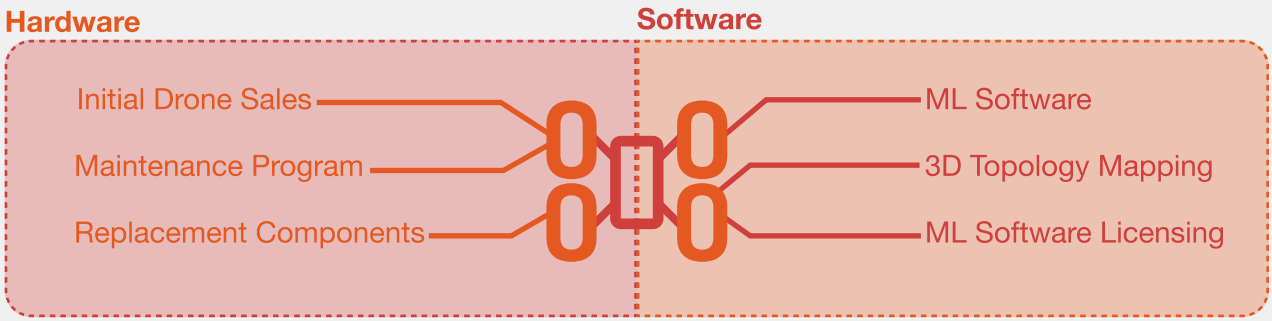
DSC via website

Defence Funds

Through investment funds connecting us to defence industry applications

Business Model, Milestones, Funding

Our '*Razor and Blade'* model sells the 'razor' (drone) relatively cheaply, whilst the 'blades' (software) providing recurring revenue stemming from subscriptions model.



Year 1:
Total funding required: £512,000.
Milestone: MVP, pilot program
Projected revenue: £30,000

Year 2:
Total funding required: £460,000.
Milestone: Functional Saas
Projected revenue: £522,500

Year 3:
Total funding required: £0
Milestone: SaaS sales, global expansion, Defence Contractors
Projected revenue: £4,500,000

Competitors

Indirect

Companies such as DJI offer a similar product; not specialised but drones equipped with thermal cameras that could be used in LP location.

Direct

Within the SaR space there are limited specialised drone competitors. Ripper Corp is a direct competitor but has several key weaknesses and lacks a robust business model. Although there are academic papers, there do not appear to be any companies using ML to locate LPs.

Replacement

As the, current, go to in the search and rescue sector; helicopters (private and public) are used to search large areas from above.

Market Opportunity

We have two markets:

SaR Market:
HaaS + Direct subscription service
Top Down SOM: £4.02 million.
Bottom Up SOM: £4.47 million.

Defence Market:
Haas and SaaS
Top Down SOM: £3.00 million.
Bottom Up SOM: £8.40 million.

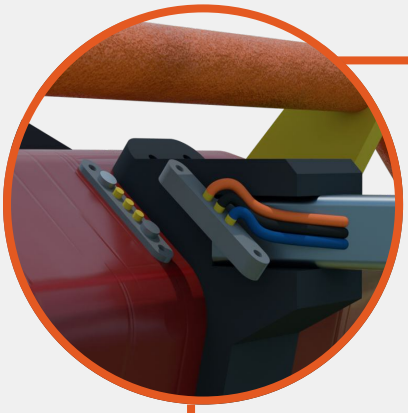
Route to Market

1. Initial Funding
2. Develop MVP
3. DASA funding for R&D
4. Pilot program
5. HaaS Rollout
6. Market Expansion
7. SaaS Rollout
8. Defence Rollout
9. Global Expansion

THE DRONE THAT FINDS YOU FASTER USING MACHINE LEARNING TO DELIVER RAPID AID

APOLLO DRONES

Product Highlights

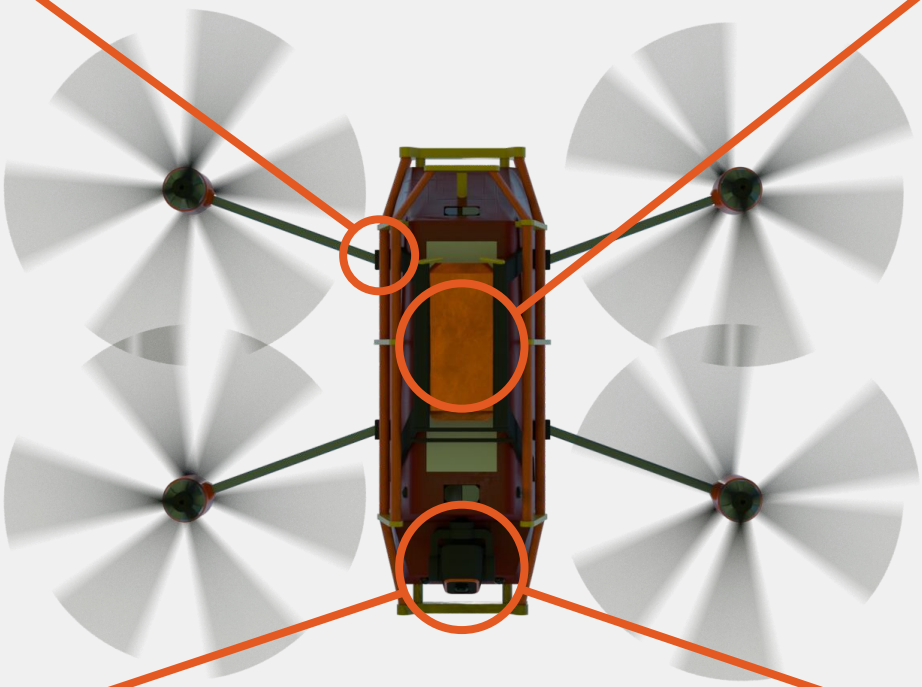


Repairability

Design for repairability ensures SaR Teams can deploy the drone knowing maintenance is as easy in the event of damage.

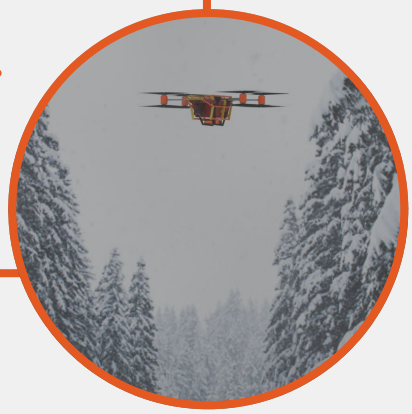
Package

The drone can take a payload with key medical supplies such as first aid kit, thermal blankets and a radio, increasing survival and decreasing stress.



Locating

Thermal and optical camera ensure fast location of lost person and detection and mapping of undocumented/ improvised paths that hiker could have used.



Machine Learning

Through the use of a machine learning algorithm, the locating of users can be optimised depending on the topology, the person, whether conditions etc.

Primary Research

Outreach

67 Resort Operators: Costs of running a resort? Rescue process timeline? Who are shareholders?
40 Search and Rescue Teams: Strengths and limitations? Opportunities to improve?
2 Academics: Trends, limitations, technologies?
23 Product Companies: Current development issues? Future developments?
1 Rescued Person: What happened?

Lindsay - Assynt MRT

Drones

- 2 DJI Enterprise equipped with IR sensors, visual cameras and searchlight
- 1 Mavic pro zoom

Drone Use Case

- **Search at night**
- Provide **extra light** for response team
- Help SaR plan safe casualty access route
- Provide **mapped image** of avalanche area
- Visual **line of sight** is required when flying

"We have used the drones when conditions are marginal but we can't afford to lose a drone as we have limited funding"

Chris - Moffat MRT

"Funding is the largest factor in the equipment and training we get for our organisation "

Hesitation

- **Funding**, as they rely on donations and grants to buy equipment (his money from Scottish Mountain Rescue)
- Fear of drone fragility:
  - Vulnerable to **high winds**
  - Recounted time on Ben Nevis where victim fell into crevice and the wind blew the drone into the mountain, destroying it.
- IR cameras often pick up things like **sheep** instead of people.

Interest

Team don't have a drone but are interested:

- Main benefit would be the ability to get an **aerial shot** of uneven ground - especially useful in the highlands.

Aryan's Rescue operation

Below is information gathered from Aryan, whom Chris connected us with; A stranded hiker who was rescued by MOFFAT SaR after injuring his leg on a hike

Background

Unprepared

Aryan was **unprepared** for any adverse situations.

Circumstance

Weather **changed suddenly,** leading to minimal visibility and **navigation** problems.

Navigation

After becoming **disorientated** by weather, Aryan took a wrong turn **off the trail,** leading him into **unfamiliar territory**.

Injury

Aryan slipped on **steep ground,** injuring his ankle and was **no longer able to walk.** Stranding him on the trail.

Report

His friends quickly reported him missing when he didn't show up.

Information

His wife **informed search and rescue** teams which **trail** Aryan was on, helping the SaR teams locate him **faster**.

Rescue

SaR found Aryan in **4 hours**. He sustained a grade 3 sprained ankle making him **unable to walk.**

Glacier National Park Rescue — 09

Below is an example of a where drones could have added value to a SaR system:

May 7th 2023

Matthew's car is found at the head of the Huckleberry Lookout Trail following a call from his Mother stating he had not come home.

SaR teams launched on foot and the Two Bear Air (TBA) helicopter launched to search. As night fell both were forced to return to base.

May 8th 2023

The next morning a second TBA helicopter which was undergoing training, noticed footprints and relayed the siting to SaR.

The original TBA helicopter responded to the reports but was forced back to base due to rain. Later that evening, using thermal imaging cameras, Matthew was found by the helicopter.

Limitations

The SaR teams were unable to cover significant amounts of ground on foot and were not able to safely traverse into the glacier field where Matthew was located.
SaR teams on foot and in the air were unable to operate safely at night, forcing a return to base.

Drones

Operates at night and in adverse weather conditions.
Covers **vast areas** quickly and efficiently.
Relatively **cheap**, allowing many to be deployed by underfunded SaR teams.

Helicopters

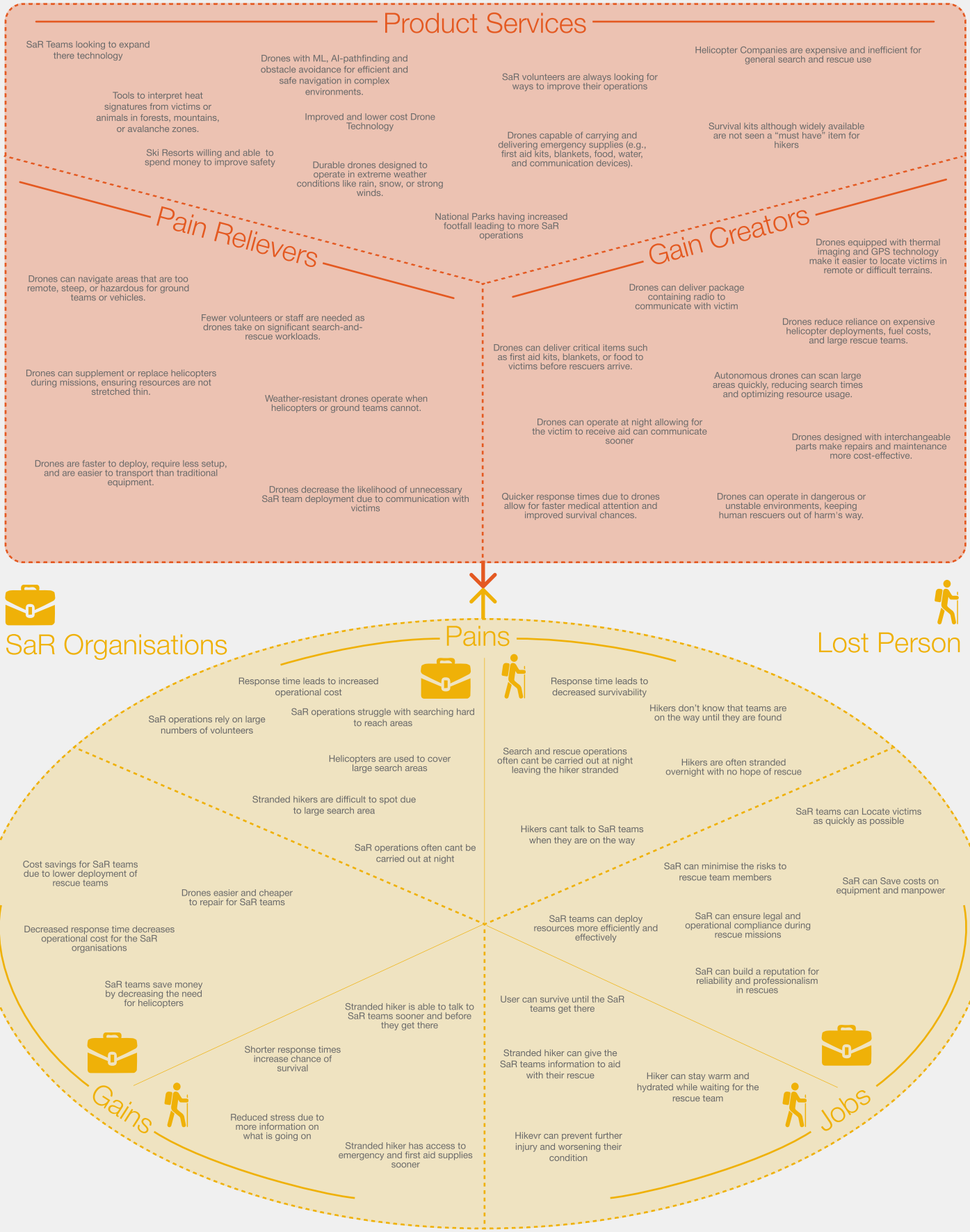
Unable to operate at night.
Unable to operate in **adverse weather** conditions.

S&R Teams

Unable to access **dangerous areas** at night.
Cover ground **slowly.**

Value Map

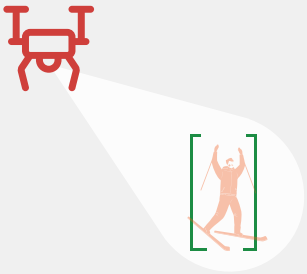
We completed a value map on our market to help demonstrate the the value of our drone and how it would improve the quality of service for the lost people and save the SaR organisations money



Initial Ideation

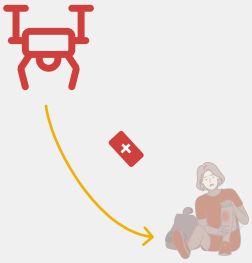
To address the challenges in **SaR**, particularly in scenarios where **victims are stranded** or **difficult to reach**, we explored several innovative solutions and evaluated them below prior to seeking expert opinions. *SWOT analysis for each idea - appendix page 10.*

1: Active tracking of users



Concept involves drones patrolling ski slopes and high-risk hiking areas to **detect unusual behaviour** and **alert** rescue teams; **cutting response times**.

2: Emergency Package



Emergency Package Drop concept delivers **essential supplies** to stranded individuals via air, bridging the gap until SaR teams arrive. Providing critical survival items, **reducing first response time**.

3: Ground Penetrating Radar

GPR locates individuals buried **under snow**, offering **faster detection**. GPR-equipped drones can **scan large areas quickly**, though require advanced radar interpretation and stable flight.

4: LIDAR

LIDAR can continuously **monitors snow depth** on mountainsides, improving avalanche prediction by **identifying hazard zones**. Proactive approach has potential for **subscription-based revenue** through long-term monitoring contracts.

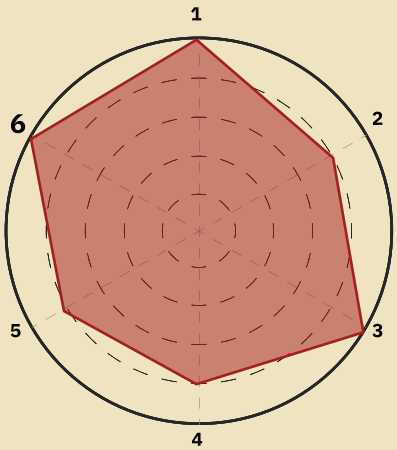
5: Machine Learning

Drone would be able to use a **machine learning algorithm** to use **different metrics** of the persons last location and **predicted movement pattern** to predict their **probable location**

Conclusion

A weighted matrix to decide on the best option and evaluate the different ideas:

1. Response time (3)
2. Reliability (2)
3. Scalability (2)
4. Flexibility
5. Low Labour (1)
6. Cost (1)



Ideation Feedback

Lindsay - Assynt MRT

Feedback

- Active tracking limited by funding.
- Emergency package great for aid and communication but would need tailored package.
- GPR is not applicable.
- LIDAR good for warning hikers but worried about the scale.
- Machine Learning could build on existing infrastructure making it good for low funding

Insight

Funding is major concern.
Reducing first response time is key.

Chris - Moffat MRT

Feedback

- Emergency delivery system could be a “game changer”, but packages must be custom.
- GPR isn’t applicable and LIDAR isn’t applicable.
- Drone swarms sounded interesting for better funded organisations, not for them.
- The Machine learning increasing the speed of the rescue without the need for an increase in equipment

Insight

Focusing on larger organisations and tailoring the drone design to each MRT. Key to improve first response time without needing to purchase multiple drones would appeal.

Aryan - Rescued Hiker

Feedback

- Emergency package with thermal blankets and phone would have been helpful during his rescue.
- GPR - He can see the need.
- Drones over a larger area would have increased the speed of his recovery.
- LIDAR sounded like a good preventative tool.
- Machine Learning would help with the speed of location without large increase in cost

Insight

Speed of rescue is most important.

“The most important thing to improve is the rescue time. The faster the rescue team can reach or contact them the better”

Conclusion

Package

After the positive feedback we received on the box we decided to ideate on the delivery system to make it as accurate as possible. This allows user to access it without needing to move. This will likely be a later addition to our product. Ideation is found in the appendix pages.

Next Steps

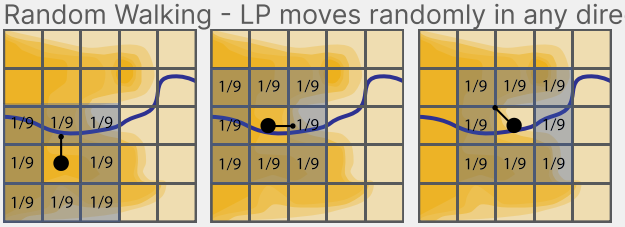
General theme was that first response time is key. Aryan’s rescue shows information is critical and after some more ideation we have decided to focus on a system to help predict the location of lost persons, to help SaR access them rapidly.

(Hitch)Hikers Guide to Machine Learning — 13 —

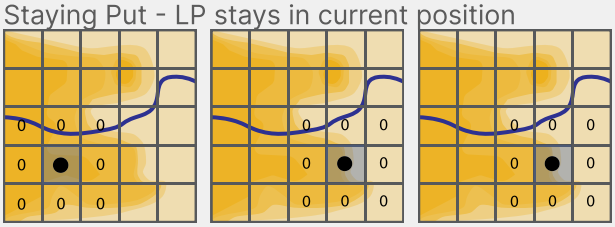
The essence of our machine learning(**ML**) algorithm is to have six behaviour strategies which our 'agent' (The lost person (**LP**)) could use over time. The six behaviours have been selected using data from ISRID, *'Lost Person Behaviour - Robert J Koester'* and *ML research published in 'Nature Scientific Reports'*. A probability function that defines the chance of the LP conforming to each of the possible behaviour strategies is based on the type of person (*demographic*) that the LP is. In short; using the demographic of the LP we can produce a 2D graphical representation of possible routes, expanding with time.

Time →

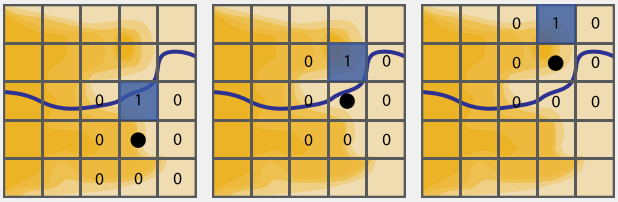
Random Walking - LP moves randomly in any direction.



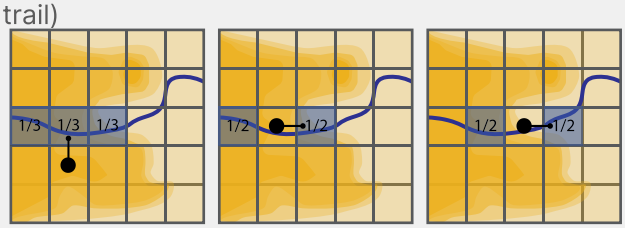
Staying Put - LP stays in current position



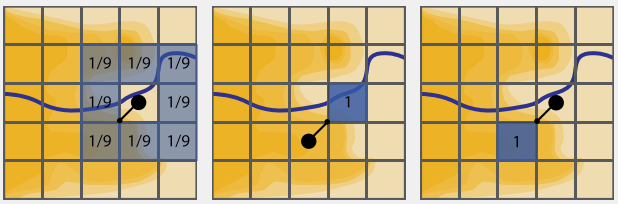
Direction Travelling - LP moves cross-country in one compass direction, ignoring trails, paths and features.



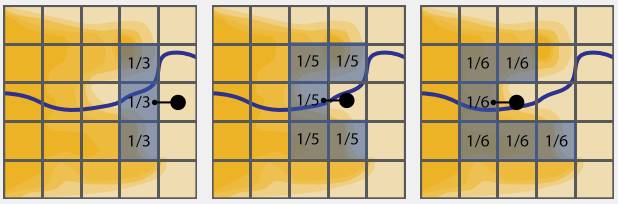
Route Travelling - LP travels on a linear feature (e.g. trail)

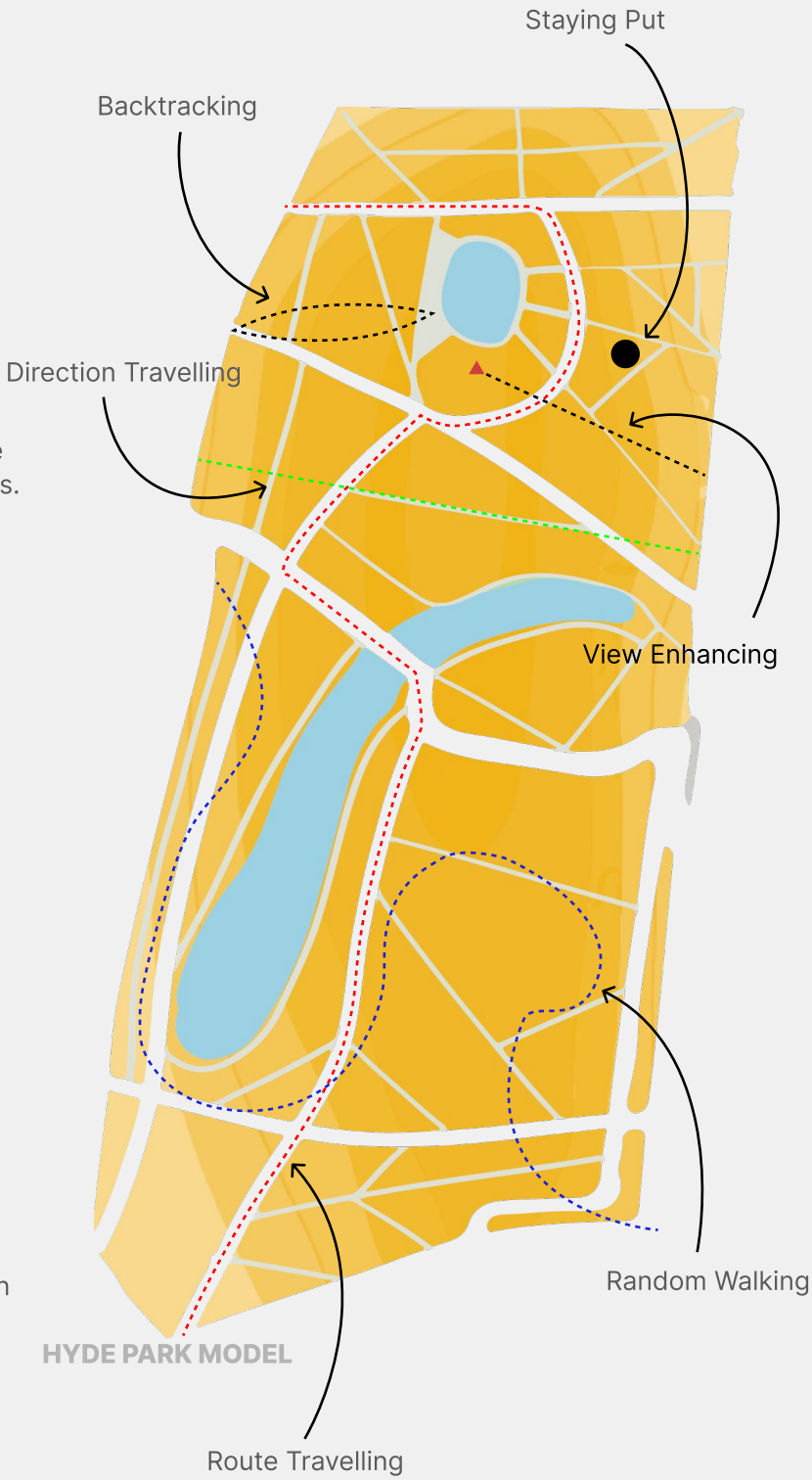


Backtracking - LP moves to previous position



View Enhancing - The LP moves to adjacent cell with higher elevation





ML Validation - Demetris Soukeras

Our ML model was validated by an expert to ensure that it had potential and to learn about computation costs.

Industry Experience:

- Chief Scientific Officer at Syvil
- Co-Founder at MitoBioTech
- Founder in residence of Entrepreneur First
- GTA and Laboratory Operations Manager at Imperial College London

Key Advice Offered:

- Feasibility of our ML Algorithm
- Improvements to the current system
- Clarification around servers and AWS

Linear regression Vs Random Forest

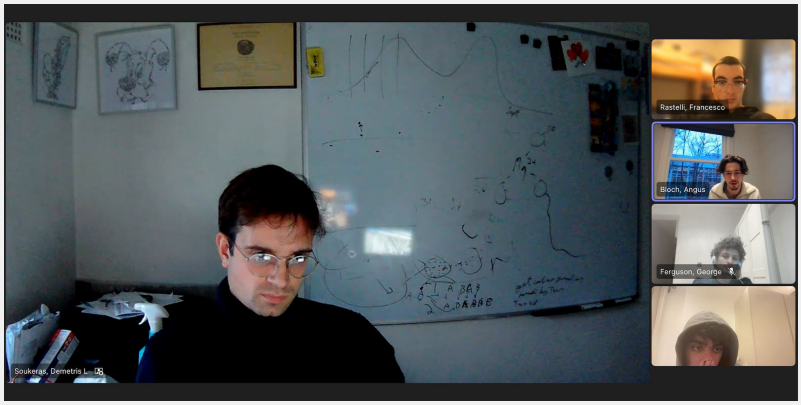
- Linear regression is not computationally intensive.
  - quick and efficient method for predictive modelling
  - only for linear relationships
- Effective for classification problems
- Struggles with rapid shifts in data at the initial stages, as it builds numerous decision trees to make predictions, which can be computationally demanding.

Neural Networks:

- For more complex patterns that cannot be captured by simpler models
- They require more computational resources
  - Higher risk of overfitting
- Graph based neural networks can be used as they work well with data structured as graphs.
- Can model relationships and interactions complexly and dynamically.

AWS EC2 Instance:

- Using AWS EC2 allows you to pay only for the compute time you use
  - More cost-effective
- There are tools provided by AWS to help determine the cost based on the usage.
  - Estimate for $1000 - $3000 for 1 month.
- We could run our model only when new data inputted (1-2 time a month)



Teams call with Demetris

Big O Notation:

- This is used to describe the time complexity and space complexity of algorithms.
- It can helps us to predict how processing time or space used grows as the size of the input data increases.

— Image Classification & Mapping – 15 —

Software will have a second component. Due to the large number of undocumented paths it is likely that while looking for the lost hiker that the drone will come across an “improvised” path. This is an un-documented route that appears like a path. The LP will see the “improvised” path as a route and so algorithm must assign a weighted probability to ‘route travelling’ behaviour.

How It Works

Database

Images of different improvised paths are marked.

Training

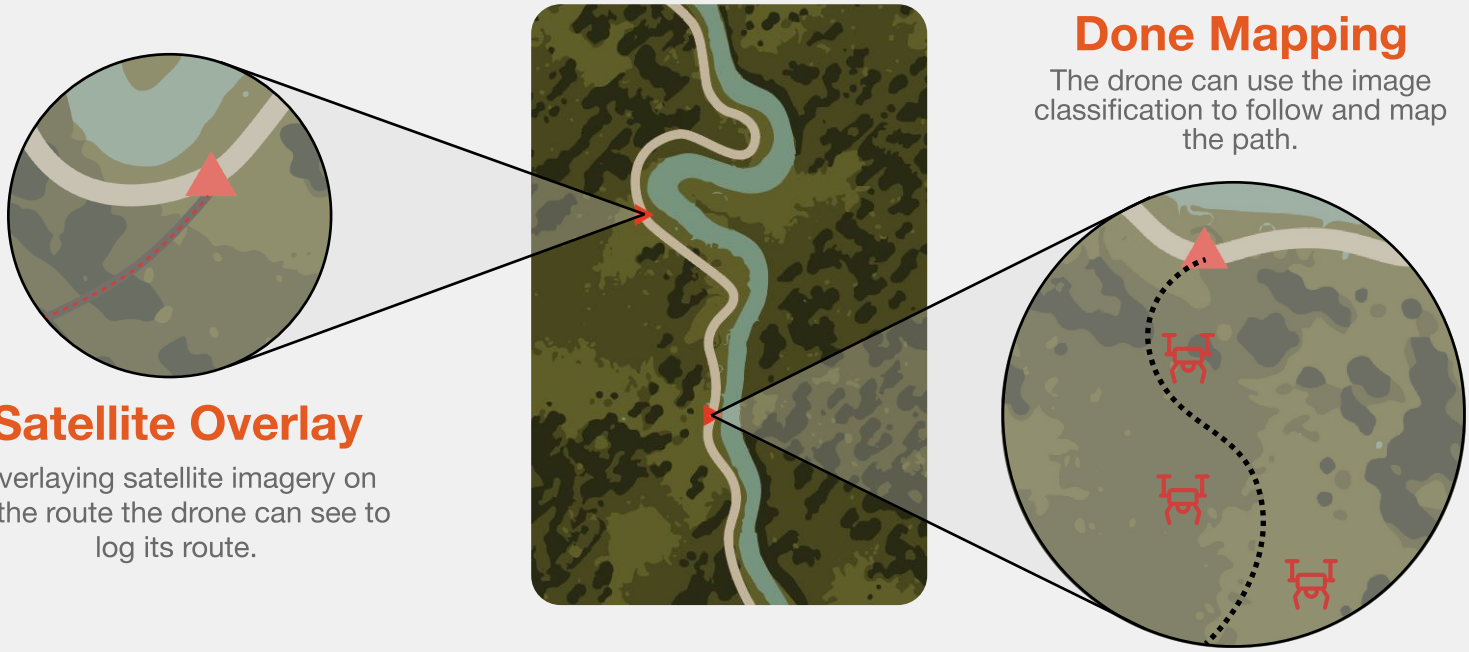
CNN uses these images to train the binary model to identify path/no-path.

Mapping

Various mapping methods are implemented

Algorithm

The new path data is put into the database and the algorithm is re-run



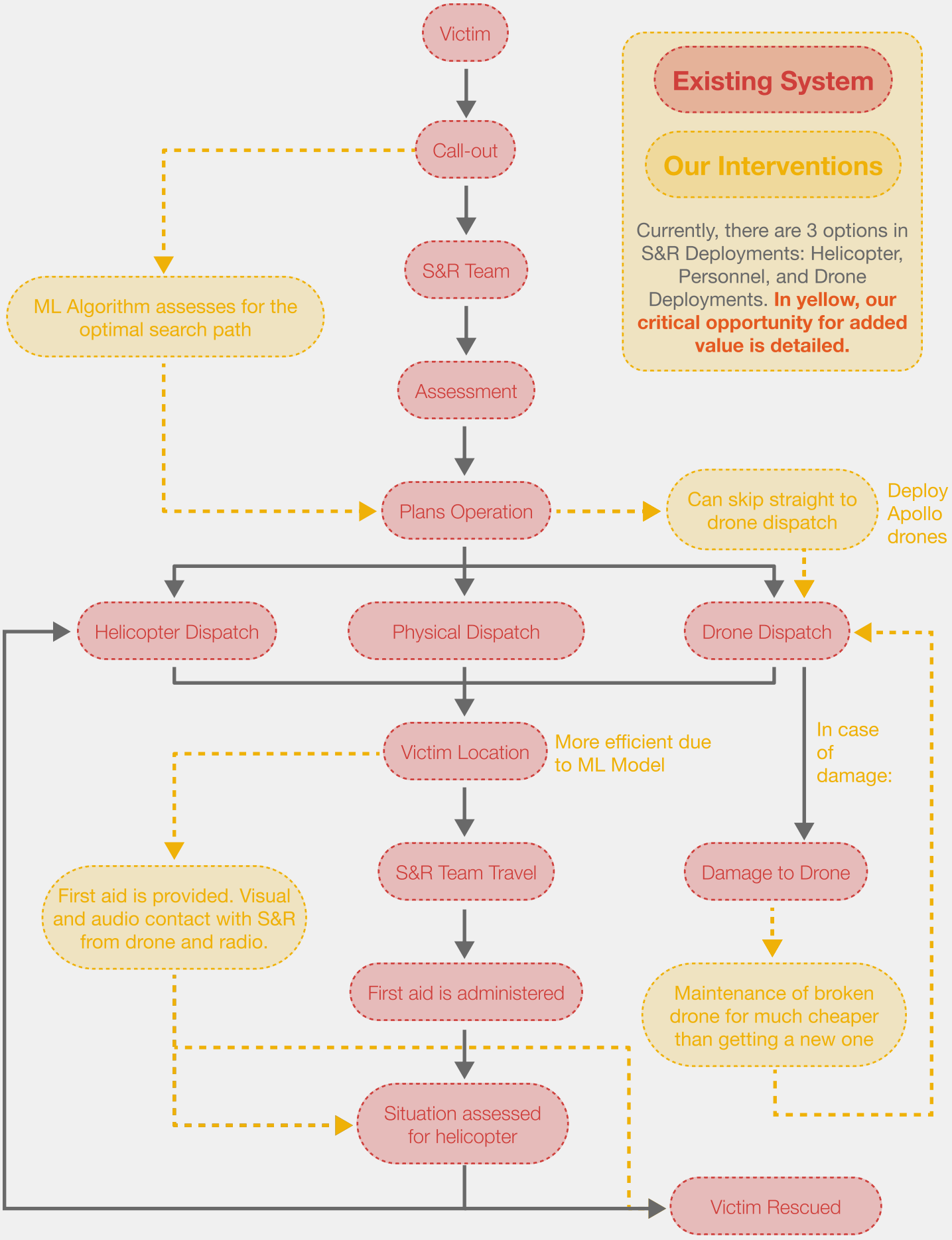
Satellite Overlay

Overlaying satellite imagery on to the route the drone can see to log its route.

Done Mapping

The drone can use the image classification to follow and map the path.

Value Proposition: Our Offering



Physical Design

Redundancy achieved via 8 propellers, multiple antenna and safety via collision avoidance sensors. Thermal camera to aid in location, package dropping to lower operational costs, and repairability so as to appeal to groups with lower funding all whilst an integrated **machine learning** algorithm runs.

Package
Stored underneath the drone and dropped using simple release mechanism within landing gear

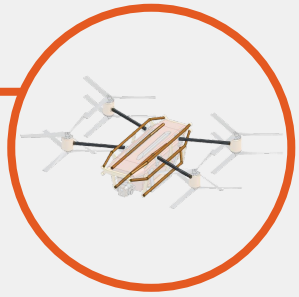
Roll cage
Protects the red plastic shell which is hte only part you can't replace yourself. Plus it looks cool.

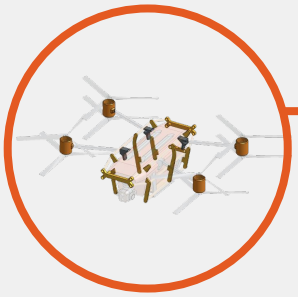
Sensors
Around the body ensure collisions are avoided and regulations met

IR Camera
Thermal and visual cameras provide clear location of victims

Repairability

Standardisation
Standard 10mm box pieces for the arms 10mm carbon tubing for the roll cage made it easy to design. However, the plastic components are more likely to break and will be the key parts to replace.





3D Printing
Components like roll cage attachments, drone arm joints, and motor hubs are 3D printable, allowing S&R teams to easily replace them if damaged.

Pogo Pins
All components are connected via Pogo Pins, letting users replace them without soldering. In a crash, the Pogo Pins also allow components to detach without wire damage.

Value Proposition: ML Algorithm UI

Flight Statistics

Key metrics displayed in red. Utilised by the pilot to ensure they are flying safely.

Inputs

The 3 Inputs for flight are the camera settings, (between thermal/normal), speed and altitude.

Battery

Battery health is displayed with estimated range.

Map

Map shows probability matrix overlayed.

Second Path

Second most likely path a victim would have taken, in case the optimal path fails

Possible Match

Algorithm can detect people in the frame and work out if they are the victim

Optimal Path

The optimal path shows where a victim is most likely to go

The Situation

Skier has fallen and broken their leg in the off country. They are cold and need to get warm ASAP. Darkness is falling and bad weather prevent Ski Patrol or Helicopter Rescue from saving him until the morning.

Assess

ML algorithm curates probability matrix based on LPT and IPP.

Deploy

The drone is deployed by SaR and image classification updates map.

Find

Drone **travels to most probable areas searching for the LP.**

Service

Assess, **Deploy**, **Find**. The product does not perform the rescue itself but allows SaR to **cover** large areas, **assess** the situation and **save** the LP.

Save

User **is provided with emergency package and location relayed to SaR.**

Assess

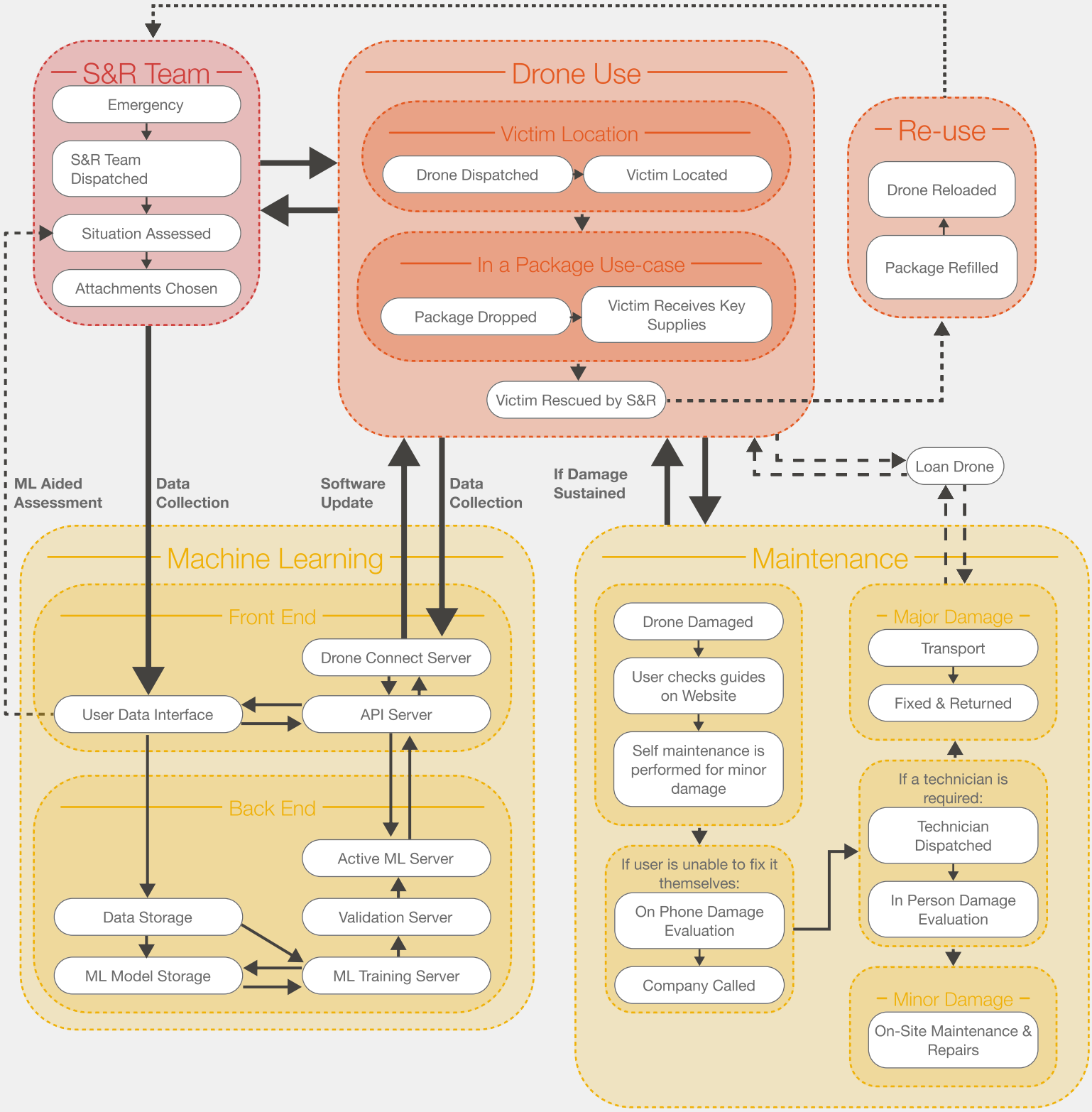
Status of the patient, weather conditions and capabilities are **assessed**.

Rescue

Emergency services **rescue** the patient.

System Diagram

System diagram outlines how SaR would utilise our system in case of emergency Leading on to how the package would be reused, fixed and what would happen depending on the degree of damage to the drone. Another key factor is the collection of data to improve our ML algorithm for optimal location of victims depending on conditions + LPTs.



Private Equity - Phillipos Kassimatis — 21

Our ML model was validated by an expert to ensure that it had potential as well as to learn about server/ computation costs.

Industry Experience:

- Co-Founder at Maven Global
- Visiting Researcher at Imperial Business School
- Deputy Director, Centre for Innovation and Disruption at Bayes Business School
- Managing Director, Global CoHead of FX Structuring at Barclays Capital

Key Advice Offered:

- Product feasibility.
- Viability and market edge.
- Early investments.
- Pricing and market analysis.

Change in business model:

The original business focused on a £3000/unit BtB.
He suggested a subscription model:

- Position the drone as part of a service offering. HaaS
- Shift the focus from selling hardware to delivering ongoing value.

Defence Funding Opportunities:

Phillipos highlighted defence-related R&D Funding

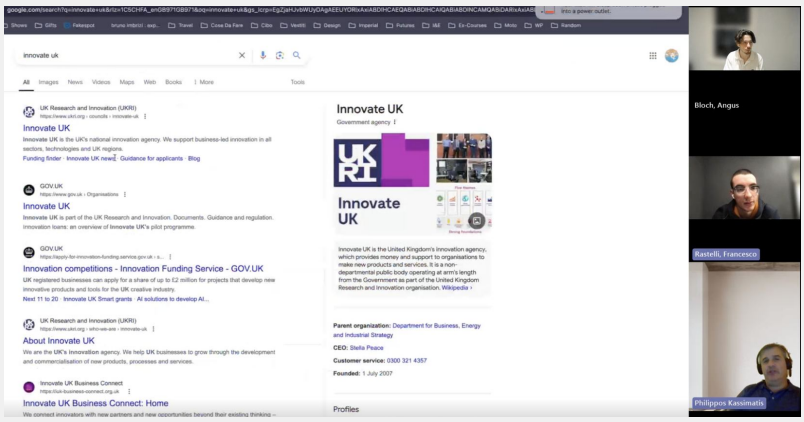
- Explore grants targeted at dual-use technology.
  - Innovate UK and Horizon Europe

Position the technology as multi-use.

- Ensure compliance with government funding requirements.

Scale and Valuation for Investors:

- Demonstrate potential for scalability and high valuation,
  - Recommended having a 1st year valuation of £3M-10M
- Hardware sales alone wouldn't achieve this; value-added service model is essential.
- Align with current government research trends to secure R&D grants.
  - Universities increasingly receive defence funding



Teams call with Phillipos

Strategic Positioning:

Advised on not focusing on hardware as a competitive advantage as

- Hardware, like drones is becoming a commodity.
- Emphasise Imperial's expertise in High end software.
- Drones are enablers for more advanced technological solutions .

Business Model Canvas 22

Key Partenrships

- Drone **components** and electronics suppliers
- Emergency gear and rations suppliers
- **S&R organisations & Teams**
- Legal certifications with the **CAA**
- Server management and maintenance companies

Key Activities

- Assembly and manufacture of drones and packages
- **R&D** into ML Algorithm
- Running of ML Algorithm
- **Maintenance** of drones and components
- Direct interaction with S&R teams for both **sales** and to gain **feedback**

Key Resources

- Initial **funding** required to set up the business and conduct R&D
- **In house assembly** of the product
- Servers for the ML
- Patents
- **Legal certificates** from the CAA to allow our drone to fly

Value Propositon

- To Improve S&R operations by a drone designed to tackle their specific needs, with an integrated machine learning algorithm to give the most probable position of a victim.

Problem Solutions:

- Rapid location and communication with victims.
- Reducing costs by **stopping** unnecessary helicopter deployments.
- Reduces medical liability **risks** to council.

Needs Solutions:

- Allowing SaR Teams to fly the drones in **bad weather**, potentially damaging them, without worrying about repair costs.v
- Reducing first response **time,** delivering key medical supplies, saving lives.

Customer Relations

- **Get**: Companies with direct contact to teams.
- **Keep**: Subscription service with iterative improvement of ML and **maintenance** of the drones.
- **Grow**: Promoting SaR events and direct contact to teams.

Channels

- Pilot Programs
- Conferences, Industry events & Trade Shows
- Referrals and Partnerships

Customer Segment

- SaR teams want to reduce first response **time** and reduce overall **costs.**
- **Casualties** of SaR situations receive better level of care and increased chance of survival.
- Defence industry for API integration.

Cost Structure

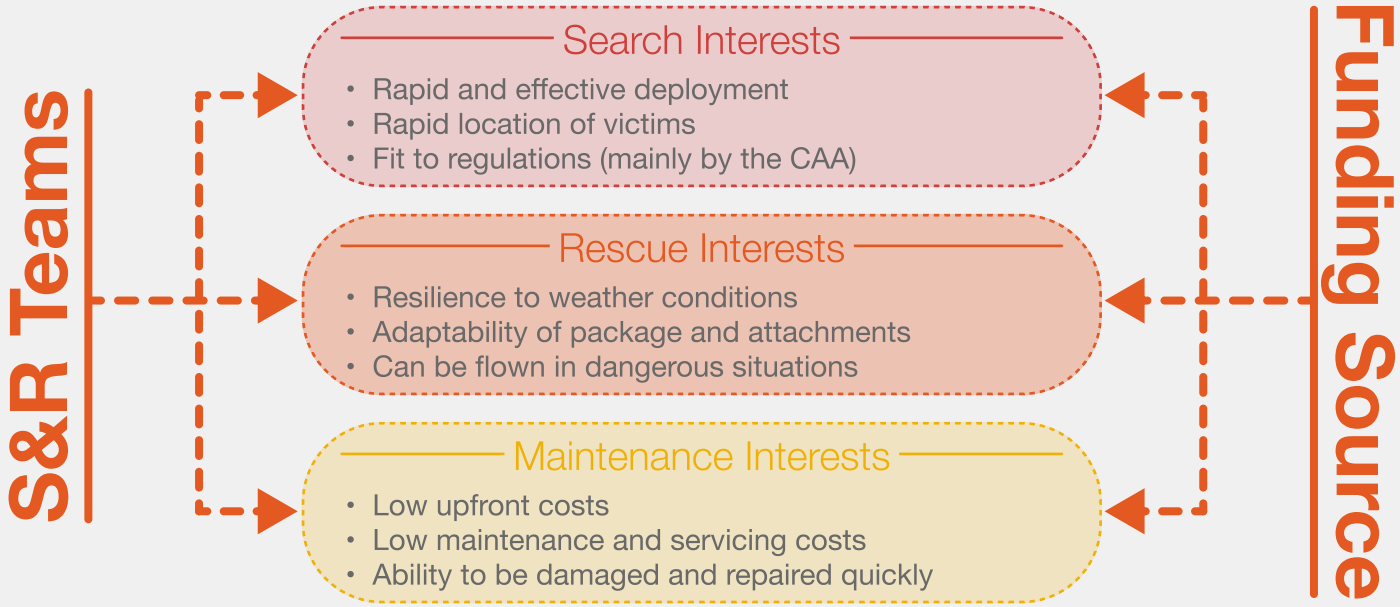
- **Fixed costs:** Office and workshop rental, server, legal, insurance.
- **Variable costs:** Manufacture and materials due to JIT manufacturing. Includes marketing, training and basic utilities.

Revenue Streams

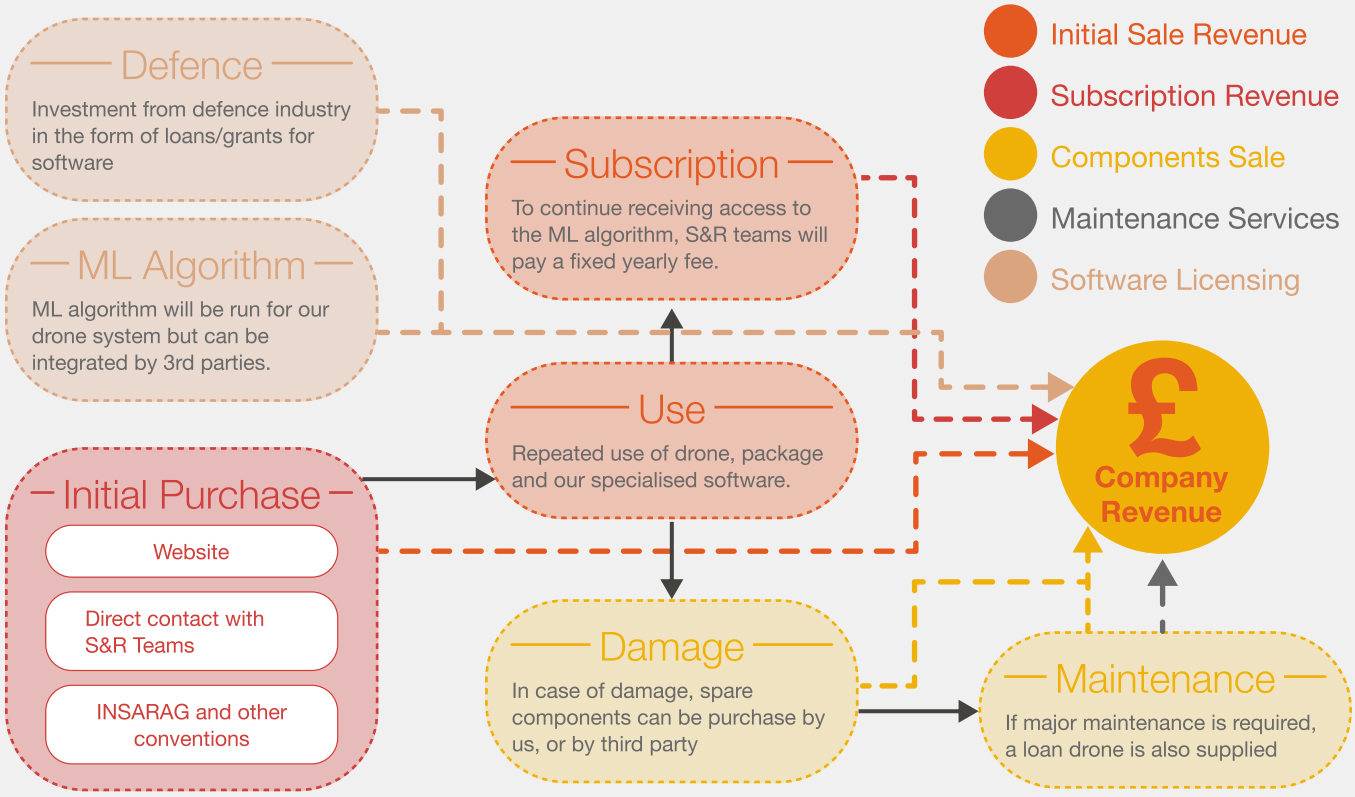
- Direct sales - **B2B / B2G.**
- **Subscription model**
- Diversification into defence industry.
- **Multiple purchases** of the drop box
- **Maintenance.**

Customer Segmentation

The vast majority of SaR teams are run by **charity** and **government funding**.Their focus is less investing in new technologies, resulting in profit and growth; but instead, **keeping costs low** to save money, allowing their limited finance to be used efficiently.

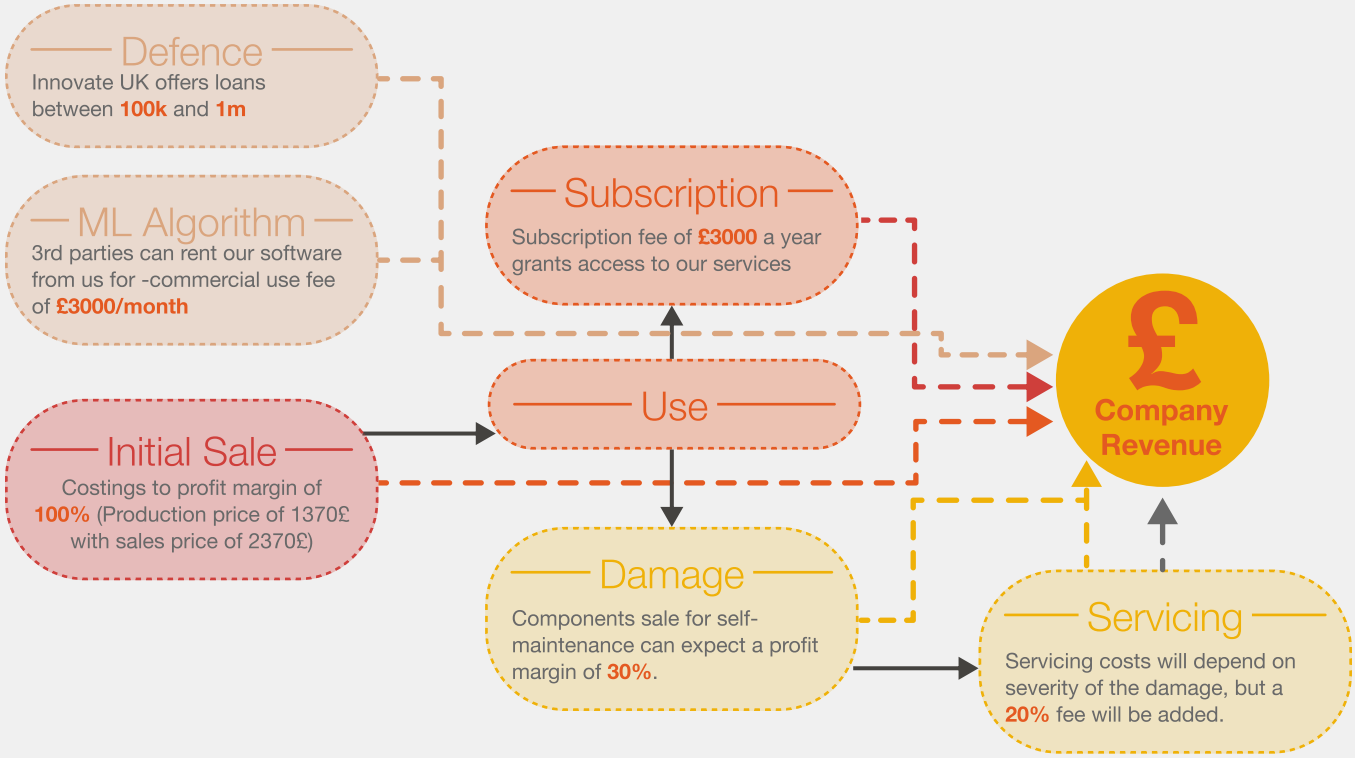


Subscription Based Revenue System

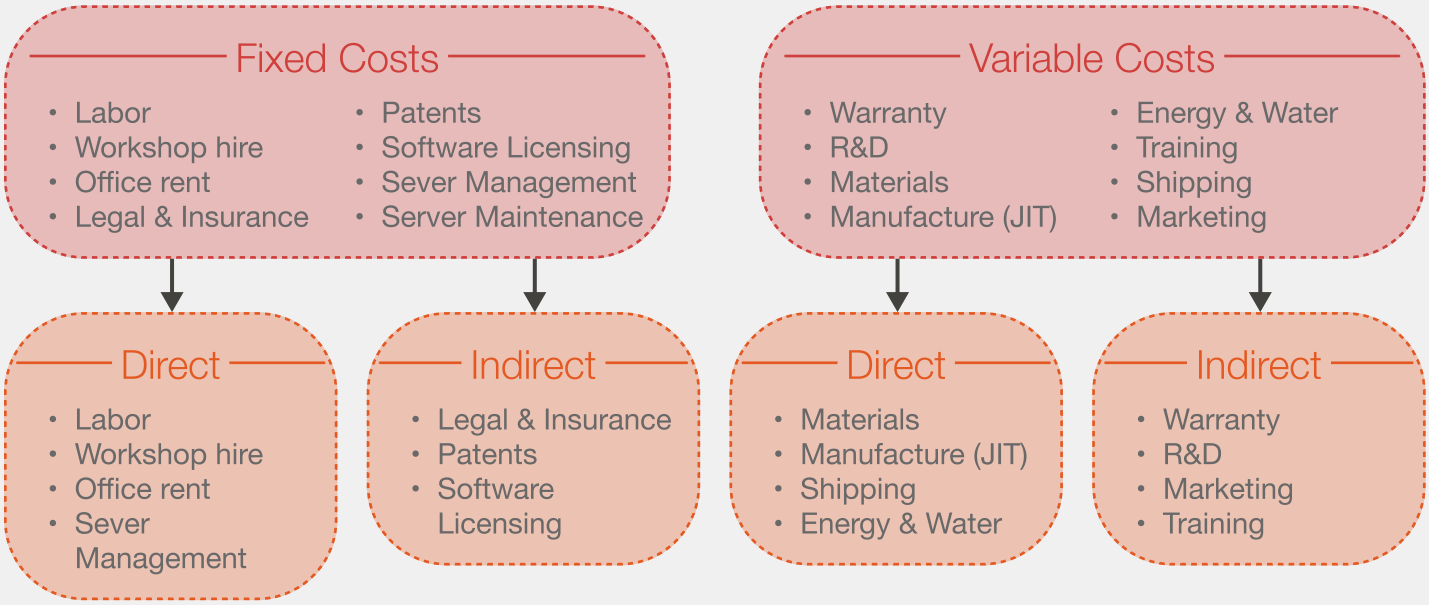


Our Business Model

Revenue Generation Strategy



Cost Structure



Production & Distribution

Just In Time Production leads to majority of costs being variable regarding materials and manufacture. Development of software will change costs

Operational

Manufacture costs vary on the number of units being produced, with only some costs remaining constant. However, server costs will expand through development

Marketing

Marketing will mostly involve costs around venue hire during events, and labor to contact S&R teams

Long Term

Costs around maintaining patents to ensure our company is protected and R&D Development

Diversification Into Defence — 25

Scalability and Growth

To scale and grow, out company will both use different sales channels and diversify into the target markets we look at. Key sales channels are Pilot Programs, which can directly demonstrate the viability to S&R Teams; Conferences and Trade Shows which demonstrate our product to a much larger market, and finally referrals and partnerships which is a quick and easy way to grow our business organically.

Secondary Business Model

One other key industry that would greatly benefit from our product whilst generating a large revenue is the defence industry. While both being able to gain a new revenue stream, it also brings with it large funding for the R&D of our product. *More information can be found in Appendix pg. 08.*

Defence Applications

- **Haas Value:**
  - Military S&R missions for natural disasters
  - Military Helicopter S&R around the UK
    - £34M spent every year
  - Location of a downed pilot
    - 39 downed pilots a year
  - Active tracking and prediction of hostiles
- **Saas Value:**
  - Access to the ML model for in-house use, allowing teams to perform independent development with their own datasets
  - Access to our existing database and topological data

Funding Opportunities

- MoD spends £3B every year on R&D, often as investments in outside companies
- **Innovate UK:**
  - Covers 70% of initial R&D costs
  - Good when starting off, as sales to early adopters can be covered outside of it
- **DASA:**
  - Grant for development once an MVP is produced

Politics

- The two largest drone manufactures are **Skydio** in the US and **DJI** in China.
- Currently, they are two companies which are opposite ends of the political spectrum, with DJI banned in the US and Skydio in china for governmental uses
- As such, placing a supply chain around China is risky, so our company will have it's manufacturing facilities based around Europe.

SWOT Market Analysis

SWOT Analysis

S

- **Expanding Market Demand:** The drone industry is growing significantly, with applications in agriculture, urban planning, disaster management, and environmental conservation driving demand.
- **Technological Advancements:** Increasing sophistication in AI, machine learning, and image recognition is enabling drones to perform complex tasks, including accurate path mapping.
- **Regulatory Support:** Many governments are introducing frameworks to integrate drones into commercial operations, boosting market growth.
- **Diverse Applications:** The ability to map and classify paths is relevant for industries like forestry, infrastructure, and emergency services.

W

- **Regulatory Constraints:** Despite progress, drone use is still restricted in many regions due to airspace regulations and privacy concerns.
- **High Initial Costs:** Drone hardware, AI development, and data integration systems can be expensive, which might deter smaller organizations.
- **Data Processing Challenges:** Mapping requires significant computational power and data storage, creating bottlenecks in resource-constrained markets.
- **Fragmented Market:** The market is saturated with numerous players offering varied solutions, making it difficult to establish a clear competitive advantage.

O

- **Environmental Applications:** Growing awareness of sustainability and climate monitoring can drive demand for path mapping in remote and ecologically sensitive areas.
- **Rising Need for Precision Mapping:** Sectors like autonomous vehicles, smart cities, and logistics require accurate mapping and navigation solutions.
- **Emerging Markets:** Developing countries are adopting drone technology for infrastructure development and environmental monitoring, offering untapped potential.
- **Integration with Satellite Imagery:** Combining drones with satellite data creates opportunities for comprehensive mapping solutions with broader coverage.

T

- **Cybersecurity Risks:** Increasing reliance on drones and AI-driven mapping systems makes them vulnerable to hacking, data breaches, and software manipulation.
- **Regulatory Uncertainty:** Sudden policy changes or stricter regulations could impact drone operations.
- **Competition from Alternatives:** Other technologies like satellite-based mapping or autonomous ground vehicles might compete with drone-based solutions.
- **Public Perception Issues:** Concerns about surveillance, privacy, and noise pollution could limit market adoption.
- **Rapid Technological Changes:** The pace of innovation may render current solutions obsolete, requiring constant investment in R&D to stay relevant.

Political
Economical
Social
Technological
Environmental
Legal

PESTEL MARKET TRENDS

This PESTEL chart helped us analyse the trends in relevant categories and then rank them in terms of importance. This gave ius a deeper understanding of the potential threats and opportunities future trends in the market would afford us.

Importance

P	E	S	T	E	L
Strict regulations impact drone deployment in sensitive areas.	Drones lower costs compared to manned vehicles.	Public sees drones as crucial for SAR missions.	Improvement in available data quality	Electric drones reduce carbon emissions.	Data privacy laws regulate surveillance missions.
Cross-border collaborations can support disaster response.	R&D investments in advanced tech are growing.	Bias within ML Algorithms	IoT integration enhances data sharing in missions.	Drones access remote areas without disturbing nature.	Liability for failures remains a concern.
Privacy concerns may lead to stricter drone laws.	Supply chain issues may disrupt production.	Skilled operators are in high demand.	Advances in battery life and autonomy boost efficiency.	Increase energy usage for the servers	Looser regulations allow for more autonomy
Security concerns over foreign drones	Increase energy costs usage for the servers	Ethical concerns about recording of data	Improved imaging tech aids in harsh environments.	Requirement for eco-friendly disposal	Military regulations around drone use
Rising defence budgets drive investment in SAR drones.	Global drone market is rapidly expanding.	Ethical concerns exist about drone militarization.	AI improves mapping, navigation, and obstacle avoidance.	Disasters from climate change increase drone demand.	Strict regulations impact drone deployment in sensitive areas.

Porters 5 forces Analysis

Porters Five Forces helped to gain insights into the SaR industry. The largest threat to this market place is the power the buyers hold; It is therefore crucial to get partnerships with these large SaR organisations due to the large number of alternative options for these institutions to invest in.

Threat of New Entrants

There are not that many players in the SaR drone market and there is a low risk of new entrants due to high start-up costs and the level of connections needed to get a product to a significant level of use across the industry.

Power of Suppliers

Suppliers include the survival tool manufactures as well as the manufacturers of the drone components. They have little power due to the large number of survival kit manufacturers and the general trend in decreasing prices for drone part suppliers due to current geopolitical climates. - See Appendix

· Industry Rivalry ·

There are established competitors but they have limitations that we will improve upon. This leaves gaps in the market to be exploited. Crucially, creating relationships with buyers.

Power of Buyers

The Search and rescue and governmental bodies have a high level of bargaining power. This is due to the existing infrastructure that these institutions posses. This puts pressure on the drone companies to communicate how their products will not only save money but increase the chances of survival for the victims.

Threat of Subsitutes

Relatively high risk of substitutes due to other options being well established in the SaR space. Options including helicopters and ground teams have a well established foothold. Effort will be required to shift the way teams think.

Direct/Key Competitor

Ripper Corp

- Company produces lifeguarding drones which can drop inflatable life buoys.
- Biggest player in the Drone SaR space.
- Business model works on a hire or purchase basis which is paid for by the organisation and insurance.
- Many applications with large ties to multinational companies and government agencies.

Indirect Competitor

Heli Securite

- Private French specialising in helicopter rescue operations.
- Monopoly over helicopter rescue in French Resorts.
- One time purchase for one time use but can often paid for by the insurance.
- Very expensive at $10,000 - $15000 per use case
- Targets:
  - Government Agencies

Replacement Competitor

Lifesystems

- Produces pocket first aid kits, which are easy to carry around for hikers.
- Cheapest and easiest alternative to other rescue methods.
- They rely on users to be prepared.
- Targets:
  - Mountaineers
  - Skiers
  - Adventurers

Competitor Comparison

Criteria	Ripper Corp	Heli Securite	Lifesystems	APOLLO DRONES
ML SaR Algorithm	✗	✗	✗	✓
Affordable pricing model	✗	✗	✗	✓
Direct aid provision	✗	✓	✗	✓
Diverse Use Cases	✗	✗	✓	✓
Low repair costs	✗	✗	✓	✓
Direct communication	✗	✓	✗	✓

Critical Buisness Risks 30

We identified possible internal and external risks which we could face. Issues around: Funding (investors and early adapters), legal (patents and regulations), and manufacturing/production.

Internal Risks

Funding cuts	Failure to comply with loan regulations or complete minimum viable product/working ML algorithm: Leading to cuts in funding. *[High impact - Medium probability]*	Active acceptance - Budgeting to ensure funds while situation is resolved
ML algorithm	Loss of algorithm IP secrecy due to negligence or overlap. *[Medium impact - Low probability]*	Transfer - Hiring attorney to ensure our IP is protected
CAA regulations	Failure to comply with drone regulations allowing our drone to be operated in the UK [1]. *[High impact - Low probability]*	Contain - Following CAA regulations throughout, including training of users
Leak of ML data	Leak of the ML algorithm or its data leading to the company being sued or to us losing our model *[High impact - Medium probability]*	Contain - Ensure servers are well protected and workers sign NDAs
Manufacture defects	Defect in the manufacture and assembly process leading to our drones being unusable *[Low impact - Low probability]*	Active acceptance - Rigorous quality control and product testing before drones are put in market place

External Risks

Regulations and Laws	Changes to regulations (CAA/government) leads to our drones not being able to fly *[High impact - Very low probability]*	Passive acceptance - Constantly check and change the design if needed
Market competiton	Another company enters the market/company that we weren't aware of *[Medium impact - Medium probability]*	Contain - Protect IP through an attorney and market share via customer relations
Funding constraints for SaR	Lack of funding leads to SaR not being able to purchase our product *[Medium impact - Low probability]*	Active acceptance - Pivot to a different market or location to ensure more funding
Issues around data it collects	Legal GDPR issues. User concerns around being recorded. *[Low impact - Medium probability]*	Active acceptance - Focus on how we can keep the largest percentage of the user base happy
Supplier delays	Supplier delays impact the assembly of our drone *[Medium impact - Low probability]*	Passive acceptance - Have back up suppliers in place in case of complications
Patent claims	Outside company encroach on algorithm IP, leading to our company having to protect it. *[High impact - Medium probability]*	Transfer - Hiring an attorney to ensure our IP is protected

[1] Civil Aviation Authority. (n.d.). Introduction to drone flying and the UK rules. Retrieved June 15, 2024, from https://www.caa.co.uk/drones/rules-and-categories-of-drone-flying/introduction-to-drone-flying-and-the-uk-rules/

Top Down - S&R

The top down analysis focusses on the search and rescue user bases of our product.

PAM

Global Market Figures of S&R Equipment Market were aggregated from multiple sources cross referenced into an average.
This results in a total of £102,500,000,000

TAM

For the total attainable market, we analysed a UK context, systematically breaking the s&r market down into the following categories, (budget of the respective user bases in assumptions page):

- RNLI [7]
- Mountain Rescue England and Wales [5]
- Association of Lowland Search and Rescue [4]
- Scottish Mountain Rescue [8]
- Maritime and coastguard agency [3]
- Air Ambulance Service [6]
- Military Helicopter S&R [2]
- Police force [9]
- Fire service [1]

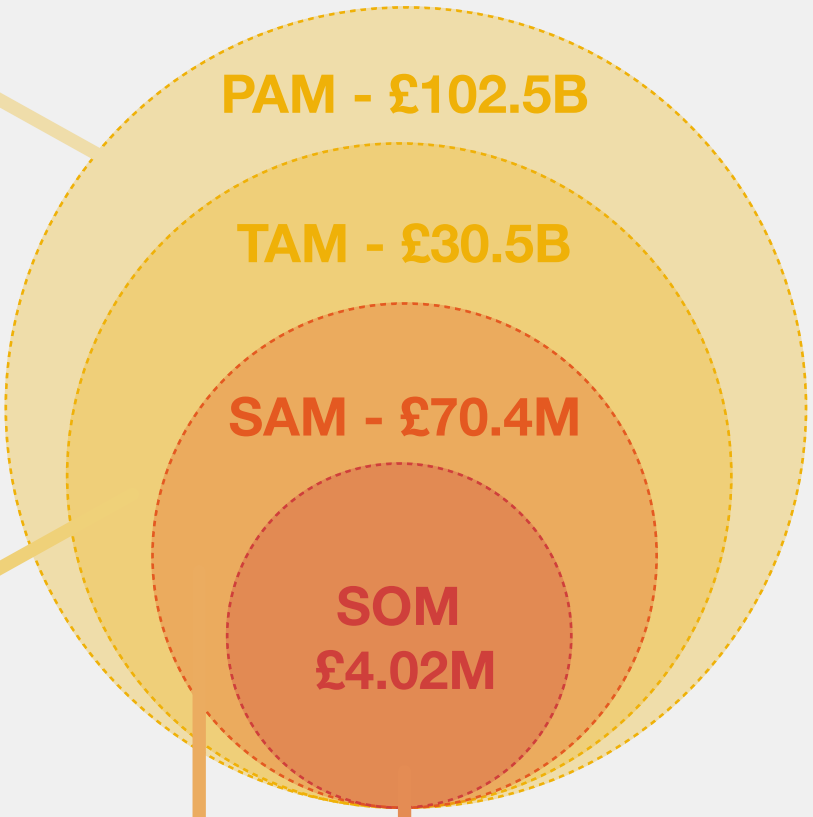
This results in a total of £30,472,400,458

SAM

Looking at the **annual reports** and financial records of the above user bases, we worked out what percentage of their budget is relevant to our product. This was generally around what they spent on locating and rescuing people.
This results in a total of £70,405,296

SOM

We assumed we would capture **5%** of the market size for the S&R companies for which our product is more of a want, but for S&R companies in which it is a need such as mountain rescue, the number was bumped up to 60% given how it covers the vast majority of existing costs. A more systematic breakdown is provided in the assumptions page.
This results in a total of £4,020,168



Bottom Up - S&R

User Base - Service:

- RNLI
- Mountain Rescue England & Wales
- Association of Lowland Search & Rescue
- Scottish Mountain Rescue
- Maritime & Coastguard Agency
- Air Ambulance Service
- UK-Based International S&R
- Military Helicopter S&R

User base - Saas:

- UK Defence Budget
- Police force
- Fire service
- Drone Research Grants
- Google Deepmind
- DJI
- Microsoft

Segmented User Base - Service:

Looking at the number of teams (735), we gave each of them an individual LTV multiplier based on their budget and annual returns. This comes to a total of around 1011 drones sold.

Segmented User Base - Saas:

For Saas, we spoke to experts to calculate a LTV and retention rate for our hypothetical product. Commercial usage had an LTV multiplier of 10 and retention rate of 0.8.

Annual Recurring Revenue - Service:

The revenue over 5 years for the service was calculated from the value theory and the number of drones sold, resulting in a total of: **£19,714,500**

Annual Recurring Revenue - Saas:

Using the LTV Multiplier and customer retention rate we got a total of : **£10,080,000**

Market share:

Assuming a market share of 15%: **£4,469,175**

Value Theory Approach

A value theory approach was created to predict how much (GBP) a user base would pay for our drone service, and software (ML) service **over 5 years.**

The full calculations are on the assumptions page.

LTV Drone Service:

£19,000

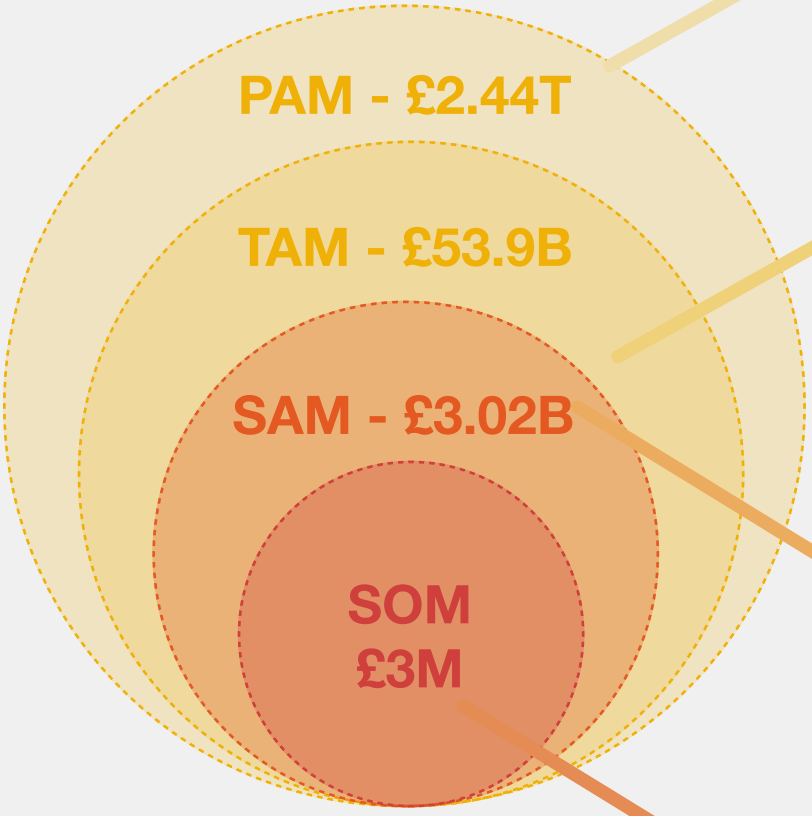
per customer

LTV S.A.A.S:

£180,000

per customer

The top down analysis focusses on the military user bases of our product. (need an etc?)



PAM

Global Market Figures of the Defense Market was obtained on the internet from Sipri and Janes, these 2 figures were aggregated and an average was taken.
This results in a total of £102,500,000,000

TAM

For the total attainable market, we obtained the UK Defense Budget for 2024 [1].
This results in a total of £53,900,000,000

SAM

Looking at the annual reports and financial records of UK Defense, we analysed and quantified how much of their budget they allocate to research and development.
This results in a total of £3,020,000,000

SOM

SOM was calculated by looking at similar investments the UK defence made in other drone companies, such as the one they made with the US based company: Marlborough Communications [2].

This gave us a feasible total of **£3,020,000**

Bottom Up - Defence

Our ML SaaS improves pilot recovery success thus reducing costs, both in failed missions and by reducing re-training. Military organisations can use our product to find downed personnel faster, removing the need for expensive and long search missions.

Revenue Generation

Our SaaS is priced at **£20,000/month** per military base.
With **35 bases** RAF bases in the UK adopting the software [1], we generate **£8.4 million annually**.

Pilot Recovery

Cost of pilot training: **£7 million per pilot**
Global Incidents [2]: **39 downed pilots annually**
Our SaaS **improves recovery success by 10%**, enabling the recovery of **~4 additional pilots annually**
This saves the military **£27,300,000**

Annual Mission Costs

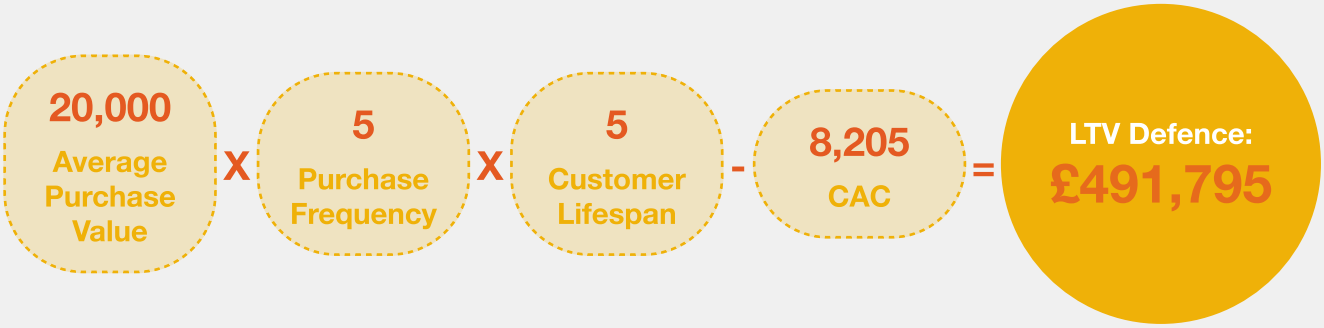
Average Mission Cost: **£50,000**
Cost to deploy Apollo / mission: **£6,154**
Total mission costs: **£1,986,000**

Total savings:

Money saved from missions - Total mission costs
£18,621,000

Value Theory Approach

A value theory approach was created to predict how much (GBP) a user base would pay for our software (ML) service **over 5 years.**



https://www.marketresearchfuture.com/reports/search-rescue-sar-equipment-market-13992

https://www.mordorintelligence.com/industry-reports/search-and-rescue-equipment-market

Assumption and Sensitivities — 35

Assumptions

Market Projections

Relied on accurate market projections from reputable sources.

Market Size %

Assumption of market size captured (5%, 60%) depending on S&R team

SaaS Customers

Assumed this based on past behaviour (churn rate is 0.2) and purchases, greatly changes based on effectiveness of ML model.

Defence Investment

Assumed investment based on current contracts defence have with other drone companies

RAF Bases Adoption

Assume all 35 RAF bases will adopt our software.

Cost Benefit - Defence

Assumed pilot recovery and average mission costs from reputable srouces

S&R Market - Drone Service

Market Penetration	Net Impact
6%	4.75M
5%	4.02M
4%	3.33M

Top Down

We have assumed a 5% market penetration which may vary

Retention Rate	Net Impact
0.9	4.66M
0.8	4.47M
0.7	4.28M

Bottom Up

We have assumed a 0.8x retention rate for the SaaS for S&R companies which may vary

Market Share	Net Impact
20%	5.95M
15%	4.47M
10%	2.99M

Bottom Up

We have assumed a 15% market share of the S&R teams

Defense Market - SaaS

SaaS Price (per month)	Total Savings
£5000	23.2M
£3000	24.1M
£2500	24.5M

Bottom Up

The price we sell our software for can vary which will affect cost savings for the military (affect on revenue projections on page 53)

Recover Success Rate	Total Savings
20%	51.4M
10%	24.1M
5%	10.4M

Bottom Up

This demonstrates that even small improvements lead to drastically higher savings, making our SaaS an essential investment.

Average Mission Cost	Total Savings
£100,000	22.1M
£50,000	24.1M
£10,000	25.9M

Bottom Up

The recovery success rate of each mission may vary which will drastically affect savings

Sales Channels

Years 1-2 Immediate Sales Channels (S&R):

Our current focus is on **selling drones to S&R teams**, emphasising our value added of the ML model which reduces search times and costs. These deployments will also allow us to gather real world data to improve our ML model.

Year 3 Future Sales Channels (Defence):

Through partnerships with DASA and the data we collect from S&R teams, our ML model will evolve into a powerful data insights tool for predicting lost persons. This data would be incredibly valuable to Defence, offering high-value, defence-specific insights to them, evolving our focus into SaaS.

Bullseye Framework

To gain access to our customer segments, we ideated through different sales channels with a Bullseye framework. ***Full justification of ICE Framework (Appendix page 24)***

ICE Framework

Referrals (Word of Mouth) = 720
Trade Shows = 567
Demos = 567
Pilot Programs = 540
Conferences & Industry Events = 448
PR Around Our Drone = 448
Partnerships with S&R Teams = 384
Business Development = 392
Public Relations (General PR) = 392
Direct Outreach = 294
Social Media Marketing = 294
Investment Funds = 336
Cross-Government Relations = 280
Speaking Engagements = 288
Offline Events = 252
Community Building = 210
SEO = 210
Target Market Blogs = 210
Engineering as Marketing = 210
Unconventional PR = 210
Newspapers = 240
Affiliate Marketing = 144

Sales Channels Strategy

Immediate Sales Focus (S&R)

Goal: Generate revenue, build credibility, and collect operational data for ML refinement

Selected Primary Sales Channels (Sales Funnels to be followed)

Pilot Programs

Builds **credibility**, gathers **data** and **proves value** of product.

→

Trade Shows

Direct access to **big sellers, live demos**, **large scale awareness**.

→

Referrals

Low-cost, **high-trust channel** leveraging customer satisfaction.

Potential Secondary Sales Channels

Direct Outreach

Why: Low CAC and relationship forming**.**
Targets: Expand into existing S&R team clusters.

+

PR/Press

Why: Amplifies credibility and results from pilots and trade shows.
Targets: Broader SaR and defence audiences **globally**

+

Partnerships

Why: Access resources (eg funding and data) and **establishes trust** with larger organisations (universities, military and defence).

How this strategy fits with our Business Model...

- SaR is a low barrier, high credibility market that helps establish trust and a track record for future sales
- The ML model improved with real world deployments and data collection setting the stage for future SaaS sales

Future Sales Focus (SaaS for Defence)

Goal: Monetize ML insights for defense applications, such as terrain analysis and disaster prediction.

Selling Lost Person Behaviour ML Model

Partnerships

Leverage relationships with DASA and other defence organisations for **contracts and scaling**.

+

Government Platforms

Use **approved procurement frameworks** (eg: MOD or NATO channels) to distribute insights

+

Direct Sales

Target high-value defence customers directly with **tailored proposals for specific data applications**.

How this strategy fits with our Business Model...

- As data insights become a high-margin revenue stream once ML capabilities are proven.
- Defence procurement relies on credibility, which will be built through DASA funding and S&R use cases.

1. Credibility Building: Pilot Programs — 38

Outline and Objectives

1. Prove Product Feasibility

- Deploy **free drones** to smaller S&R teams for **6 months** in exchange for **operational data** and **testimonials**.
- **Showcase benefits** like reduced search times and improved efficiency using ML-driven analytics.

2. Build Credibility

- Leverage testimonials and operational data from pilot teams to **establish trust** and **evidence** for other **potential customers**.

3. Collect ML data

- Collect real-world **topological** and **lost person** behaviour data to **strengthen ML model**

4. Generate Referrals and Expand Outreach

- Use successful pilot programs as a **springboard for word-of-mouth and broader marketing efforts.**

AWARENESS

Teams Contacted
15 aware teams

Action: Outreach to 50 small S&R teams via emails, LinkedIn campaigns, and direct calls.
Assumptions: 50 S&R teams targeted; assume a 30% positive response rate.
Metrics: 50 x 0.30 = 15 teams

INTEREST

Teams Interested
8 interested teams

Action: Provide detailed proposals and benefits of the pilot (e.g., free drones, data insights).
Assumptions: 50% of aware teams express interest in exploring further.
Metrics: 15 x 0.50 = 8 teams

EVALUATION

15% interest rate
6 evaluating teams

Action: Conduct personalized meetings with interested teams and showcase MVP features.
Assumptions: Assume 80% of interested teams move forward after evaluations.
Metrics: 8 x 0.80 = 6 teams

ENGAGEMENT

50% Demos Scheduled
5 engaged teams

Action: Negotiate terms of free pilot deployment (data sharing, testimonials) and finalise agreements.
Assumptions: 5 of 6 teams agree to the pilot program (one backs out due to internal barriers).
Metrics: 6 x 0.83 = 5 teams

ACTION

60% retention
3 paying customers

Action: Run 6-month pilot program, collect data and follow up with offers for paid contracts.
Assumptions: Assume 60% of pilot participants convert into paying customers.
Metrics: 5 x 0.60 = 3 teams

Revenue

£114,000
3 customers x LTV

Total Costs

£90,200
Breakdown in page x

CAC

£30,067
£90,200 ÷ 3 customers

2. Lead Generation: Conferences & Events 39

Events	Type	Details
Emergency Services Show *(Birmingham)*	Booth & Live Demos	- Live drone demos with simulated search-and-rescue scenarios - Booth with marketing materials and product showcases
DSEI (*Defence and Security Equipment International, London*)	Speaking Engagement & Booth	- Keynote presentation on ML-powered drones for defence and emergency applications - Booth for follow-up discussions
Search and Rescue Conference UK *(Sunderland)*	Case Studies & Demos	- Present case studies from pilot programs - Live demo focused on terrain analysis and photogrammetry features
Future Robotics Forum *(Manchester)*	Panel Discussion & Booth	- Participate in a panel discussion on innovation in robotics - Highlight drone features and SaaS model
Avalanche Awareness and Mountain Safety Conference *(Scotland)*	Specialised Demo	- Target mountain rescue teams with demos showcasing thermal capabilities in snowy/ remote conditions

Objectives based on chosen events:

1. **Raise awareness** and **establish credibility** of the value propositions of Apollo
2. **Generate interest** and **leads** for follow-ups
3. Conduct **live demonstrations** and provide **testimonials** and **case studies** from pilot programs
4. Convert attendees into **customers**

AWARENESS

Conference Outreach
1500 aware leads

Action: Attend 5 key trade shows
Assumptions: 2,000 attendees/event x 5 events = 10,000 attendees.
Metrics: 6,000 x 0.15 = 1500 aware leads

INTEREST

People Engaged
540 interested leads

Action: Engaging booth visitors with live demos, testimonials, and product literature.
Assumptions: 60% of aware leads visit the booth or engage with the materials.
Metrics: 900 x 0.60 = 540 interested leads

EVALUATION

Follow Ups
162 evaluating leads

Action: Hold detailed follow-ups via calls, emails, or meetings to qualify leads.
Assumptions: 30% of interested leads are serious prospects and engage in detailed discussions.
Metrics: 540 x 0.30 = 162 evaluating leads

ENGAGEMENT

50% Engagement Rate
81 engaged leads

Action: Provide pricing, use cases, and optional trial agreements for serious prospects.
Assumptions: 50% of evaluating leads move closer to making purchasing decisions.
Metrics: 162 x 0.50 = 81 engaged leads

ACTION

Final Contracts
16 paying customers

Action: Finalise contracts with paying customers post-event follow-ups.
Assumptions: Assume 20% of engaged leads convert into paying customers.
Metrics: 81 x 0.20 = 16 paying customers

Revenue

£608,000
16 customers x LTV

Total Costs

£60,000
Breakdown in page x

CAC

£3,750
£60,000 ÷ 16 customers

3. Organic Growth: Referrals

"Lead By Example"

1. Why it Works with our Strategy

It is a **low-cost, high-trust channel** that complements the other two funnels. It **capitalises on existing customer satisfaction**, making it cost-effective and reliable for lead generation and conversion.

2. Easy Scaling Through Clusters

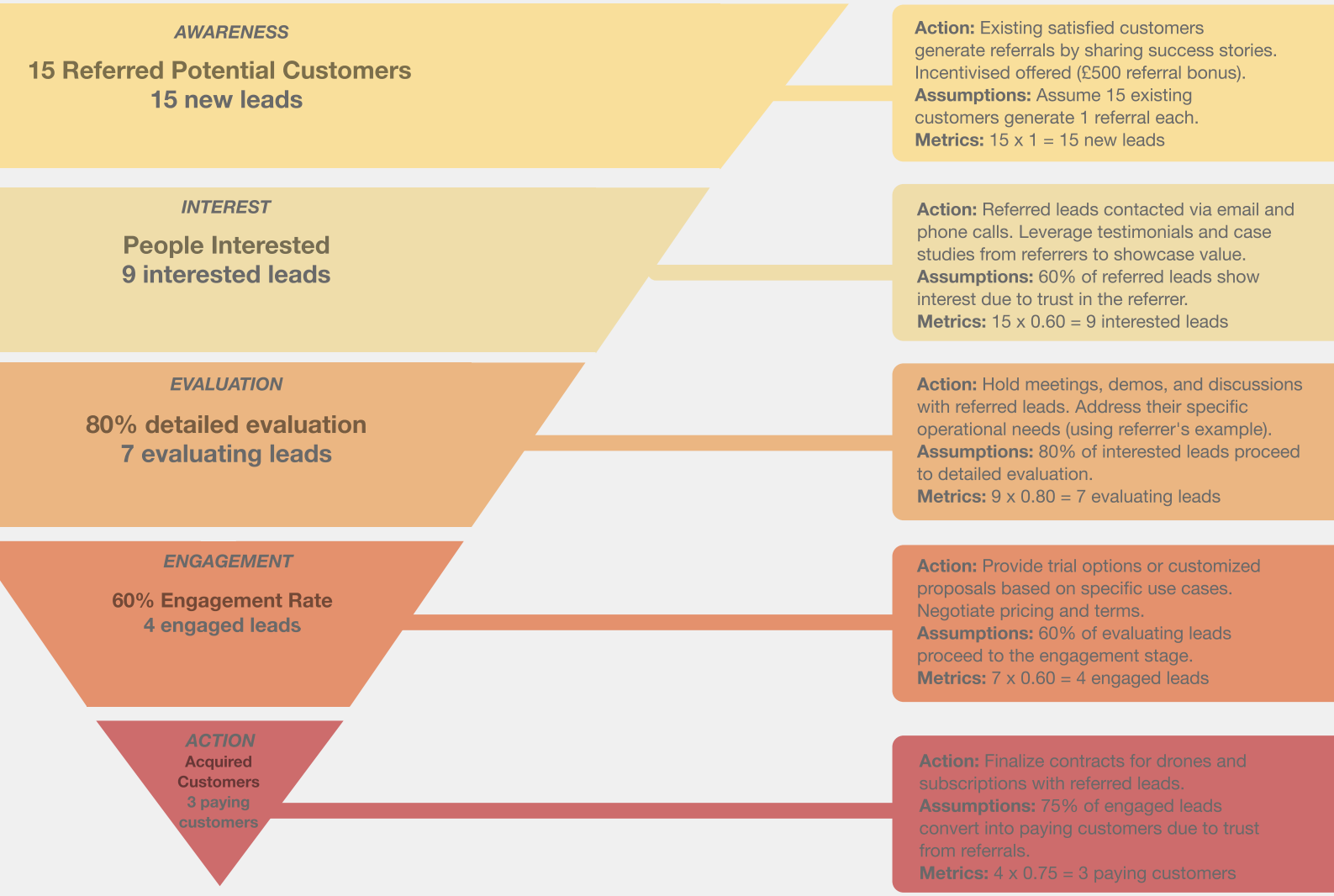
The **regional clustering of SaR teams** (eg 14 mountain rescue organisations in the UK or 25 coastal SaR teams) makes referrals even more effective. **Teams that operate in similar terrains often share best practices**, accelerating word-of-mouth growth.

3. Want Real World Validation

In the SaR industry, many organisations prefer technologies that have already been **proven in the field due to life and death**, making referrals a natural channel for scaling.

4. Past Trends

A 2019 Frost & Sullivan study on EMS technology adoption found **over 60% of procurement decisions for new technologies were influenced by recommendations** between agencies.



Revenue
£114,000
3 customers x LTV

Total Costs
£6,500
Breakdown in page x

CAC
£2,167
£6,500 ÷ 3 customers

Route to Market Strategy

The following pages qualitatively outlines our business strategy from pre-seed to scaling globally incorporating assumptions made. *(Full Systematic Breakdown of Costings follow route to market pages)*

Initial Funding (Months 1-3)

Objective

Secure funding to develop a **Minimum Viable Product (MVP)** that demonstrates core functionality and provides a foundation for pitching to DASA.

Activities

1. VC and Grants Funding (70/30 Split):

-Use *Grantify* to apply for non-dilutive grants (70% of funding) [1], focusing on Innovate UK and Horizon Europe
-Raise 30% of funding (£x) from *VCs* which will be dilutive in nature and help develop relationships for further investing later on.

2. Build MVP:

-Develop a functional proof of concept drone with ML capabilities for initial demonstrations. ***(expanded on page x)***
-Ensure the MVP can perform **thermal imaging** and **basic terrain analysis** using ISRID topographical data integration.

Costs

- **MVP Development: £40,000 (materials, engineering, and basic ML training).**
- **VC & Grant Total: £177,700 (from the earlier cost breakdown).**

Output

- **A functioning MVP drone.**
- **Prepared technical material (data analysis, use cases) to approach DASA.**

DASA Funding for R&D (Months 3-5)

Objective

Leverage DASA's funding and resources to enhance the product's technical capabilities and **build multiple drones for the pilot stage.**

Activities

1. Apply to DASA for R&D Grants

Pitch MVP results to DASA to secure funding for: full drone development, scaling the ML model to handle complex datasets, enhancing product durability and capabilities for military and civilian use.

2. Expand ML Model:

Integrate real-world data from DASA's network into the ML training process.
Focus on improving: target accuracy and photogrammetry and terrain mapping capabilities.

3. Build Pilot Ready Drones:

-Use DASA funding to manufacture 4 drones for the pilot stage.
-Include redundancy systems to reduce catastrophic failure risks.

Costs

- **R&D Funding Needs: £100,000 from DASA for advanced development.**
- **Drones: £1,307 x 5 = £6,850 (manufacturing).**

Output

- **5 fully functional drones ready for pilot testing.**
- **Enhanced ML model with improved accuracy.**

Pilot Programs (Months 6–12)

Objective

Test the product in real-world environments, collect operational data, and **build evidence to attract additional VC funding for scaling.**

Activities

1. Pilot Programs:

-Deploy 4 drones to 4 S&R teams, offering free use for 6 months
-Receive back: operational data and testimonials/case studies for marketing.
-Evaluate success metrics: reduced search time and cost savings.

2. Collect Data For Scaling:

-Use pilot data to refine the ML model further.
-Gather operational insights to improve drone hardware (e.g. battery efficiency, durability).

3. Secure Additional Funding (Post-Pilot)

-Use pilot success as evidence to approach: VCs for growth funding (£50,000–£100,000)
-Potential partnerships with defence contractors or military stakeholders.

Costs

- **Pilot Deployment: £17,200 (maintenance + pilot salaries + outreach).**
- **Additional VC Funding Needed: ~£50,000 for scaling and conferences.**

Output

- **Strong case studies and real-world testimonials.**
- **Evidence for VC pitches and marketing campaigns.**

S&R Market Expansion (Months 9-24)

Objective

Showcase pilot success at conferences and events while **conducting direct outreach** to convert medium-sized S&R teams into paying customers.

Activities

1. Attend Trade Shows and Events

-Use pilot case studies to highlight faster search times and lower costs.

1. Direct Outreach to Medium-Sized S&R Teams:

-Contact 10–15 medium-sized teams with tailored pitches.
-Offer trial periods (2–3 months) for first-time paying customers.

3. Publish Case Studies:

-Publish 2–3 case studies in industry publications (e.g., Wired, Defense News)
---Issue press releases to build industry credibility.

Costs

- **Events: £15,000/event x 3 = £45,000.**
- **Marketing & Outreach: £4,000 (direct outreach materials).**
- **Publishing: £10,000 (press releases, case study creation).**

Output

- **Generate 1,500 leads from events.**
- **Convert 10% (150 leads) into paying customers.**

SaaS Global Expansion (Months 24-36)

Objective

Establish a fully-fledged business by scaling operations in the UK, targeting larger S&R teams and NATO partners **and expanding globally.**

Activities

1. Target Larger S&R Teams:

-Incentivise referrals with current customers with discounts or bonuses
-Use testimonials/referrals to get contracts with bigger S&R teams.

2. Expand to NATO Partners:

-With expanded dataset and strengthened ML model, use connections with DASA and UK military endorsements to sell to NATO partners.

3. Leverage Word-of-Mouth for Growth:

-Encourage smaller and medium sized S&R teams to use Apollo as it becomes a trusted industry standard.

4. Increase SaaS Sales:

-Begin selling operational data insights to military organisations (£20,000/month per contract).
-Secure 3 military SaaS contracts within the first year of global expansion

Costs

- **Scaling Operations: £50,000 (team expansion, manufacturing).**
- **Marketing & PR: £20,000 (global campaigns and trade shows).**

Output

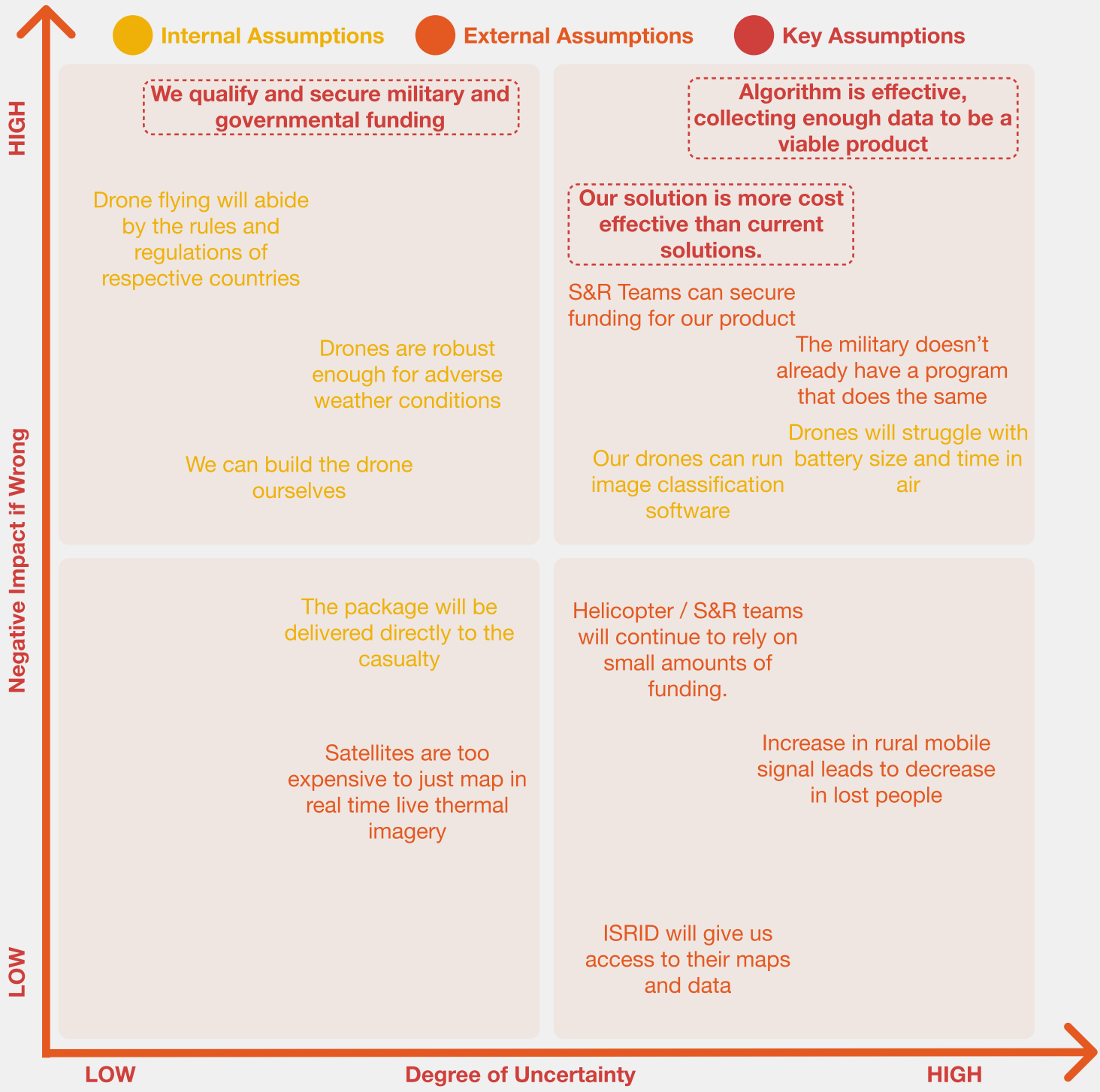
- **Add 20–30 new customers annually.**
- **Reach £1.5M ARR within the first year of scaling.**

Summary

Stage	Key Activities	Timeline	Outputs
1. Initial Funding	Secure VC (30%) + grants (70%), build MVP.	Months 1-3	MVP drone built, funding secured
2. DASA Funding	Use DASA to fund R&D and build 4 drones.	Months 3-6	Drones ready for pilot programs
3. Pilot Programs	Deploy 4 drones, collect data, secure VC growth funding.	Months 6-12	Pilot success metrics and testimonials
4. Market Expansion	Attend events, outreach to S&R teams, publish case studies.	Months 12-24	Paid contracts with >10 new customers.
3. Global Scaling	Expand to larger S&R and NATO partners, increase SaaS sales.	Months 24-36	£1.5M ARR and 25 new customers globally.

Our Assumptions

The following assumptions were taken from **the whole document** around the research, ideation and development of both the product and business side.

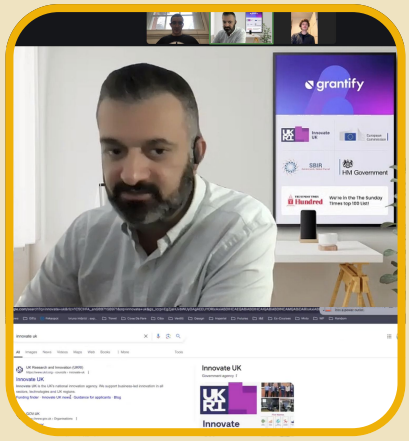


Assumption Validation 45

Funding

Neel Sarna - Grantify Regional Head:

- Initial response: able to qualify for Innovate UK R&D Grant
- After securing this, we would be able to other grants such as DASA
  - Grantify have secured this with a drone based company before
- This aligned with the the advice we received from Phillipos.



Research:

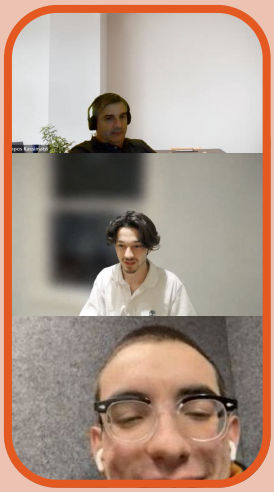
Aligns with Innovate UK's priorities of:

- Combing cutting-edge technology for real-world applications
- Improving natural disaster response by identifying accessible routes in disaster-stricken areas for rescue and relief operations
- Having a scalable approach allowing expansion to other use cases

Cost Effectiveness

Phillipos Kassimatis

- Validated the cost effectiveness by looking at the recovery value of assets.
- Pivoting to a subscription based service allows us to be cost effective to SAR services even with limited funding
- Stated that by working with the UK government to develop the algorithms it is likely that the government will subsidise use of it in SAR organisations



Research

- Looked at the value saved within the Military in the aspect of saving the life of a downed pilot:
  - Given it costs £7 Million to train a new pilot, just saving 10% more pilots in a year would save the military 28 million pounds.
- For S&R, given the current DJI drones they use cost them between 6,000 to 8,000 pounds, ours is a much cheaper yet more effective solution.

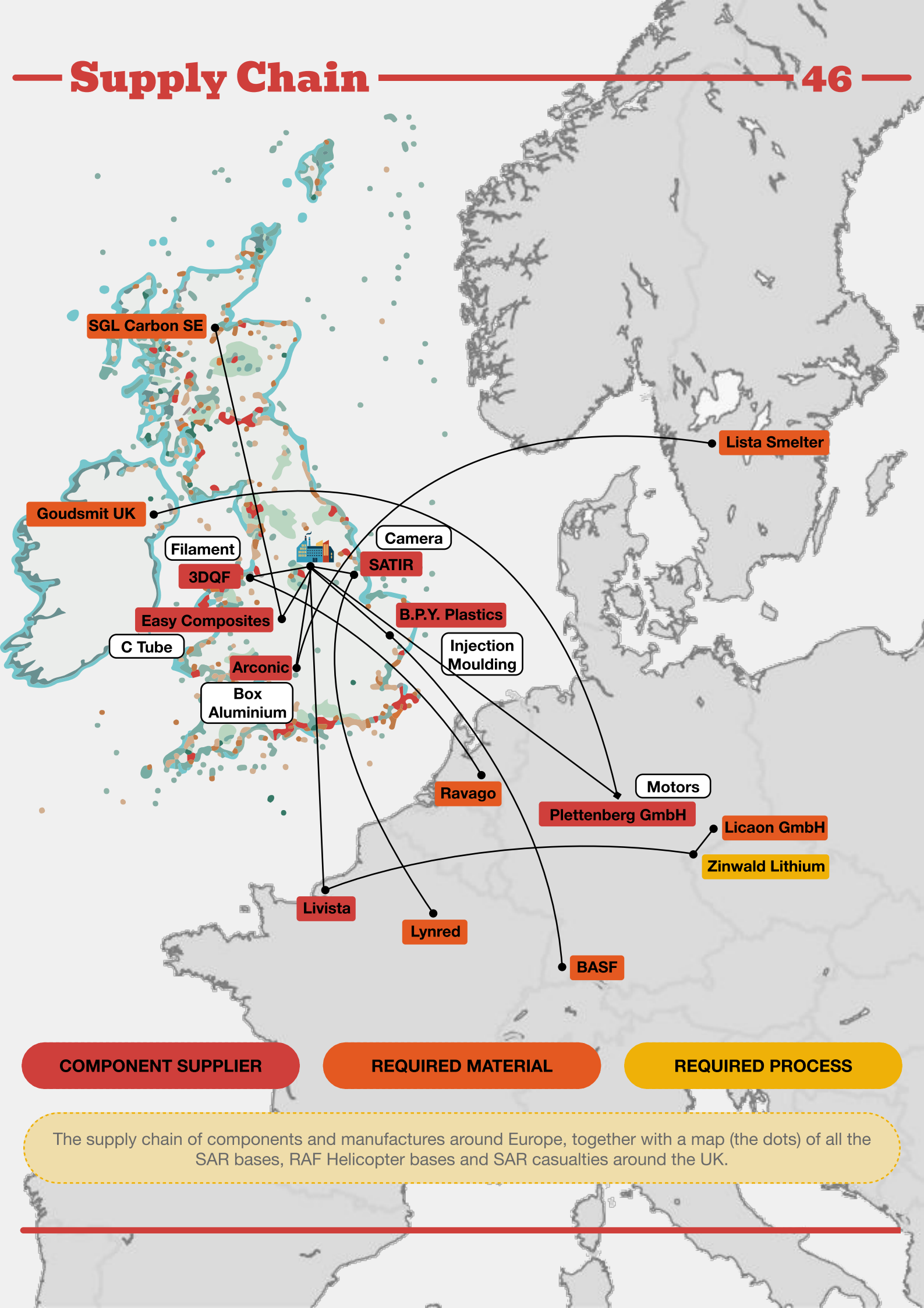
ML Algorithm

Demetris Soukeras

- By looking at existing research papers and confirming their logic, he validated:
  - ML algorithm design structure
  - Use of the image classification for undocumented path detection and mapping
  - The model's ability to give an optimal search area to locate the stranded individual faster using different LP's.
- Operational stress is heavy for testing, but when the model has been run, the outputted algorithm will be able to run on the drone.

Research

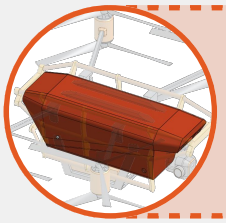
- Models with similar applications have been seen to work in real world testing in academic papers
- Image classification software is improving with publicly available tools being available that can recognise trails, maps and other landmarks in real time.
- Satellite maps are accurate enough to recognise undocumented paths.



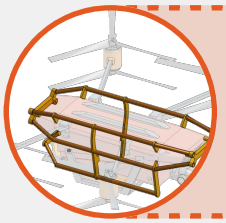
COMPONENT SUPPLIER | REQUIRED MATERIAL | REQUIRED PROCESS

The supply chain of components and manufactures around Europe, together with a map (the dots) of all the SAR bases, RAF Helicopter bases and SAR casualties around the UK.

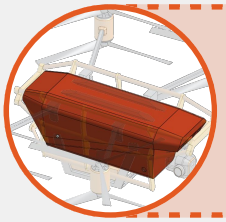
Components Costs



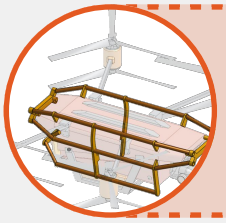
Main Shell £5.68
Injection Moulded with PA66 With Glass Fibre
B.P.Y Plastics
BASF



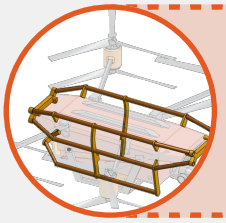
Crash Cage £10
3D printed and cut with ABS and Carbon tubes
3DQF
SGL Carbon SE



Landing Gear £1.29
3D printed with ABS
3DQF
Easy Composites



Drone Arms £10
Cut to length 10mm aluminium box section
Arconic
Lista Smelter



Motor Holders £10
3D printed with ABS
3DQF

Pre-Manufactured Components

T-Motor MN3150
£140.32 for 8

Flight Controller
£39.95 for 1

Skyroid C12
£299.99 for 1

Radio Module
£25.90 for 1

Pogo Pins
£9.60 for 8

Radio Antennae
£40.94 for 1

Cube Autopilot System
£289.99 for 1

Transmitter
£62.99 for 1

LIDAR Sensor
£120.12 for 4

GPS Module
£15.70 for 1

LIPO Battery
£146.75 for 1

Lights
£10.96 for 1

All of these numbers and figures were calculated using various websites and quotes that can be seen on the costings assumption page in the appendices.

Server Costs

To train our algorithm, we must rent servers and update once a month (having to run for circa 24 hours at a time. Cost based on AWS Bedrock's model [1]:

Cost of running the model for 1 min: **£0.062**

Expected annual run time: **17280 min**

Total: **£1,089.72**

Timings between re-runs changing based on model improvement.

TOTAL

£1089.72
Server costs / year

£1370
Cost / drone

Sales Channels Costs

Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Trade Shows	Cost of attending 3 trade shows annually (booth, travel, materials).	54,000	£18,000 per event; 3 events per year.	Based on average trade show costs (industry norms for booths and travel).
Referrals	Incentives for referred customers and PR for word-of-mouth campaigns.	6,500	£500/customer + £5,000/year for PR/case studies.	Typical referral bonuses and PR costs in B2B industries.
Pilot Programs	Cost of running 4 pilot programs annually (drones, maintenance, deployment support).	16,028	£1,370/drone + £137 maintenance + £10,000 monitoring/support.	Drone part costs calculated, typical pilot monitoring costs validated through industry.
Direct Outreach	Tools (e.g., LinkedIn Sales Navigator) and labor for personalized outreach efforts.	24,000	£4,000 tools + 1 sales staff 50% time = £20,000/year.	Based on LinkedIn pricing and B2B outreach best practices.

Employee Costs

Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Engineers (x2)	Salary for software and drone engineers working on product development and maintenance.	100,000	£50,000/year/ engineer.	Typical engineer salary based on industry averages in the UK.
Data Scientists (x2)	Salary for data scientists managing ML model updates and operational data.	90,000	£45,000/year/ data scientist.	Salary range aligns with data roles in R&D-focused startups.
Sales & Marketing (x2)	Salary for staff managing sales channels (e.g., trade shows, outreach).	80,000	£40,000/year/ staff member.	Consistent with average pay for sales in tech startups.
Administrative Staff (x1)	General HR, legal, and operational support staff.	35,000	£35,000/year.	Standard admin salaries validated against industry norms.
Lab Space	Lab Space	Lab Space	Lab Space	Lab Space
Office Space	Office Space	Office Space	Office Space	Office Space

Product Costs

Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Drone Assembly	Cost of assembling a drones.	1,370	£1,370 per drone	Based on a previous breakdown of drone part costs.
Drone Maintenance	Annual maintenance cost for 4 drones.	548	£137/unit	Maintenance cost validated as 10% of hardware value annually (standard hardware upkeep).
Consumables	First Aid Kits and shipping costs for drones.	20	£20/kit	Consumable pricing based on bulk costs of medical kits and average shipping costs.

Operational Costs

Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Server Costs	Costs of running servers for monthly ML model updates and data integration.	1,089.72	Page 51	Pricing sourced from AWS and cloud server providers.
Legal Consultations	Legal and regulatory compliance consultations (1x per month).	3,000	£250/month for solicitor consultations.	Industry averages for solicitor fees.
Accounting Consultations	Monthly tax and accounting support services.	2,400	£200/month for accounting services.	Aligned with small business accounting service costs.
Regulatory Compliance	Licensing and certifications for drone operations.	500	£500/year estimated for drone and ML software compliance.	Based on average licensing fees for tech businesses in regulated industries.
Lab/Office Space	Co-working space for R&D and operations.	18,000	£1,500/month for shared office space.	Validated against average co-working lab and office costs.
Warehousing	Storage space for drones, parts, and consumables.	2,400	Office £200/month for small warehouse space.	Industry benchmarks for small-scale warehousing costs.

Customer Acquisition Costs (CAC)

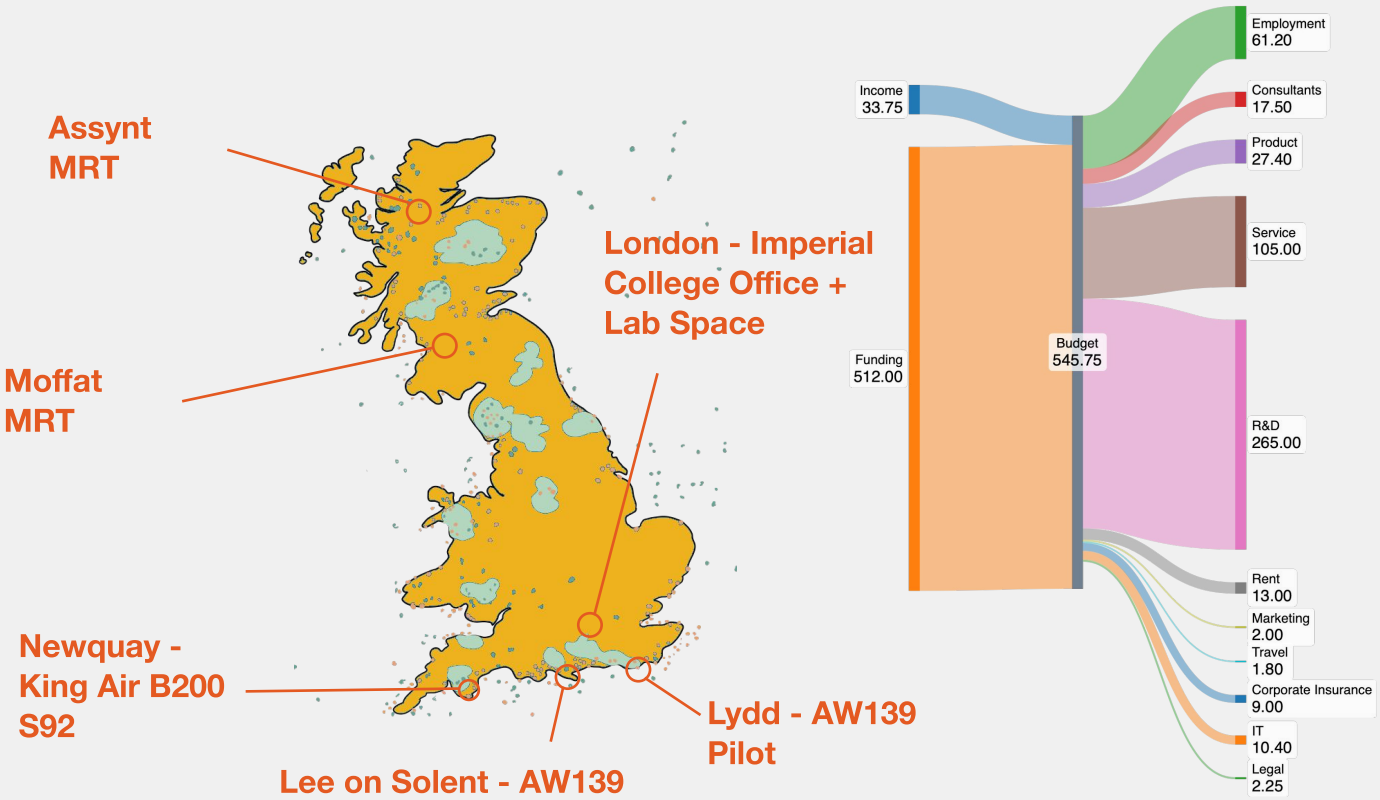
Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Total Sales Channel Costs	Total costs to acquire new customers in a year	100,528	Previous Page	Breakdown of different sales channel costs
Sales Work Costs	Total costs to hire sales staff and sell our product to clients	80,000	Previous Page	Consistent with average pay for sales in tech startups.
CAC	Calculated for the 22 expected customers in the	8,205	Calculated for the 22 expected customers in the sales funnels	Based on the assumptions in those pages

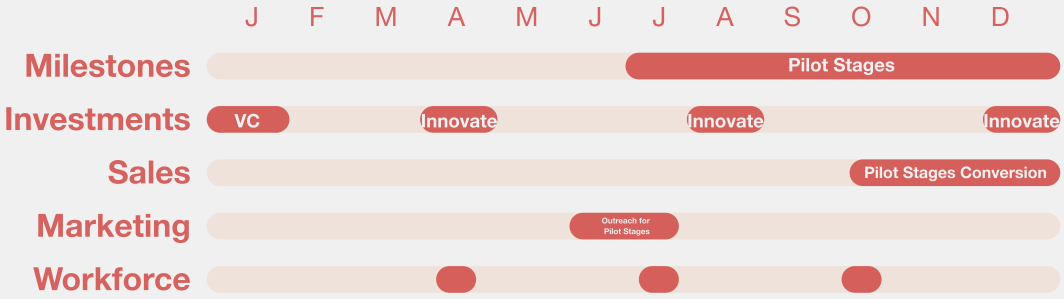
Lifetime Value (LTV)

Cost Type	Explanation	Value (£)	Assumption	Assumption Validation
Initial Product Sale	Initial deposit for the drone	2000	£1370 production cost per drone	Covers production costs - look at breakdown of product costs
Subscription Revenue	Yearly subscription for the maintenance and access to software	3000	Fits within SAR teams funding	Aligned with commercial software service pricing.
Maintenance Cost	Expected costs for major damages per year	500	Most likely components to break are propellers, motors and plastics	Calculated via the cost breakdown of components
LTV	Calculated for the expected 5 years they'll use our product for	11295	Expected 5 years of subscription minus the CAC	Average time a user will use our drone before upgrading to a newer model

Year 1

Goal: Develop an MVP, receive VC and grants. Receive DASA funding and develop university connections, implement pilot programs



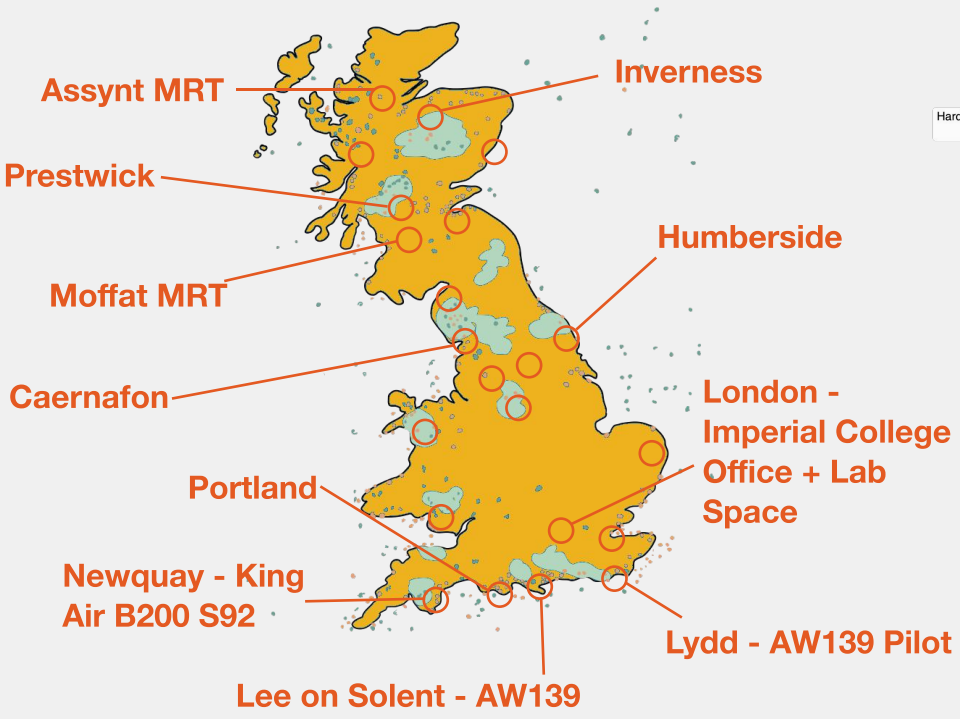


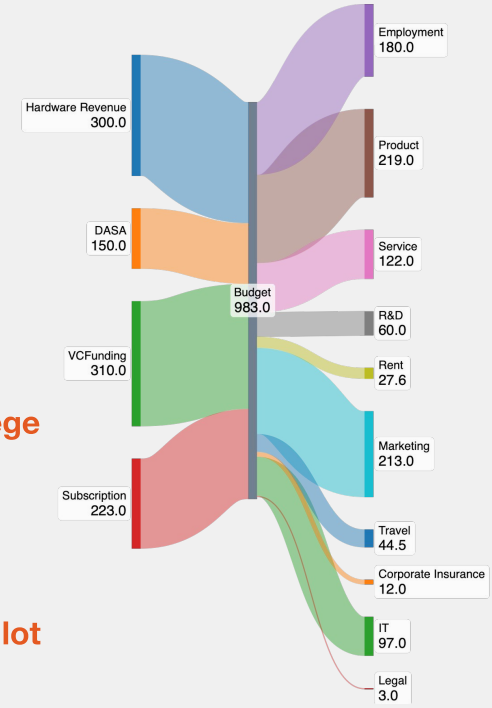
Income	**£545,814**
Investments	£512,064
Own	£33,750
Costs	**£515,161**
Employment	£78,700
Product	£316,156
Service Cost	£120,305
Drones In Deployment	**10**

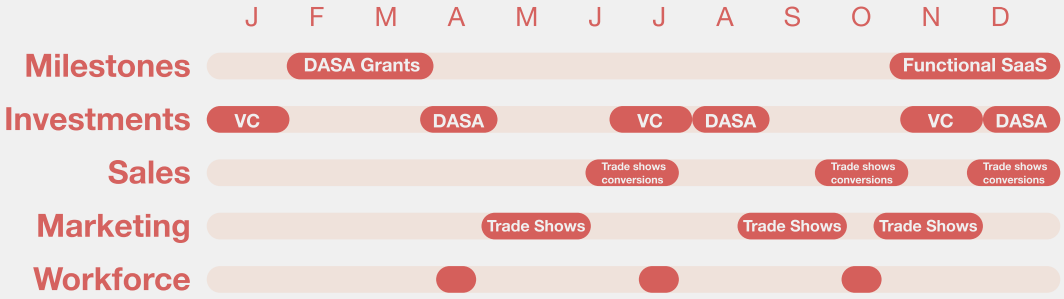
Cumulative Profit ***£30,653***

Year 2

Goal: Market Expansion; showcase pilot success at conferences and events while conducting direct outreach to convert medium-sized S&R teams into paying customers.





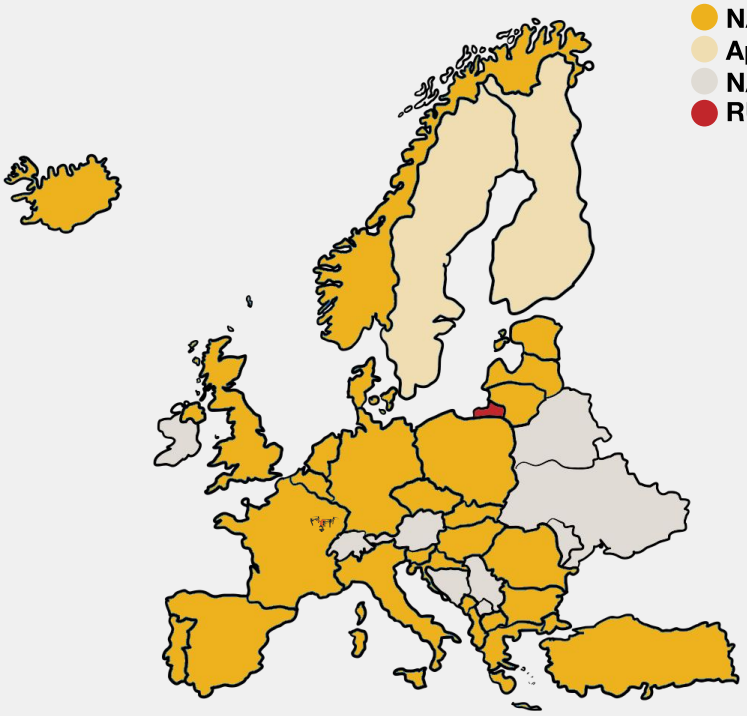


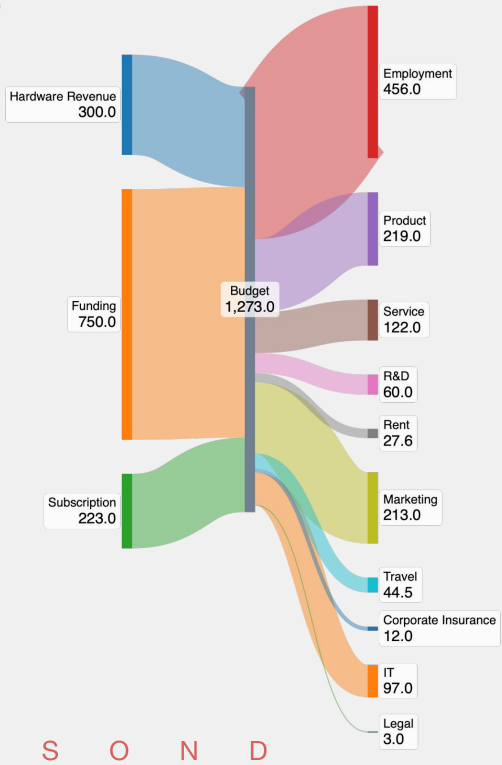
Income	**£982,500**
Investments	£460,000
Own	£522,500
Costs	**£977,734**
Employment	£179,583
Product	£403,721
Service Cost	£394,430
Drones In Deployment	**160**

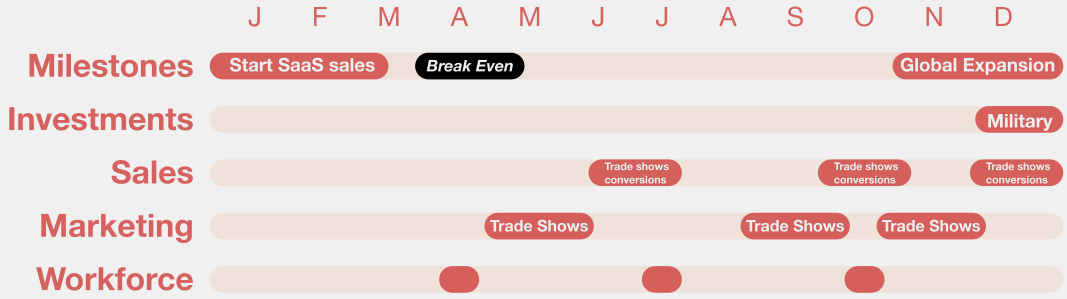
Cumulative Profit ***£35,419***

Year 3

Goal: Establish a fully-fledged business by scaling operations in the UK, targeting larger S&R teams and NATO partners and expanding globally.





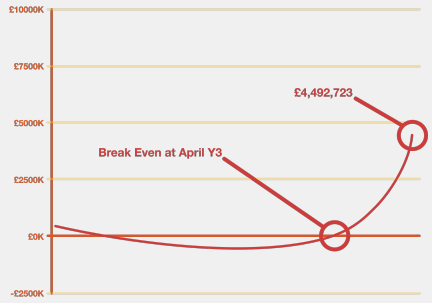


Income	**£6,267,500**
Investments	£3,000,000
Own	£3,267,500
Costs	**£1,810,196**
Employment	£472,917
Product	£659,109
Service Cost	£678,170
Drones In Deployment	**310**

Cumulative Profit ***£4,492,723***

Expected case

Based around all of our previous assumptions and projected PnL

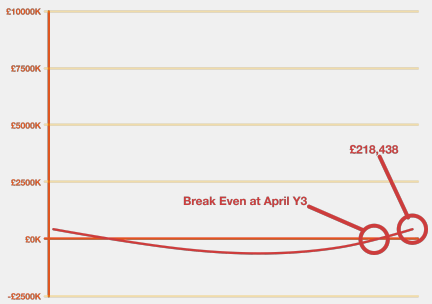


Expected Case

	Number	Timeline	Assumption
Sales/Year	150 / Year	Y2/Y3	Based on pilot program
DASA Investments	100k	Y1 / Y2	Based on average investment
SaaS Sales	10 / Month	Y3	Based around user base LTV
Defence Contract	3M	Y3	Based on Top Down Model
SaaS Price	£3000 / month	Y3	Based on sensitivity analysis

Worst case

Based around changes to the projected PnL based on lower sales and prices

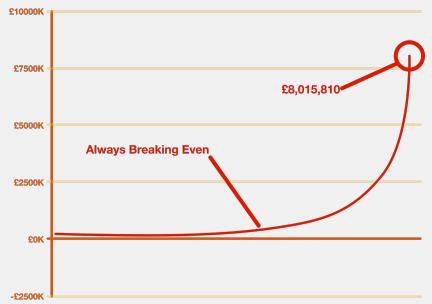


Worst Case

	Number	Timeline	Assumption
Sales/Year	100 / Year	Y2/Y3	50% of expected sales
DASA Investments	50k	Y1 / Y2	Based on lower than average investment
SaaS Sales	5 / Month	Y3	50% of expected sales
Defence Contract	0	Y3	Failure to get contract
Saas Price	£2000 / month	Y3	Based on sensitivity analysis

Best case

High sales and investments, together with a higher sales price



Best Case

	Number	Timeline	Assumption
Sales/Year	200 / Year	Y2/Y3	150% of expected sales
DASA Investments	200k	Y1 / Y2	Based on average investment
SaaS Sales	20 / Month	Y3	150% of expected sales
Defence Contract	5M	Y3	Failure to
Saas Price	£5000 / month	Y3	Based on sensitivity analysis

Financial Forecast Evaluation — 54

Expert Opinion:

We went back to Phillipos Kassimatis, co-founder of a Private Equity firm, to discuss and **evaluate the details of our P&L statements and business plan**

Year 1

Key Considerations and Risks:

- **Funding Reliance:** VCs or grant approvals could be delayed which could stall MVP development.
- **Grantify Costs:** Fees are non refundable even for unsuccessful applications.
- **Matching Funds:** Grants require 30% self funding which increase pressure to get VC funding
- **Technical Risks:** Challenges in ML dev would negate the value proposition for our whole business.

Mitigation:

- **Funding Contingency:** We are to ensure timelines for grant applications are realistic, and explore contingency funding options and carefully assess the most likely grants to succeed.

Year 2

Key Considerations and Risks:

- **Rising Costs:** Workforce and marketing costs (£68,000) strain budgets and result in modest profits on year 2, but are critical for showcasing product success
- **Marketing Dependency:** Low pilot conversions and use cases of drones could impact revenue growth.
- **Competition:** Established S&R companies could slow down our customer acquisition.

Mitigation:

- **Pilot Tracking:** Create and monitor KPIs to address performance issues early.
- **Targeted Marketing:** Use testimonials and case studies for trade show campaigns.
- **ML Differentiation:** Continuously strengthen our ML model on database using pilot data.

Year 3

Key Considerations and Risks:

- **Revenue Diversification:** SaaS (about 50% income) reduces hardware reliance, but slow adoption due to long time to strengthen ML database poses risks to revenue
- **Scaling Costs:** Workforce and SaaS infrastructure drive higher costs.
- **Regulations:** Drone and SaaS laws may delay scaling in new markets.
- **Geopolitical Risks:** Defence clients are sensitive to international politics.

Mitigation:

- **Credibility:** Leverage DASA and UK endorsements to build trust within global community
- **Clear SaaS Use Cases:** Highlight very clearly of our predictive analytics and defence applications.
- **Compliance:** Localise solutions to meet regional regulations.

Moving Forward Year 4+

We evaluated 5 different paths that we could take after successfully completing 3 year plan:

1 Expand SaaS Globally

Plan: Scale SaaS sales to global defence organisations, leveraging DASA and NATO military endorsements.
Considerations: Geopolitical risks (ie hostile countries to NATO) and regulatory hurdles may slow adoption.

Alignment: High
Feasibility: High
Profitability: High
Risk: Moderate

2 Diversify into Civilian Markets

Plan: Target civilian industries like mining, oil and gas, and disaster management by repurposing ML and drone capabilities for logistics, safety, and terrain analysis.
Considerations: Opens up massively untapped markets, reducing reliance on defence budgets.

Alignment: High
Feasibility: High
Profitability: High
Risk: High

3 Public Ownership Strategy

Plan: We take the company public to raise capital for global scaling and aggressive R&D investment. Use these funds to expand SaaS, enhance drone technologies and enter new domains.
Considerations: Offers substantial capital, brand credibility, and founder/investor liquidity but shareholder pressures that may not align with Defence.

Alignment: Moderate
Feasibility: High
Profitability: High
Risk: Moderate

4 Exit Strategy: Strategic Acquisition

Plan: Acquisition by a larger defence, aerospace or tech companies. Our selling points are novel ML capabilities strong S&R track record and SaaS potential.
Evaluation: Leverages acquirer resources for scaling. However, we receive a loss of control and integration challenges post-acquisition. Provides significant returns for us as founders and investors in case we want to exit-focus.

Alignment: High
Feasibility: High
Profitability: High
Risk: Moderate

5 Develop Adjacent Technologies

Plan: Invest heavily into R&D autonomous drones, satellite analytics, or advanced photogrammetry technologies. This enables us to expand ML applications to broader EMS and logistics domains.
Evaluation: We try to positions Apollo as a tech leader. However, high R&D costs and uncertain ROI make this a very long term play.

Alignment: High
Feasibility: Moderate
Profitability: High *Long Term*
Risk: High

APOLLO DRONES

APPENDIX

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Phillipos 1 - Business Model Expert — 01

Background

Industry Experience:

- Partner & Co-Founder at Maven Global
- Visiting Researcher at Imperial Business School
- Deputy Director, Cass-X, Centre for Innovation and Disruption at Bayes Business School
- Managing Director, Global CoHead of FX Structuring at Barclays Capital

Key Advice Offered:

- Feasibility of our product
- Ways to make it more viable and give it a greater market edge
- Ways to get early investments
- Ways to structure our pricing and market analysis

1st Interview

Change in business model:

The original business model was made from a 3000-unit price point which required significant production volume to make the business viable. As such, a large-scale operation is necessary to sustain profitability.

As such, he suggested considering a subscription model:

- Position the drone as part of a service offering, including software updates, maintenance, and data integration.
- This shifts the focus from selling hardware to delivering ongoing value to potential customers.

Strategic Positioning:

Advised on not focusing on hardware as a competitive advantage as

- Hardware, like drones is becoming a commodity.
- Instead, emphasise Imperial's expertise in machine learning, mapping, and network systems.
- Position drones as enablers for more advanced technological solutions and services.
- This then brought us into the machine learning aspects

Defence Funding Opportunities:

Phillipos highlighted massive funding shifts into defence-related R&D both in the UK and the EU.

- He suggested exploring grants and funding targeted at dual-use technology such search and rescue for military and civilian applications
- He told us to look into potential funding like Innovate UK and Horizon Europe

As such, we could position the technology for natural disaster management, such as early wildlife fire detection in Greece and Italy and for potential military applications such as rescuing a downed pilot

- As such, we should ensure compliance with government funding requirements, such as non-China manufacturing for defence use

Phillipos 2- Business Model Expert — 02

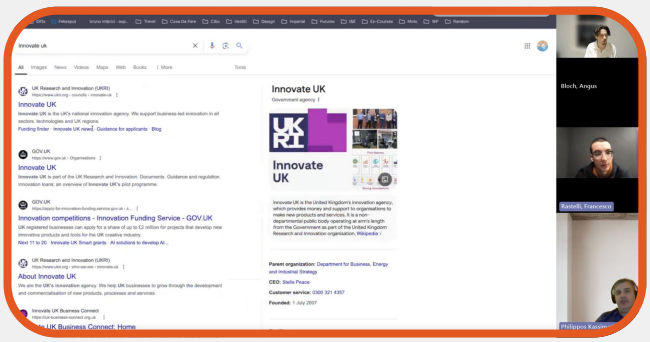
Scale and Valuation for Investors:

To attract investors, we need to demonstrate potential for scalability and high valuation, without reaching a ridiculous number, as such, he recommended having a 1st year valuation of £3M-10M

- Pure hardware sales alone wouldn't achieve this; a value-added service model is essential.

As such, he suggested we align with current government research trends to secure R&D grants.

- Universities are increasingly part of defence-related R&D funding, making this a pragmatic choice.



2nd Interview

Defining the Total User Base:

- Identify the total number of potential users, or the unsegmented user base, by looking at various sectors like government, private, and non-profit organisations that would utilise our drones and SaaS for search and rescue, military, and tourism.
- From this unsegmented user base, identify how many are in your direct target market (e.g., 735 S&R teams) and expand from there to include related groups who may not be currently considered but show potential for usage.

Market Segmentation:

- Split it into S&R Commercial and the Defence Market
- Evaluate each segment by potential budget, need for drones and SaaS, and unique requirements.
- Consider creating bespoke offerings for each segment that cater to their specific needs and budget levels.

Competitive Analysis:

Investigate the current solutions available in the market, especially those provided by competitors in the defence sector.
Understanding the value proposition of these competitors, especially those offering high-value products, will help position your offering effectively.

Value Proposition and Pricing:

Discuss the value of assets and lives saved or supported by your technology, particularly in high-value contexts like military operations, or in our case, for cost of downed pilots.
For the SaaS, pricing could range based on the complexity and necessity of the service:

- Around £30,000 for standard software
- Up to £100,000 for specialised military-grade software.

Business Model Considerations:

Subscription Models:

- A lower-priced drone subscription for basic services paired with higher-tier offerings for advanced needs can attract a broad customer base.

Direct Sales vs. Outsourced Operations:

- Explore both selling your product directly and operating it as a service, where you manage drone operations for clients.

Demetris - Machine Learning Expert — 03

Background

Industry Experience:

- Chief Scientific Officer at Syvil
- Co-Founder and Chief Technology Officer at MitoBioTech
- Founder in residence of Entrepreneur First
- GTA and Laboratory Operations Manager at Imperial College London

Key Advice Offered:

- Feasibility of our ML Algorithm
- Ways our algorithm could work on the drone
- Improvements to the current system
- Clarification around servers and AWS, how to you use it, etc.

Interview

Linear regression Vs Random Forest

- Linear regression is not computationally intensive. This makes it a quick and efficient method for predictive modelling when dealing with data that has a linear relationship.
- This algorithm can handle a large set of features and is effective for classification problems. However, it struggles with rapid shifts in data at the initial stages, as it builds numerous decision trees to make predictions, which can be computationally demanding.

Neural Networks:

- For more complex patterns that cannot be captured by simpler models like linear regression, neural networks, especially larger ones, are used.
- They require more computational resources and we need to be careful in our design to avoid overfitting.
- Graph based neural networks can be used as they work well with data structured as graphs. They can model relationships and interactions complexly and dynamically.

AWS EC2 Instance:

- Using AWS EC2 allows you to pay only for the compute time you use, which can be cost-effective for running large-scale computations.
- There are tools provided by AWS to help determine the cost based on the usage, with an estimate that running an EC2 instance continuously for a month could cost between $1,000 and $3,000.
- For our software, we'd only run it when new data would be inputted, around once, maybe twice a month.

Big O Notation:

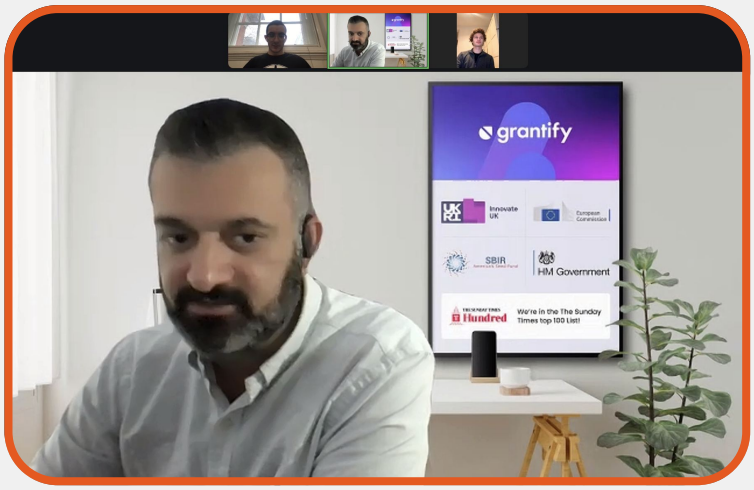
- This is used to describe the time complexity and space complexity of algorithms.
- It can helps us to predict how processing time or space used grows as the size of the input data increases.

Neel - Initial Funding Expert — 04

Background

Industry Experience:

- Regional Head of Growth at Grantify
- Head of Product at Grantify
- Senior Fine Art Consultant
- Senior Investment Sales Consultant at CW
- Client Services Manager at The First Group



Key Advice Offered:

- Grants route
- Requirements for grants applications
- Information on Innovate UK
- Information on Grantify

Interview

Registering a company

- Registering the company does not have to be with university.
- We need to be registered as a company before being able to apply to funding.
- Looks good that we have a military aspect as it can be used in both public and private sector

Questions about our product:

- What similar competition?
- How much funding would be required?
- What grant funding have we looked at?

Advices for Innovate UK smart grant:

- Least competitive round for a startup. 30-50% less applications in this round.
- Purely R&D based government grant. (Looka t Opex and Tapex).
- No money given for sales cost etc.
- Usually give 100M a year, on a quarterly basis. 14M acquired last year by Grantify.
- 70% og R&D costs were covered.
- We cover the first quarter - then at the end we put in everything we've spent and we claim back the 70%.
- We need to tell them where the 30% is going to come from in the application.
- First get smart grant with innovate - then move on to DASA
- Innovate needs to be done by 22nd January

Grantify Information:

- 15% of all applicants walked with 72% of funding
- Has won 170M of grants since they started it
- 46th fastest growing company in the UK
- Founders are smart grant winners as well
- They do military funding - Won drone based grants with DASA.
- Works by first having a one hour consultation. Then moves onto their platform to develop their application. Has a trained AI and previous applications to help with the application - their team will edit it and give suggestions for changes to check the application. Cost of service is just under 5000£ with VAT. Split into half before, half after. Gratify operates with money back guarantee - if any red flags appear and they aren't confident, they will give us back the money and cancel it.

Aryan - Rescued Hiker 05

Background

Industry Experience:

- Was stranded in Highlands after injuring leg on hike
- Experienced hiker
- Volunteers with MOFFAT MRT for fundraising events

Insights Offered:

- What went wrong and caused him getting stranded
- What it is like to be stranded/rescued
- Feedback on how our different ideas would have helped with his situation.

1st Interview

During the first interview we wanted to get some information on how and why he got stranded and on what happened with the search and rescue effort.

What Went Wrong:

- After starting the hike on a clear day the weather suddenly turned and visibility became minimal due to fog and rain
- He wasn't prepared for this change in the weather and so was very cold and tired
- He became disorientated at the top of a peak and took the wrong path to get down
- The terrain got steeper and he slipped and fell got a grade 3 twisted ankle meaning he was unable to walk
- He was meant to be meeting some friends after the hike and when he didn't show they got worried and they called ASSYNT MRT.
- Luckily he had told his wife the route he was taking which dramatically decreased the time taken for the team to reach him

What was it Like Being Stranded

- He again mentioned how cold he was as he did not anticipate the change in weather
- He said how scary it was as even though there were people out looking for him he did not know they were on there way
- He was stranded for over 4 hours and it was mentally very tough not only due to the feeling of helplessness but he would go over how stupid it was that he had not been better prepared for the hike.
- There was a huge feeling of relief when he was found not only because he was safe but because he was able to get some much needed food, water and warm clothes.

2nd Interview

After we had completed some ideation into different drone technologies we could use with the search and rescue operations we again spoke to him regarding him thoughts on each idea

Ideation feedback:

- As someone who was rescued, I can't overstate how much it would've helped to have earlier contact with rescuers. The idea of using drones for emergency delivery and communication is fantastic. If I'd been able to talk to someone sooner, it would have helped me stay calm and provide clear details about my situation. Plus, getting essential supplies—like water, a thermal blanket, or even a simple first aid kit—could've made a huge difference while I was waiting.
- The concept of using ground-penetrating radar might not apply to every situation, like mine, but I can see how it'd be a lifesaver in avalanche zones or places where someone might be trapped out of sight. It's not something I'd have needed, but for people in those areas, it could be the difference between life and death.
- What stood out to me in my rescue was how quickly help arrived. If drones can cover the area faster than traditional methods, I think they'd be a major upgrade. But of course, it has to be affordable—resources are always a challenge, I get that.
- LIDAR for monitoring avalanches sounds like a great preventative tool. Even though it wouldn't have applied to my situation, I can see how warning hikers about dangerous areas could prevent emergencies altogether. The challenge of covering a large area and needing line-of-sight makes sense, though—it's something to weigh carefully.
- If adding drones to your team helps make rescues faster and safer, I'm all for it. Even with a limited budget, starting small with a few drones could still make a big difference.

Lindsay - Drone Pilot 06

Background

Industry Experience:
- Drone pilot for Assynt Mountain rescue team
- 5 years of experience in Assynt MRT

Insights Offered:
- When drones are used in S&R operations
- Advantages of drone usage in their S&R operations
- Current limitations with drones in the S&R domain
- Feedback on Ideation phase

1st Interview

During this first interview we looked to get a background in how and when drones were used as well as their limitations

When are drones used:
- Drones are used in cases where for both daytime and night-time rescue operations but truly excel in low light rescue operations.
  - they are able to provide more light in low light rescue operations due to large spotlights fitted.
  - Infra-red cameras allow for locating victims in dark due to the body giving off heat.
- Drones are used to search hard to reach areas quickly as they can fly above tough terrain that would take hours to search using traditional methods.
- The drones can quickly map areas after the occurrence of an avalanche without the dangers associated with the unsettled snow.

Current Drone limitations
- They have limited funding so they cannot fly the drones in adverse conditions as they are unable to deal with the loss of the asset.
  - There are operations that could have benefited from the use of a drone but we have been unable to use the drones in this case due to the marginal conditions
- The drones still require line of sight in operation due to the limited control and again the limited funding meaning they cannot afford to lose the drone
- Although the drones can help find the casualty there is nothing they can do to aid or help retrieve the hiker.

2nd Interview

After we had completed some ideation into different drone technologies we could use with the search and rescue operations we again spoke to her regarding her thoughts on each idea

Ideation feedback:
- Active tracking would not be applicable to our needs due to the limited number of drones we possess and the large footfall that we have in our operation area.
- The emergency package would be great to get supplies to the casualty as well as be able to talk to the person and gather more information on their condition before we get there.
  - We would want to tailor the package to our own needs as we feel that know what would help the person in most cases in our search area.
  - being able to talk to the person sooner would be a big help in rescue operations. If nothing else it would allow us to keep the person calm
- Ground penetrating radar would be less applicable to our needs but we could see the need in higher risk areas for avalanches.
  - how fast would it be able to cover the search area?
  - would it be faster than traditional methods with our limited number of drones
  - would we be able to afford it?
- LIDAR for avalanche monitoring would allow us to warn hikers of more hazardous areas but again we are not sure if we would be able to use this to cover our large search area especially since we need line fo sight for operation.
- With our current funding we would not be able to afford the drone swarms that you are talking about but it may be useful for better funded organisations.

Chris Huntley - MRT Secretary — 07

Background

Industry Experience:

- Secretary of search and rescue team in MOFFAT MRT
- 12 years experience in MOFFAT mountain rescue

Insights Offered:

- Funding for S&R teams
- thoughts on drone usage
- What similar technologies he has hear of
- Feedback on Ideation phase

1st Interview

During this first interview we looked to get a background in how and when drones were used as well as their limitations

How do Mountain Rescue Teams get Funding

- Many rescue teams are limited by budget constraints as they rely on donations and grants for specific equipment. This often involves applying for funds from donor organizations to purchase drones or other high-cost items.

Thoughts on Drone Usage and Other Technologies

- There is a growing Interest in Drones for Rescue: There is a growing interest within Scottish Mountain Rescue in using drones, with some teams acquiring and training with them.
- Drones provide an aerial view of rugged terrain, which helps locate casualties that might be obscured by uneven ground. Infrared cameras on drones can assist in locating warm bodies, though they also pick up other heat sources like animals.
- Drones can be fragile and vulnerable to weather conditions, especially high winds. Previous attempts to use drones in rescue operations, such as in Ben Nevis, were unsuccessful due to drones being blown off course.
- Some rescue teams, like Patterdale, have used optical technology to detect ground irregularities, which has been effective in finding casualties, although this was not necessarily drone-based.
- We were then told to check out the Scottish Mountain Rescues Team's website

2nd Interview

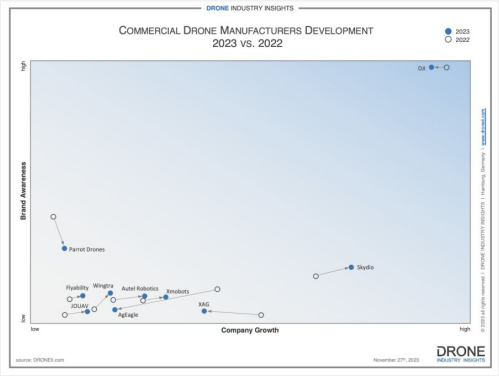
After we had completed some ideation into different drone technologies we could use with the search and rescue operations we again spoke to him regarding him thoughts on each idea

Ideation feedback:

- Given that drones are not yet part of our operations, we need to carefully consider how they could be used to address our specific challenges and ensure they provide value within our constraints.
- Introducing an emergency delivery system could be a game-changer. Drones could deliver essential supplies to individuals in distress and establish communication with them. This capability would allow us to gather critical information about their condition and provide reassurance until rescue teams arrive. Customizing the emergency packages to meet the most common needs in our area would further enhance their effectiveness.
- Ground-penetrating radar, while useful in high-risk avalanche zones, may not be as relevant to our operations. However, it could be considered for future use in areas where such risks are more prevalent.
- Speed and efficiency would be crucial in deciding whether drones are suitable for our needs. We'd need to assess whether they could cover search areas faster than traditional methods and whether this advantage justifies the investment. Budget constraints are a key consideration, so we would need to evaluate cost-effective drone options that align with our funding.
- LIDAR technology could offer valuable insights for avalanche monitoring, helping us warn hikers about hazardous zones. However, we must consider the challenges of covering large areas and the line-of-sight requirements for drone operation.
- While drone swarms may be an option for well-funded organizations, we need to focus on scalable, affordable solutions that can deliver meaningful results with a smaller number of drones. Starting with a few versatile, multi-purpose drones could provide a solid foundation for expanding our capabilities over time.

Politics and Defence Funding — 08

Economic and Political Effects



The two largest drone manufacturers/developers are the United States and China. Their key companies are **Skydio** and **DJI** respectively. They are two companies in the same space; controlled by opposite ends of the political spectrum.

DJI are the most well known drone supplier in the world, in March 2021 recorded as commanding 76% of the market share based on volume - *Statista*.

As part of their 'anti-china' campaign; in early 2024 the **US voted to ban DJI drones** within 'sensitive' situations such as communication analysis - with the clear intention of damaging their company brand throughout the west. They also increased spending and support for US based drone companies such as Skydio.

Key components from Skydio were previously sourced from Dongguan Poweramp but the US House's decision to allow Skydio to provide attack drones to Taiwan led to **Beijing placing sanctions** on the company.

The volatile political climate, especially surrounding drones means that basing a supply chain analysis around China would be risky. We will focus away from the current supply chains of Garmin and DJI in China, instead opting to follow the near-shoring approach of tech companies such as LG and Samsung. For now, we will focus on a European manufacturing facility to cater for the UK/Alpes/Pyrenees.

To form an accurate cost analysis for our product we needed to analyse potential supply chains. The nature of our sector requires investigation into current macro-political events. Justification for this claim found below:

Cheap, extremely advanced drones, have become common place on the 21st centuries battlefield. With the ability for a £300 drone to wipe out a £10 million tank; they are widespread. At any time it is reported that 10,000 drones are active in Ukraine. As with all technology, the military application has meant that defence spending has poured in and improvements become daily. The result is drones and their components are cheaper and more advanced than ever. Equally, the defence aspect means that the supply chain can be disrupted more commonly by macro political issues.

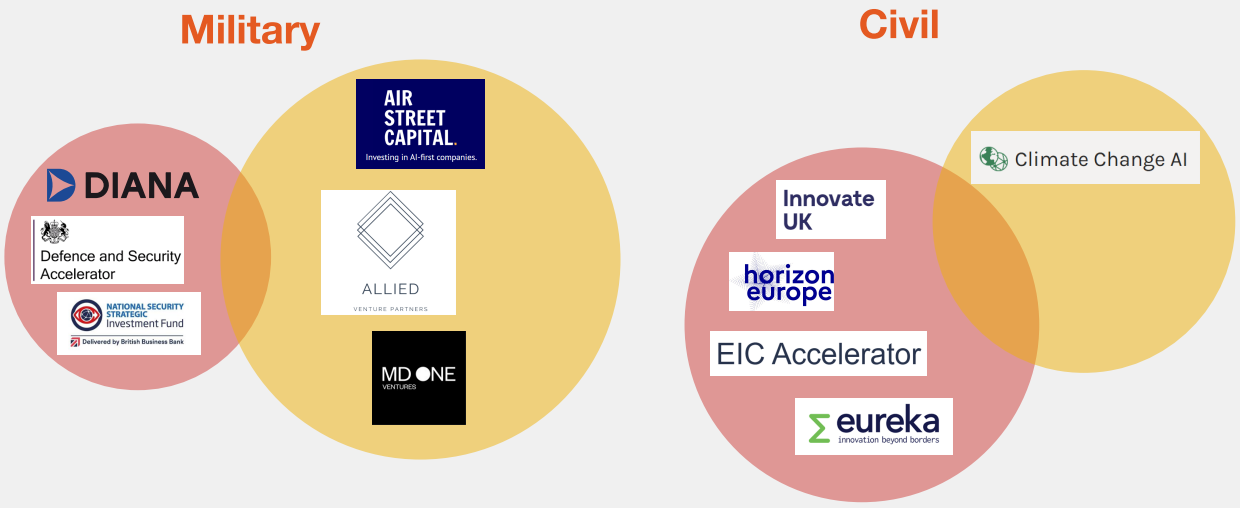
The military interest has opened up a host of potential funding opportunities that could be used to develop our product with the safety net of a government.

DIANA

Defence and Security Accelerator

First Year Funding

Our initial funding path focuses on military and academic research. Looking at private vs funding research paths.



The graph above shows potential investors as well as potential grants that our company would be applicable for showing the large number of options and opportunities that are available to us for first year funding

Military Funding

Drones have become a transformative technology in the defence sector. this has led to major investment in the drone sector from a military standpoint. In the UK, the Ministry of Defence (MoD) is the largest provider of public funding for research and development , accounting for 69.98% of government R&D expenditure. The combat air sector, the area applicable to our product, plays a major role. In 2022 they generated £7.9 billion in revenue. £1.6 billion of this was allocated to R&D projects. This shows a clear focus on advancing military technologies in our area. This presents strong evidence for securing funding and support from these defence and military organisations.

In addition to public funding, Private military contractors and defence companies such as the ones listed above can play a significant role in funding for the R&D for drone and image classification in military applications. These organisations often are backed by multi-billion-dollar budgets are constantly seeking investments that align with their goals. Partnering with these entities would help us not only access funding but their high quality datasets and assets that would dramatically help with the training of our models and software.

Whether through public or private investment there is significant potential to tap into these well funded resources. This would mot only help with the funding and development of our product but position our system in the a well funded rapidly evolving sector.

Civil funding

Civil funding also offers a multitude of different opportunities to fund and aid the research and development of our product. University research grants which are often backed by research councils and private institutions would be able to provide financial support. As well as this these institutions often have access to other benefits such as societal insights and environmental and natural disaster data allowing us to access more data to help train our models. Collaborating with academic institutions also often open the door to national and international research initiatives such as Innovate UK and horizon Europe.

Venture capital firms may be drawn in by the scalability and versatility of our product offering the possibility of investment for equity or preferential market deals. these well funded firms would help accelerate the development and commercialisation. The possibility of an MoD R&D investment would increase the likelihood of support from these VC funds due to the robust safety net provided by government backing

Ideation SWOT

Emergency Package Drop

STRENGTH

- Immediate assistance to stranded individuals
- Increases survival chances

WEAKNESS

- Limited payload capacity
- Dependence on accurate location data

OPPORTUNITY

- Collaboration with drone technology companies
- Potential for rapid deployment in a variety of rescue scenarios

THREAT

- Environmental factors like wind and snow may impact delivery
- High initial development costs

Machine Learning Algorithm

STRENGTH

- Cost effective method to reduce rescue time increasing chance of survival

WEAKNESS

- Relies on the data it uses in the model to be as accurate as possible
- will require training

OPPORTUNITY

- Not currently seen on the market and would work for both well funded and poorly funded organisations

THREAT

- The Algorithm will be difficult and costly to build making R&D a larger part of the product

Ground Penetrating Radar

STRENGTH

- Allows for detection of people buried under snow, which is vital in avalanche scenarios

WEAKNESS

- Requires specialised, costly equipment.
- May have limitations in detection depth and precision

OPPORTUNITY

- Advancements in radar technology could improve detection
- Application in other emergency fields

THREAT

- Technical limitations in certain snow conditions lead to inaccuracies.
- Issues with battery life in harsh environments.

LIDAR

STRENGTH

- Allows for precise measurement of snow levels.
- Identify avalanche-prone areas.

WEAKNESS

- Limited to preemptive measures instead of direct rescue
- May require expert knowledge

OPPORTUNITY

- Could be integrated with weather data to improve avalanche forecasting.

THREAT

- Dependence on favourable weather conditions for accurate readings.
- Costs

Drone Swarms

STRENGTH

- Can cover a larger area faster than single drones
- Reduces the manpower required

WEAKNESS

- Complexity with coordinating
- High potential for collisions

OPPORTUNITY

- Advances in swarm AI
- Expand to applications beyond mountain rescues.

THREAT

- Battery limitations in cold climates
- Regulatory restrictions on deployment

Active Tracking of Users

STRENGTH

- Continuous monitoring of individuals on ski slopes provides real-time data

WEAKNESS

- High costs for implementation and maintenance.
- Privacy concerns for individuals

OPPORTUNITY

- Partnerships with ski resorts
- Integration with machine learning algorithms to enhance accuracy

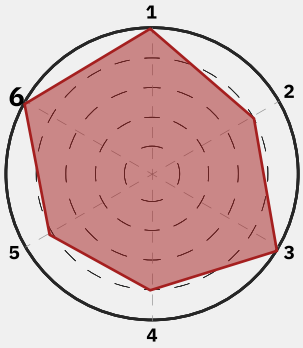
THREAT

- Weather conditions or interference in mountainous areas
- Regulatory challenges regarding drone use

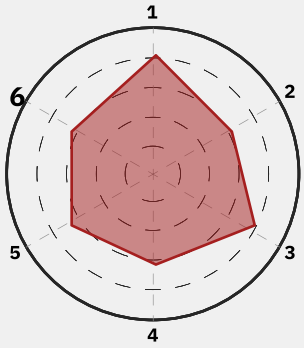
Weighted Matrix

Below are the weighted matrices and calculations of the different ideas for the drone technology. We used the initial interviews and feedback from the ideation stage to weight each criterium accordingly

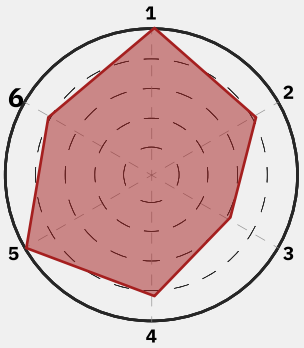
1. Response time (3)
2. Reliability (2)
3. Scalability (2)
4. Flexibility
5. Low Labour (1)
6. Cost (1)



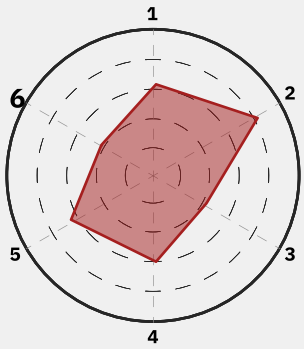
Machine Learning



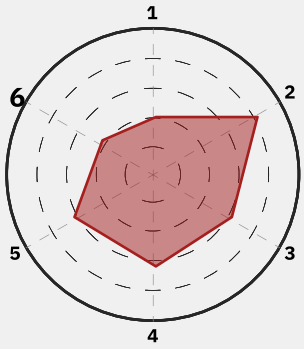
Active Tracking



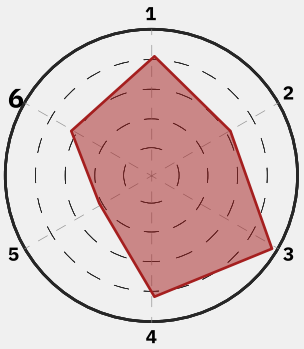
Emergency Package



Ground Penetrating Radar



LIDAR



Drone Swarms

Weighted Matrix

1. **Response Time (Weight: 3) –** Speed of deployment and victim reach.
2. **Reliability in Varied Conditions (Weight: 2) –** Consistent performance in different weather and terrains.
3. **Scalability (Weight: 2) –** Feasibility of deploying across various locations (remote or high-demand areas).
4. **Operational Flexibility (Weight: 2) –** Adaptability to different rescue scenarios or potential for repurposing.
5. **Low Labor Intensity (Weight: 1) –** Minimising the need for human resources.
6. **Cost (Weight: 1) –** Affordability in terms of initial setup and ongoing operations.

Criteria	Weight	Active Tracking		Emergency Package Drop		GPR		LIDAR		Drone Swarms		Machine Learning	
		Score	Total	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total
Response Time	3	4	12	5	15	3	9	2	6	4	12	5	15
Reliability	2	3	6	4	8	4	8	5	10	3	6	4	8
Scalability	2	4	8	3	6	2	4	3	6	5	10	5	10
Feasible	2	3	6	4	8	3	6	3	6	4	8	4	8
Labour	1	3	3	5	5	3	3	3	3	2	2	4	4
Cost	1	3	3	4	4	2	2	2	2	3	3	5	5
Total			40		46		34		36		43		50

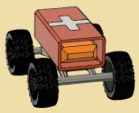
Drone Delivery System

For part of our ideation process we decided to focus on designing a package delivery system that minimises- the user's need for movement. This concept focuses on scenarios of injured hikers that have been immobilised. This design ensures that they are still able to access the essential supplies.

Secondary Drone

A supplementary drone capable of repositioning the package after its initial drop, ensuring precise delivery to the user.

The secondary drone is dropped



Drone then travels to the injured hiker

Can manoeuvre around objects to get to the hiker

Targeted Rope Launcher

A mechanism that propels a rope directly from the package towards the user, enabling easy retrieval.

The drone has a rope launcher on the bottom

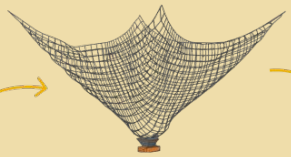
The rope can be launched to the hiker

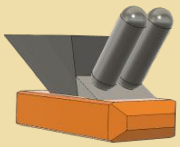
Hiker can pull the emergency supplies to them

Net Launcher

A system designed to fire a net in multiple directions upon landing, increasing the likelihood of reaching the user regardless of their position.

Net launches out in all directions



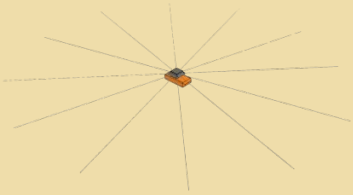


Net launcher is on the top of the package

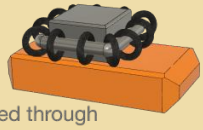
The hiker can grab the net and reel in the package

SMAs

Utilising SMA properties to deploy components that expand upon landing, creating a structure similar to an umbrella to enhance usability.



package with SMA coils is lowered via winch



Current is passed through the nitinol coils initiating their SMA properties

coils spread out and let the hiker grab them and pull in the package

Primary Physical TESTING:

We bought and tested some Nitinol to evaluate whether it could be used. The high power draw meant that it was not feasible.

Diversification & Expansion 13

Scalability and Growth

To ensure we keep growing our user base, and thus avoiding stagnation, we must **diversify and expand** as the business grows. Ranging from diversification, to expanding to new markets and opportunities.

Starting with the current business model, we would begin within Mountain Rescue England and Mountain Rescue Scotland as our initial market and expand to the teams within this region. This would allow us to conduct both RaD within these regions to further develop our product and to create a more effective distribution model as the distance between customers is much smaller.

Secondary Business Model

Our secondary business models focus on getting funding to **offset costs** of R&D. This means expanding into two possible markets. Either going down the **defence route**, for locating lost pilots, or to aid in national SaR operations. Or we go down the **climate change funds route** to aid in both early fire detection and SaR during natural disasters.

Innovate UK Loan [1]

100k up to **1m £** in loans

7.4% intrest

Loans are intended to support the commercialisation of innovations relevant to defence, covering up to 100% of project.

This would be around the location of downed pilots or military use in their own SaR operations during natural disasters.

Evaluation Process:
- Stage 1: DASA looks at the innovation's desirability, feasibility, and viability. Innovate UK conducts a credit evaluation
- Stage 2: Innovate UK conducts a full credit evaluation for loan suitability

Application Requirements:
- Value proposition statement
- Detailed technical project plan with milestones
- Business model and expected financials

Horizon Europe [2]

€95.5b budget

70% budget for SMEs

This fund from the European Union is focused around the development of products to tackle climate change.

Our algorithm could both be used during SaR situations given the rise in natural disasters, but also with applications around early fire detection systems in places like Greece and south Italy.

Five key mission areas:
- **Adaptation to climate change**
- Cancer
- Ocean health
- Climate neutral cities
- Soil health and food

Mandatory Open Access to publications meaning our research would not be private.

[1] UK Government. (n.d.). Competition: Defence Innovation Loan. Retrieved December 4, 2024, from https://www.gov.uk/government/publications/competition-defence-innovation-loans/competition-document-defence-innovation-loan

[2] European Climate, Infrastructure and Environment Executive Agency (CINEA). (n.d.). Horizon Europe. Retrieved December 4, 2024, from https://cinea.ec.europa.eu/programmes/horizon-europe_en

Product Diversification

A few diversification ideas born from brainstorms.

Physical Product Diversification

Different Attachments/Cameras; customised to fit specific needs. Increasing usability and potential sales opportunities.

Strava Heat Map

Using Strava heat maps to help with route tracking. QJIS, KOMOOT

AI Integration

Using AI to judge the call out from people to get the process started ASAP. i.e. monitoring emails or calls etc

Shift In Manufacture

Shift from buying pre made components, which often come from China and so aren't viable in the American market, to building our own in house. This would both lower long term costs for us, while reducing reliance on foreign suppliers and opening us up to new markets

Market Diversification

Move into different markets, such as sea rescue, by adapting our product to their specific needs. Allows it to diversify so it's not as reliant on a specific and small market. RNLI for example has a much larger budget of £242m [3]

R&D Into A Drone Fleet

R&D to develop our product into an autonomous drone fleet to sweep large areas at once. This is especially useful in disaster zones to find and aid multiple people in need. Thus we can find a new market to enter

3rd Party Drone Integration

Allowing 3rd party drone integration into our software. This would go further than the API integration as it would incorporate the machine learning model into the 3rd party drone camera.

Satellite Tracking

Using live satellite feeds to help build accuracy of machine learning and real time weather changes.

Bullseye Framework Justification | 15

Inner Circle

Partnership with S&R Organisations

Partner with organisations that currently have sufficient funding to afford our drones. Currently S&R Organisations such as Mountain Rescue England are the ones who approve the funding and spending of the teams on equipment like this. Pitch Apollo as a cheaper, life saving drone compared to current methods. Combine the drone services with SaaS to give them added value (our competitive advantage) to sell the product to them.

Conference and Industry Events:

Live demos of Apollo's features and capabilities. Emphasise case studies during the pitches as proof that it works, including advertising it's different applications. Allows us to connect with both S&R Teams but much more in the military field.

Grants and Public Funding:

This will both allow us to both cover initial set up and R&D Costs, while gaining access to military channels to be able to better target investors and potential users for both our Saas and Haas.

Middle Circle

Government Listings:

Military organisations normally get solutions through approved government platforms. As such we can register Saas on platforms like GSA (USA) or MOD contracts portal (UK). Ensure our Saas meets the compliance and certification standards.

Word of mouth:

S&R Teams and organisations are always in contact with each other. As such, news about our drone and it's values can spread quickly gaining large intrest.

PR around our drone:

Building up a positive PR in military-focused outlets/publications to build up credibility and leads. Publish PRs that focus on our case studies and savings for current customers. Focus on defence or tech publications such as Defense News, Wired etc.

Outer Circle

Social media ads and affiliate content ads:

Post case studies and success stories on social media platforms, such as onYouTube: "How Apollo saved 2 lives in Scotland disaster". Run targeted ads at S&R workers and EMS teams around our product.

Partnerships with hardware supplier:

Partner with manufacturers for cobranding on our drones, or provide a package deal for S&R teams to allow for a focus on the software side of things on our end.

Community building with existing teams:

Create a religion in which our drone reigns supreme. This will spread across the S&R Teams until all of them use our products and we gain majority in the market

Competitor Overview

	Direct	Indirect	Replacement	Us
Company	Ripper Corp.	HELI SECURITE FRENCH HELICOPTER AIRLINE	LIFESYSTEMS® GARMIN™	APOLLO DRONES
Optimed Route Pattern?	• Hire or purchase basis • Paid for by organisation and insurance	• One time purchase for one time use • Paid for by insurance	• Subscriptions for Satellite usage • Paid for by users	• One time purchase of drone • software on subscription
Requires Personnel?	• Naval Search and Rescue • Asset and Infrastructure surveying	• Ski resort and mountain Search and Rescue	• Satellite phones • Survival kits and supplies	• ML Algorithm for faster location • Image classification for improved mapping
Location	• Australia	• France	• Global	• UK
Strenghts	• Many applications • Large ties with multinational companies and government	• Has a monopoly when it comes to helicopter rescue in French Resorts	• Relatively cheap alternative to other solutions	• Too many
Weaknesses	• Heavily based in Australia • Not widely used elsewhere	• Very expensive at $10,000 - $15000 per use case	• Relies on users to be prepared • Can be forgotten/ lost by user	• Is only as good as data used to train it • subscription service may put
Competitive Advantages	• Biggest player in the Drove Search and Rescue game	• Is currently gold standard in the mountain search and rescue operations in France	• Cheapest and easiest alternative to other methods of rescue • if used properly erases the need	• Only company offering this technology for locating lost hikers • Can asses the condition of
Target Customers	• Government agencies and multinational institutions	• Government Agencies	• Mountaineers • Skiers • Adventurers	• Mountaineers • Skiers • Adventurers

Key Competitor

Value Added

Mountain Rescue Capability:
- One limitation in the ripper drone is its lack of mountain rescue capability.
- The company currently had no drones designed for mountain applications.

Accurate Package Delivery System:
- Another limitation is its lack of an accurate package delivery system.
- It requires the stranded individual to move in order to collect the floatation device.

ML Algorithm:
- Being able to track and accurately predict where a lost person will be
- Ripper corp need someone to locate them first

Maintenance Service:
- Offering a maintenance service within our subscriptions allows user to always have a running drone
- Designed for repairability

Ripper Corp SWOT Analysis

Strength
- Good market coverage with regards to other applications for drones
- Already partnered with large governmental and multi national corporations meaning they have good future proofing

Weakness
- Ripper drone is very heavily based in Australia
- They are struggling to break into the global market when it comes to governmental bodies due to their limited and existing relationships with Australian organisations

Opportunity
- Does have a section on the website for custom drone design and implementation meaning that any company could approach for their own custom system

Threat
- Having so many different systems and types of drone service can make managing and affording repairs difficult as company may be overexerting itself

Apollo Drones SWOT Analysis

Strength
- Innovative solution for unmapped path identification and mapping
- Combines AI and machine learning for scalable, real-time applications.

Weakness
- High technical complexity and cost of implementation.
- no market standing or reputation

Opportunity
- Growing demand for sustainable development and remote mapping.
- Collaboration potential with GIS organizations and SaR groups.

Threat
- Competition from other drone companies.
- Data reliability and cybersecurity risks.

Drone Regulations UK

It is important to consider the current drone regulations and potential future trends. These regulations contribute to mandatory training and legal costs.

Drone regulations are relatively similar throughout Europe. We will focus on the UK as our example.

Our drone would not fall under the 'open category' as it would not be limited to recreational use. As such operators would need to obtain **Operational Authorisation** from the CAA which involves submitting a risk assessment and demonstrating adequate pilot competence.

As the drone will not be limited to line of sight, the operator will need to obtain a **Operating Safety Case** (OSC) and **General Visual Line of Sight Certificate** (GVC) that explains how you will mitigate risks associated whilst flying **Beyond Visual Line Of Sight** (BVLOS).

Drones flying BVLOS must include avoidance system (in our case LiDar) and emergency procedures in case of loss of link (e.g. return to home). There must be built in systems to stop drones entering Flight Restriction Zones (FRZs) of airports etc without prior permission. BVLOS operations are generally approved for Search and Rescue operations as they provide 'clear public or commercial benefits'.

To obtain the BVLOS permits a pilot would have to spend roughly £10,000.

Know Your Client (KYC)

Although KYC is usually focusing on money laundering - We would require some kind of due-diligence for/ during sale. This could be a potential barrier/irritation to sale but ensuring that the client offers some kind of warranty that they have (or are planning to obtain) the necessary license to operate is essential. Within our sales prospectus we would have to make it clear that it was the purchasers responsibility to comply with all regulations.

e.g.:

"In agreeing to purchase this item you, the purchaser, warrant and represent that you have all necessary licences and permits to operate the products in question and agree to indemnify the company against any and all liability arising from misuse of the products in question"

GDPR Drone Rules

Under the UK's General Data Protection Regulation (GDPR) drone operators are required to comply with certain provisions while capturing images or video. Number plates, identifiable faces, etc mean that the operator is bound by these GDPR provisions.

The Civil Aviation Authority (CAA) regulates drones based on staying away from sensitives sites, 50m away from people and buildings and mandating the notification of personnel if personal data is being captured.

Non compliance can lead to fines or legal action. These fines we would not expect to be 'on the hook' for.

Mountain Rescue England & Wales

3462 call-outs

2775 deployments

79% casualties

21% missing

Factors

403 Weather

836 Human error

489 Inexperience

144 Poor visibility

141 Lack of equipment

Transport

Transported by train: **7**

Transported by helicopter: **248**

Transported in team vehicle: **328**

Walked off: **393**

Stretcher: **712**

Equipment

Advice given: **704**

Wound dressing: **78**

Limb splint age: **431**

Spinal immobilization: **341**

CPR: **157**

80 head injuries

105 arm, wrist and hand injuries

108 knee injuries

108 lower leg injuries

343 ankle and foot injuries

[1] Mountain Rescue England and Wales. (2024). Annual Review 2024. Retrieved from https://www.mountain.rescue.org.uk/wp-content/uploads/2024/05/AnnualReview2024.pdf

Template Colours

Typography

Title

Subtitle

Highlight

Body

[1] References

Statistics

Section Text

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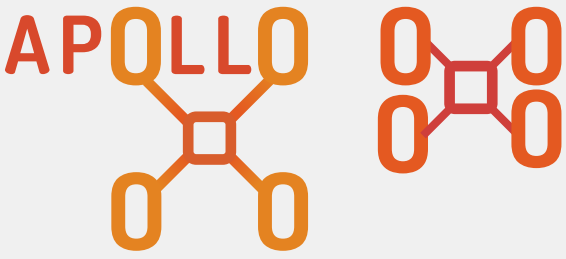
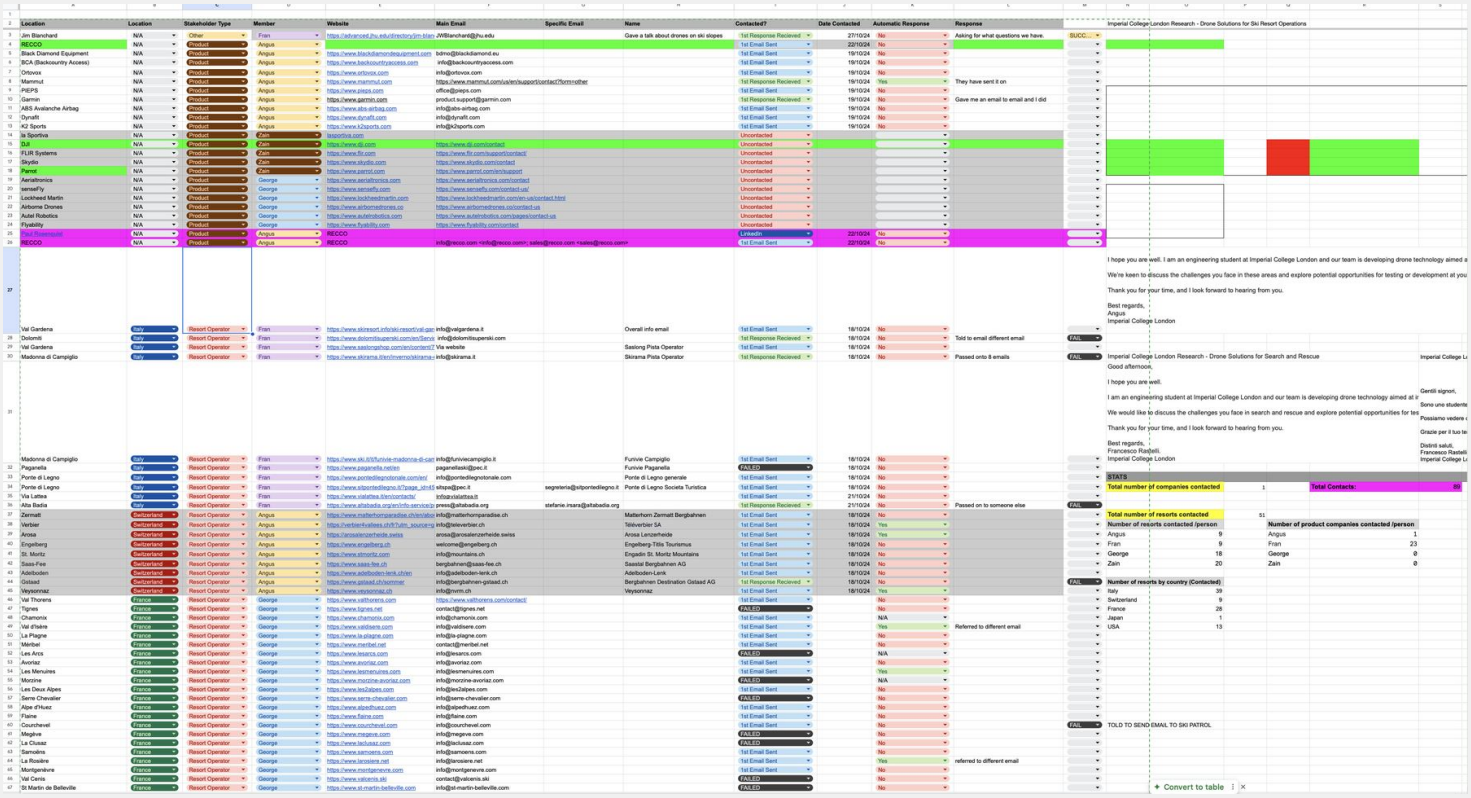
Branding Tests



Reach-Out Sheets 21

Expert Interview Requests



Location	Location	Stakeholder Type	Member	Website	Main Email	Specific Email	Name	Contacted?	Date Contacted	Automatic Response	Response
Jim Blanchard	N/A	Other	Fran	https://advanced.jhu.edu/directory/jim-blanchard	JWBlanchard@jhu.edu		Gave a talk about drones on ski slopes	1st Response Recieved	27/10/24	No	Asking for what questions we have.
RECCO	N/A	Product	Angus					1st Email Sent	22/10/24	No	
Black Diamond Equipment	N/A	Product	Angus	https://www.blackdiamondequipment.com	bdinv@blackdiamond.eu			1st Email Sent	19/10/24	No	
BCA (Backcountry Access)	N/A	Product	Angus	https://www.backcountryaccess.com	info@backcountryaccess.com			1st Email Sent	19/10/24	No	
Ortovox	N/A	Product	Angus	https://www.ortovox.com	info@ortovox.com			1st Email Sent	19/10/24	No	
Mammut	N/A	Product	Angus	https://www.mammut.com	https://www.mammut.com/us/en/support/contact?form=other			1st Response Recieved	19/10/24	Yes	They have sent it on
PIEPS	N/A	Product	Angus	https://www.pieps.com	office@pieps.com			1st Email Sent	19/10/24	No	
Garmin	N/A	Product	Angus	https://www.garmin.com	product.support@garmin.com			1st Response Recieved	19/10/24	No	Gave me an email to email and I did
ABS Avalanche Airbag	N/A	Product	Angus	https://www.abs-airbag.com	info@abs-airbag.com			1st Email Sent	19/10/24	No	
Dynafit	N/A	Product	Angus	https://www.dynafit.com	info@dynafit.com			1st Email Sent	19/10/24	No	
K2 Sports	N/A	Product	Angus	https://www.k2sports.com	info@k2sports.com			1st Email Sent	19/10/24	No	
la Sportiva	N/A	Product	Zain	lasportiva.com				Uncontacted			
DJI	N/A	Product	Zain	https://www.dji.com	https://www.dji.com/contact			Uncontacted			
FLIR Systems	N/A	Product	Zain	https://www.flir.com	https://www.flir.com/support/contact/			Uncontacted			
Skydio	N/A	Product	Zain	https://www.skydio.com	https://www.skydio.com/contact			Uncontacted			
Parrot	N/A	Product	Zain	https://www.parrot.com	https://www.parrot.com/en/support			Uncontacted			
Aerialtronics	N/A	Product	George	https://www.aerialtronics.com	https://www.aerialtronics.com/contact			Uncontacted			
sensefly	N/A	Product	George	https://www.sensefly.com	https://www.sensefly.com/contact-us/			Uncontacted			
Lockheed Martin	N/A	Product	George	https://www.lockheedmartin.com	https://www.lockheedmartin.com/en-us/contact.html			Uncontacted			
Airborne Drones	N/A	Product	George	https://www.airbornedrones.co	https://www.airbornedrones.co/contact-us			Uncontacted			
Autel Robotics	N/A	Product	George	https://www.autelrobotics.com	https://www.autelrobotics.com/pages/contact-us			Uncontacted			
Flyability	N/A	Product	George	https://www.flyability.com	https://www.flyability.com/contact			Uncontacted			
Elio Rosenblatt	N/A	Product	Angus	RECCO				LinkedIn	22/10/24	No	
RECCO	N/A	Product	Angus	RECCO	info@recco.com <info@recco.com>; sales@recco.com <sales@recco.com>			1st Email Sent	22/10/24	No	
Val Gardena	Italy	Resort Operator	Fran	https://www.skiresort.info/ski-resort/val-gardena/	info@valgardena.it		Overall info email	1st Email Sent	18/10/24	No	
Dolomiti	Italy	Resort Operator	Fran	https://www.dolomitisuperski.com/en/Servizi	info@dolomitisuperski.com			1st Response Recieved	18/10/24	No	Told to email different email
Val Gardena	Italy	Resort Operator	Fran	https://www.sastongshop.com/en/content/7	Via website		Sastong Pista Operator	1st Email Sent	18/10/24	No	
Madonna di Campiglio	Italy	Resort Operator	Fran	https://www.skirama.it/en/riviverna/skirama/	info@skirama.it		Skirama Pista Operator	1st Response Recieved	18/10/24	No	Passed onto 8 emails
Madonna di Campiglio	Italy	Resort Operator	Fran	https://www.ski.it/it/funivie-madonna-di-campiglio.html	info@funiviecampiglio.it		Funivie Campiglio	1st Email Sent	18/10/24	No	
Pagarella	Italy	Resort Operator	Fran	https://www.pagarella.net/en	pagarellaski@pec.it		Funivie Pagarella	FAILED	18/10/24	No	
Ponte di Legno	Italy	Resort Operator	Fran	https://www.pontedilegnotonale.com/en/	info@pontedilegnotonale.com		Ponte di Legno generale	1st Email Sent	18/10/24	No	
Ponte di Legno	Italy	Resort Operator	Fran	https://www.sliponteidilegno.it/?page_id=6	slisapi@pec.it	segreteria@sliponteidilegno.it	Ponte di Legno Societa Turistica	1st Email Sent	18/10/24	No	
Via Lattea	Italy	Resort Operator	Fran	https://www.vialattea.it/en/contacts/	info@vialattea.it			1st Email Sent	21/10/24	No	
Alta Badia	Italy	Resort Operator	Fran	https://www.altabadia.org/en/info-service/p	press@altabadia.org	stefanie.insam@altabadia.org		1st Response Recieved	21/10/24	No	Passed on to someone else
Zermatt	Switzerland	Resort Operator	Angus	https://www.matterhornparadise.ch/en/abo	info@matterhornparadise.ch		Matterhorn Zermatt Bergbahnen	1st Email Sent	18/10/24	No	
Verbier	Switzerland	Resort Operator	Angus	https://verbierfestivaleu.ch/?utm_source=g	info@televerbier.ch		Televerbier SA	1st Email Sent	18/10/24	Yes	
Arosa	Switzerland	Resort Operator	Angus	https://arosalenzerheide.swiss	arosa@arosalenzerheide.swiss		Arosa Lenzerheide	1st Email Sent	18/10/24	Yes	
Engelberg	Switzerland	Resort Operator	Angus	https://www.engelberg.ch	welcome@engelberg.ch		Engelberg-Titlis Tourismus	1st Email Sent	18/10/24	No	
St. Moritz	Switzerland	Resort Operator	Angus	https://www.stmoritz.com	info@mountains.ch		Engadin St. Moritz Mountains	1st Email Sent	18/10/24	No	
Saas-Fee	Switzerland	Resort Operator	Angus	https://www.saas-fee.ch	bergbahnen@saas-fee.ch		Saastal Bergbahnen AG	1st Email Sent	18/10/24	No	
Adelboden	Switzerland	Resort Operator	Angus	https://www.adelboden-lenk.ch/en	info@adelboden-lenk.ch		Adelboden-Lenk	1st Email Sent	18/10/24	No	
Gstaad	Switzerland	Resort Operator	Angus	https://www.gstaad.ch/sommer	info@bergbahnen-gstaad.ch		Bergbahnen Destination Gstaad AG	1st Response Recieved	18/10/24	No	
Weissensz	Switzerland	Resort Operator	Angus	https://www.weissensz.ch	info@hvtm.ch		Weissensz	1st Email Sent	18/10/24	Yes	
Val Thorens	France	Resort Operator	George	https://www.valthorens.com	https://www.valthorens.com/contact/			1st Email Sent		No	
Tignes	France	Resort Operator	George	https://www.tignes.net	contact@tignes.net			FAILED		No	
Chamonix	France	Resort Operator	George	https://www.chamonix.com	info@chamonix.com			1st Email Sent		N/A	
Val d'Isère	France	Resort Operator	George	https://www.valdisere.com	info@valdisere.com			1st Email Sent		Yes	Referred to different email
La Plagne	France	Resort Operator	George	https://www.la-plagne.com	info@la-plagne.com			1st Email Sent		No	
Meribel	France	Resort Operator	George	https://www.meribel.net	contact@meribel.net			1st Email Sent		No	
Les Arcs	France	Resort Operator	George	https://www.lesarcs.com	info@lesarcs.com			FAILED		N/A	
Avoriaz	France	Resort Operator	George	https://www.avoriaz.com	info@avoriaz.com			1st Email Sent		No	
Les Menuires	France	Resort Operator	George	https://www.lesmenuires.com	info@lesmenuires.com			1st Email Sent		Yes	
Morzine	France	Resort Operator	George	https://www.morzine-avoriaz.com	info@morzine-avoriaz.com			FAILED		N/A	
Les Deux Alpes	France	Resort Operator	George	https://www.les2alpes.com	info@les2alpes.com			1st Email Sent		No	
Serre Chevalier	France	Resort Operator	George	https://www.serre-chevalier.com	info@serre-chevalier.com			FAILED		No	
Alpe d'Huez	France	Resort Operator	George	https://www.alpedhuez.com	info@alpedhuez.com			1st Email Sent		No	
Flaine	France	Resort Operator	George	https://www.flaine.com	info@flaine.com			1st Email Sent		No	
Courchevel	France	Resort Operator	George	https://www.courchevel.com	info@courchevel.com			1st Email Sent		No	
Megève	France	Resort Operator	George	https://www.megeve.com	info@megeve.com			FAILED		No	
La Clusaz	France	Resort Operator	George	https://www.laclusaz.com	info@laclusaz.com			FAILED		No	
Samoëns	France	Resort Operator	George	https://www.samoens.com	info@samoens.com			1st Email Sent		No	
La Rosière	France	Resort Operator	George	https://www.larosiere.net	info@larosiere.net			1st Email Sent		Yes	referred to different email
Montgenèvre	France	Resort Operator	George	https://www.montgenevre.com	info@montgenevre.com			1st Email Sent		No	
Val Cenis	France	Resort Operator	George	https://www.valcenis.ski	contact@valcenis.ski			FAILED		No	
St Martin de Belleville	France	Resort Operator	George	https://www.st-martin-belleville.com	info@st-martin-belleville.com			FAILED		No	

Imperial College London Research - Drone Solutions for Ski Resort Operations

I hope you are well. I am an engineering student at Imperial College London and our team is developing drone technology aimed a

We're keen to discuss the challenges you face in these areas and explore potential opportunities for testing or development at you

Thank you for your time, and I look forward to hearing from you.

Best regards,
Angus
Imperial College London

Imperial College London Research - Drone Solutions for Search and Rescue

Good afternoon,

I hope you are well.

I am an engineering student at Imperial College London and our team is developing drone technology aimed at r

We would like to discuss the challenges you face in search and rescue and explore potential opportunities for tes

Thank you for your time, and I look forward to hearing from you.

Best regards,
Francesco Rastelli
Imperial College London

STATS

Total number of companies contacted	1	Total Contacts:	69
Total number of resorts contacted	51		

Number of resorts contacted /person		Number of product companies contacted /person	
Angus	9	Angus	1
Fran	9	Fran	23
George	18	George	0
Zain	20	Zain	0

Number of resorts by country (Contacted)	
Italy	39
Switzerland	9
France	28
Japan	1
USA	13

TOLD TO SEND EMAIL TO SKI PATROL

Convert to table

Expert Interview Requests 2

Location	Location	Stakeholder Type	Member	Website	Main Email	Specific Email	Name	Contacted?	Date Contacted	Automatic Response	Response	Im
Montgenèvre	France	Resort Operator	George	https://www.montgenevre.com	info@montgenevre.com			1st Email Sent		No		
Val Cenis	France	Resort Operator	George	https://www.valcenis.ski	contact@valcenis.ski			FAILED		No		
St Martin de Belleville	France	Resort Operator	George	https://www.st-martin-belleville.com	info@st-martin-belleville.com			FAILED		No		
Isola 2000	France	Resort Operator	George	https://www.isola2000.com	contact@isola2000.com			FAILED		No		
Risoul	France	Resort Operator	George	https://www.risoul.com	info@risoul.com			1st Email Sent		No		
Valmorel	France	Resort Operator	George	https://www.valmorel.com	info@valmorel.com			1st Email Sent		No		
Les Gets	France	Resort Operator	George	https://www.lesgets.com	info@lesgets.com			1st Email Sent		No		
Vaujany	France	Resort Operator	George	https://www.vaujany.com	info@vaujany.com			1st Email Sent		No		
Châtel	France	Resort Operator	George	chatel.com	info@chatel.com			FAILED		No		
Aspen Snowmass	America	Resort Operator	Zain	https://www.aspensnowmass.com/	contactus@aspensnowmass.com			1st Email Sent	18/10/24	Yes	some dumbass automated response	
Jackson Hole Mountain Resort	America	Resort Operator	Zain	https://www.jacksonhole.com/	info@jacksonhole.com			1st Email Sent	18/10/24	No		
Vail Ski Resort	America	Resort Operator	Zain	https://www.vail.com/	vailinfo@vailresorts.com			1st Email Sent	18/10/24	Yes	respond within 3 working days	
Alta Ski Area	America	Resort Operator	Zain	https://www.alta.com/	info@alta.com			1st Email Sent	18/10/24	No		
Big Sky Resort	America	Resort Operator	Zain	https://www.bigskyresort.com/	https://www.bigskyresort.com/contact		contacted via website portal	1st Email Sent	18/10/24	Yes	acknowledgement	
Palisades Tahoe (formerly Squaw Valley)	America	Resort Operator	Zain	https://www.palisadestahoe.com/	info@palisadestahoe.com			1st Email Sent	18/10/24	Yes	response within 48 hours	
Breckenridge Ski Resort	America	Resort Operator	Zain	https://www.breckenridge.com/	breckinfo@vailresorts.com			1st Email Sent	18/10/24	Yes	respond within 3 working days	
Steamboat Resort	America	Resort Operator	Zain	https://www.steamboat.com/	hr@steamboat.com			1st Email Sent	18/10/24	Yes	acknowledgement	
Snowbird	America	Resort Operator	Zain	https://www.snowbird.com/	info@snowbird.com			1st Email Sent	18/10/24	No		
Deer Valley Resort, Utah	America	Resort Operator	Zain	https://www.deervalley.com/	jobs@deervalley.com			1st Email Sent	18/10/24	Yes	acknowledgement	
Killington Ski Resort, Vermont	America	Resort Operator	Zain	https://www.killington.com/contact-us	info@killington.com			1st Email Sent	18/10/24	No		
Winter Park Resort, Colorado	America	Resort Operator	Zain	https://www.winterparkresort.com/	wpinfo@winterparkresort.com			1st Email Sent	18/10/24	No		
Heavenly Mountain Resort, California/Nev	America	Resort Operator	Zain	https://www.skiheavenly.com/	info@vailresorts.com			1st Email Sent	18/10/24	Yes	respond within 3 working days	
Hakuba Valley	Japan	Resort Operator	Zain	hakubavalley.com	https://ticket.hakubavalley.com/en/contact/		contacted via website	1st Response Recieved	18/10/24	Yes		
PINZOLO	Italy	Resort Operator	Zain		info@funiviepinzolo.it		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
FOLGARIDA-MARILLEVA	Italy	Resort Operator	Zain		sales@ski.it		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
PEJO	Italy	Resort Operator	Zain		sales@skipejo.it		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
ALPE CIMBRA: Folgaria-Lavarone	Italy	Resort Operator	Zain		commerciale@alpecimbra.it		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
PAGANELLA SKI	Italy	Resort Operator	Zain		skipass@paganella.net		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
MONTE BONDONE	Italy	Resort Operator	Zain		funvive@montebondone.it		follow up from Segreteria Generale Skirama	1st Email Sent	22/10/24	No		
Dolomiti	Italy	S+R Contractor	Fran	https://aiut-alpin-dolomites.com/en/contact/	info@aiut-alpin-dolomites.com			1st Response Recieved	21/10/24	Yes	Contact: giorgia@altabadiaskipass.it	FAIL
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	fabian.oberbacher@gmail.com		Soccorso Alpino Alta Baida	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	sanmartindetorr@soccorsoalpino.org		Soccorso Alpino Antermoia	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	donat.norbert@yahoo.com		Soccorso Alpino Appiano	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	bassaatesina@soccorsoalpino.org		Soccorso Alpino Bassa Atesina	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	bolzano@soccorsoalpino.org		Soccorso Alpino Bolzano	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	bressanone@soccorsoalpino.org		Soccorso Alpino Bressanone	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	funes@soccorsoalpino.org		Soccorso Alpino Funes	FAILED	21/10/24	No		FAIL
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	richard.pezzei@rolmail.net		Soccorso Alpino La Valle	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	armin.plangger@me.com		Soccorso Alpino Melago	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	merano@soccorsoalpino.org		Soccorso Alpino Merano	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni/stz	renon@soccorsoalpino.org		Soccorso Alpino Renon	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	resia@soccorsoalpino.org		Soccorso Alpino Resia	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	marebbe@soccorsoalpino.org		Soccorso Alpino Marebbe	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	info@soccorsoalpino-sesto.com		Soccorso Alpino Sesto	FAILED	21/10/24	No		FAIL
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	reinstadler@gmail.com		Soccorso Alpino Solda	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	bergrettung.stuls@gmail.com		Soccorso Alpino Stules	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	gotti.ratschiller@rolmail.net		Soccorso Alpino Val Martello	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	trafoi@soccorsoalpino.org		Soccorso Alpino Trafoi	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	wolfgangpunsch@gmail.com		Soccorso Alpino Tubre	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	aaron@monoder.bz.it		Soccorso Alpino Val Gardena	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	vipiteno@soccorsoalpino.org		Soccorso Alpino Vipiteno	1st Email Sent	21/10/24	No		
Dolomiti	Italy	S+R Team	Fran	https://www.soccorsoalpino.org/stazioni.htr	speleo@soccorsoalpino.org		Soccorso Speleologico	1st Email Sent	21/10/24	No		
Assynt Mountain Rescue	Scotland	S+R Team	Fran	https://www.assyntmountainrescue.co.uk/c	teamleader@assyntmountainrescue.co.uk			1st Email Sent	6/11/24	No	Lindsey the goat	SUCC...
Trossachs	Scotland	S+R Team	Fran	https://trossachs-sar.com/contact-us/	secretary@trossachs-sar.com			1st Email Sent	6/11/24	No		
Aberdeen	Scotland	S+R Team	Fran	https://amrt.org.uk/	secretary@amrt.org.uk			1st Email Sent	6/11/24	No		
Arran	Scotland	S+R Team	Fran	https://www.arranmountainrescue.com/	arranmrt@icloud.com			1st Email Sent	6/11/24	No		
Arrochar	Scotland	S+R Team	Fran	https://www.arrocharmrt.org.uk/	info@arrocharmrt.org.uk			1st Email Sent	6/11/24	No		
Dundonell	Scotland	S+R Team	Fran	https://www.dmrt.org.uk/team-contacts.htm	secretary@dmrt.org.uk			1st Email Sent	6/11/24	No		
Braemar	Scotland	S+R Team	Fran	https://www.braemarmountainrescue.org.u	secretary@braemarmountainrescue.org.uk			1st Email Sent	6/11/24	No		
Glenelg	Scotland	S+R Team	Fran	https://glenelgmountainrescue.org.uk/	jo@glenelgmountainrescue.org.uk			1st Email Sent	6/11/24	No		
Hebrides	Scotland	S+R Team	Fran	https://www.hebmrt.org/	contact@hebmrt.org			1st Email Sent	6/11/24	No		
Killin	Scotland	S+R Team	Fran	https://www.killinmountainrescue.scot/	secretary@killinmountainrescue.scot			1st Email Sent	6/11/24	No		
Lomond	Scotland	S+R Team	Fran	https://www.lomondmrt.org.uk/	enquiries@lomondmrt.org.uk			1st Email Sent	6/11/24	No	Dickheads	FAIL
Moffat	Scotland	S+R Team	Fran	https://www.moffatmrt.org.uk/	secretary@moffatmrt.org.uk			1st Email Sent	6/11/24	No		
Torridon	Scotland	S+R Team	Fran	http://www.torridonmrt.org.uk/contact/	chair@torridonmrt.org.uk			1st Email Sent	6/11/24	No		
Tayside	Scotland	S+R Team	Fran	https://taysidemrt.org/	teamleadership@taysidemrt.org			1st Email Sent	6/11/24	No		
	England &...	S+R Team	Zain					1st Email Sent	6/11/24	No		
	England &...	S+R Team	Zain					1st Email Sent	6/11/24	No		
	England &...	S+R Team	Zain					1st Email Sent	6/11/24	No		

Bottom Up/Top Down + Sales — 22

Top Down, Bottom Up, LTV (x2) SaR

TOP DOWN	TAM		SAM					SOM		
Product Service	Budgets	% Relevant to our product	£ Relevant to our Product		why they buying?	Assumptions	Appendix justification	% We can try and capture	£ We can try and captu	Links
RNLI	231,000,000	0.12	£27,720,000.00		get there quicker, non-critical cases, help ppl	12% spent on locating people		0.04	£1,108,800	https://register-of-charities.charitycommr
Mountain Rescue England and V	912,000	0.73	£665,760.00		get there quicker, non-critical cases, help ppl	% spent on locating / aiding people based on research		0.6	£399,456	https://register-of-charities.charitycommr
Association of Lowland Search a	176,459	0.6	£105,875.21		Percentage of money spent on locating			0.6	£63,525	
Scottish Mountain Rescue	312,000	0.44	£137,280		get there quicker, non-critical cases, help ppl	% spent on locating / aiding people based on research		0.6	£82,368	https://www.scottishmountainrescue.or
Maritime and costguard agency	372,000,000	N/A	£195,881		get there quicker, non-critical cases, help ppl	Exact spent on reaching pepople		0.04	£7,835	https://assets.publishing.service.gov.uk
Air Ambulance Service	34,000,000	0.51	£17,340,000		51% are non-critical cases: need to find a way to stop this inefficiency	% spent on reaching people who don't need to be relocated		0.04	£693,600	https://register-of-charities.charitycommr
Military Helicopter S&R	34,000,000	0.41	£13,940,000		41% of call outs our product could have been used in	Replacement of useless helicopter call outs		0.04	£557,600	https://assets.pu https://assets.publishi
Police force	27,300,000,000	0.4	£5,300,500			Research Funds		0.04	£212,020	
Fire service	2,500,000,000	0.002	£5,000,000			Research Funds		0.04	£200,000	
Total:	**£30,472,400,458.68**		**£70,405,296.21**					0.04	**£3,325,204**	https://dronewars.net/british-drones-ar
Defence (Saas)	**Budgets**	**% Relevant to our product**	**£ Relevant to our Product**			**Assumptions**	**Appendix justification**	**% We can try and capture**	**£ We can try and captu**	**Links**
UK Defence Budget	53,900,000,000		£3,020,000,000			Research Funds	Looking at similar investments	0.001	£3,020,000	https://dronewar https://assets.pu https

BOTTOM UP PRODUCT

Service	LTV Multiplier	# of Teams	Expected Revenue (£)	Number of Drones
RNLI	1	238	4,641,000	238
Mountain Rescue England & Wal	3	50	2,925,000	150
Association of Lowland Search &	3	36	2,106,000	108
Scottish Mountain Rescue	3	24	1,404,000	72
Maritime & Coastguard Agency	1	352	6,864,000	352
Air Ambulance Service	1	21	409,500	21
UK-Based International S&R	5	4	390,000	20
Military Helicopter S&R	5	10	975,000	50
Total		**735**	**£19,714,500.00**	1011

+ Convert to table

Number of Buisnesses	LTV Multiplier	Retention Rate	Expected Revenue (£)	15% Market share
7	10	0.8	**£10,080,000.00**	**£2,979,450.00**

LTV PRODUCT

Component	Description	Price (php)	Years	Calculation	Value (£)
Hardware Profit	Initial product sale	2000	1	2000	
Subscription Revenue	5 years of subscriptions (£3000/year)	3000	5	15000	
Maintenance Cost	Yearly maintenance cost for 5 years	500	5	2500	
Total LTV	**Lifetime Value per Customer**			**19500**	
Software Licensing	Buisness / Month	3000	5	£180,000.00	

27,300,000,000	0.4	£5,300,500		Research Funds	0.05	£265,025		
2,500,000,000	0.002	£5,000,000		Research Funds	£0	£250,000	£0	
		0.4	£5,300,500				£0	https://www.gov.uk/government/statistic
		0.32	£800,000,000				£0	https://freerangeuk.uk/national-overview
Drone Research Grants	78,000,000	1	78,000,000					https://www.nesta.org.uk/press-release
Google Deepmind	1,500,000,000		£5,000,000					
DJI	10,000,000,000			Money spent over external R&D				
Microsoft	166,000,000,000							
UK AI Company Investments	1,200,000,000		£2,181,818	Split over 550 companies gives average investment			£2,181,818	
Total:	**£324,950,400,491.08**		**£890,482,318.18**				**£6,345,204**	

BOTTOM UP DEFENCE (SAAS)

Variable	Value	Description	Sources
SaaS Price (Per Month)	£20,000	Monthly subscription cost per customer.	
Annual Subscription (Auto)	£240,000	Annual SaaS cost (calculated).	
Number of Customers	35	Number of RAF Bases adopting the SaaS.	https://www.raf.mod.uk/our-organisation/stations/
Recovery Success Rate (%)	0.1	Improvement in recovery success due to SaaS.	
Cost to Train a Pilot (£)	£7,000,000	Average cost to train and deploy a pilot.	
Annual Downed Pilots	39	Number of pilot incidents globally.	https://www.globalsecurity.org/military/library/report/2020/ncmas_final_report_20201201.pdf?utm_source=chatgpt.com
Missions Saved (%)	0.1	Reduction in failed mission costs.	
Average Mission Cost (£)	£1,000	Average cost per search mission.	
Cost to deploy Apollo	£6,154		
Metric per annum	**Formula**	**Description**	
Total Annual SaaS Revenue	**£8,400,000**	**Annual revenue from all customers.**	
Pilots Recovered (Extra)	3.9	Additional pilots recovered due to SaaS.	
Savings from Recovery (£)	£27,300,000	Total cost savings from additional recoveries.	
Total Mission Cost (£)	£279,000	Total cost of all missions annually.	
Total Savings for military(£)	**£18,621,000**	**Combined recovery and mission savings.**	

Top Down, Bottom Up, LTV (x2) Defence

EVENTS				
people	**calculation**	**value**	ASSUMPTION	SOURCE
Total Attendees	2000	2,000	average attendees to these events	
Relevant Audience to our business	0.15	300	In large-scale events, only a subset of attendees aligns with your target market (e.g., SAR team leads, procurement officers). Industry standards suggest 10–20% is realistic for B2B-focused expos.	
Engaged Leads (Booth/demo visitors)	0.6	180	For interactive booths and live demos, engagement can be high. However, this figure may be optimistic without robust pre-event marketing.	
Qualified Leads (Serious interest)	0.3	54	This percentage represents those who are seriously interested after seeing your demo. A 20–40% qualification rate is typical in B2B settings, depending on the industry and presentation quality.	
Potential Customers (Follow-up success)	0.3	16.2	A 20–30% follow-up success rate is standard for B2B industries, assuming prompt, personalized communication.	
Converted Customers (Conversion rate)	0.1	1.62		
Cost Element	**calculation**	**value**	**assumption**	
Event Booth Setup	15000	15000	Booth space, materials, staff	
Travel & Accommodation	3000	3000	Travel and lodging for 3 staff	
Marketing Collateral	2000	2000	Brochures, banners, digital materials	
Total Cost Per Event		20000		
Cost per customer (CAC)	***cost per event/convert customer***	***12345.67901***		
Revenue Stream	**cost (per month)**	**multiplier**	explanation	value
SaaS Revenue	3000	36	3 years (36 months)	108000
Drone Revenue	2000	2	2 drones per customer	4000
Total AAC				112000
AAC FROM EVENT				***181440***
ROI				13.69664
GOV GRANTS				
Step	**calculation**	**value**	**Assumptions**	
Grant Programs Targeted	50	50	Estimated number of disaster response-focused grant programs globally.	
Applications Submitted	0.5	25	based on the requirements that we can fit into (professional standard etc) and based in uk, and etc etc	
Grants Awarded	0.1	2.5	5% success rate for well-written grant proposals with strong ROI focus.	
Deals Closed	0.5	1.25	Grants directly subsidize purchases for SaR organizations.	
Pilots				
Step	**People**	**Calculation**	**Assumptions**	
Teams Contacted	110	110	Small S&R teams targeted for free pilots.	
Interested Teams	0.1	11	Positive response rate for free drones.	
Pilot Participants	0.5	5.5	Final participants for 6-month pilots.	
Converted to Paid Contracts	0.6	3.3	High conversion rate post successful pilots.	
alterantive				
Step	**People**	**Calculation**	**Assumptions**	**Sources**
Targeted Organizations	110	110	Based on ~1,000 SaR-focused organizations globally, including national parks, coast guards, and NGOs.	FEMA, International Association of Emergency Managers (IAEM).
Responses from Outreach	0.15	16.5	10% response rate is typical for cold email campaigns targeting B2B audiences.	Mailchimp benchmark for B2B email campaigns (10-15% response rate).
Leads Qualified (Initial Call)	0.5	8.25	50% qualification rate for B2B after initial contact.	HubSpot reports average qualification rates for B2B as 30–60%.
Demo Scheduled	0.7	5.775	70% conversion to demos from qualified leads is achievable with targeted outreach.	Outreach.io B2B benchmarks show 60–80% demo scheduling rate with effective follow-up.
Deals Closed	0.3	1.7325	30% demo-to-close conversion rate.	Industry standard for B2B SaaS and hardware products in niche markets.

AWS Cost

Models	Number of models	Cost of running 1 model / min	Expected running time (min)	Cost of storing 1 model / year	Total Cost
Active (10% of drones)	1	0.062	720	18.36	£63.00
R&D Models	1	0.062	17280	18.36	£1,089.72
					£1,152.72

https://aws.amazon.com/bedrock/pricing/

Models	Number of models	Cost of running 1 model / min	Expected running time (min)	Cost of storing 1 model / year	Total Cost
Active	1	0.062	512640	18.36	£31,802.04
API	1	0.062	512640	18.36	£31,802.04
Traing & Validation	1	0.062	512640	18.36	£31,802.04
					£95,406.12

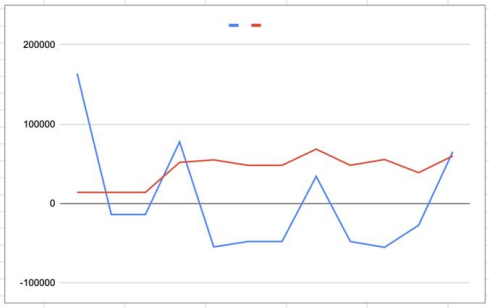
Grant Funding Min

	Success Rate	Source	Average Grant Given Min	Average Grant Given Max	Source
Innovate UK	0.1	https://tbat.co.uk/knowledge/how-competitive-is-grant-funding/#:~:text=Only%20top%2Dscoring%20applications%20are,a%20good%20chance%20of%20success.	£100,000	£1,000,000	
DIANA					
Horizon Europe	0.025	https://tbat.co.uk/knowledge/how-competitive-is-grant-funding/			
National Security Strategic Fund					
Air Street Capital					
Allied VC			£100,000	£350,000	
MD One VC			£500,000	£1,500,000	https://www.tedcomd.com/funding/venture-funds
EIC Accelerator				£10,000,000	https://eic.ec.europa.eu/eic-funding-opportunities
Climate Change AI				£150,000	https://www.climatechange.ai/calls/innovation_gr

Year 1

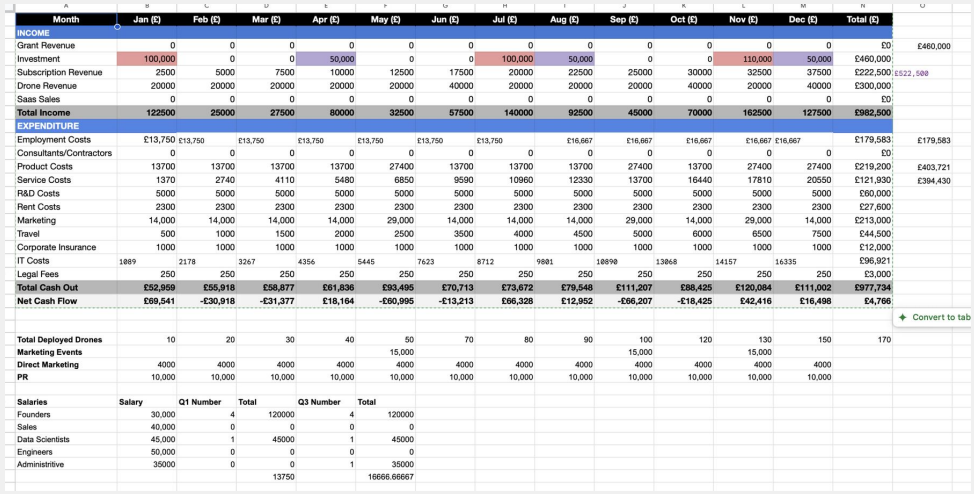
Month	Jan (£)	Feb (£)	Mar (£)	Apr (£)	May (£)	Jun (£)	Jul (£)	Aug (£)	Sep (£)	Oct (£)	Nov (£)	Dec (£)	Total (£)	
INCOME					PILOT	PILOT	PILOT	PILOT	PILOT	PILOT				
Grant Revenue		0	0	29364.3	0	0	0	52500	0	0	0	52500	£134,364	£512,064
Investment	177,700	0	0	100,000	0	0	0	50,000	0	0	0	50,000	£377,700	
Subscription Revenue	0	0	0	0	0	0	0	0	0	0	1250	2500	£3,750	£33,750
Drone Revenue	0	0	0	0	0	0	0	0	0	0	10000	20000	£30,000	
Saas Sales	0	0	0	0	0	0	0	0	0	0	0	0	£0	
Total Income	**177700**	**0**	**0**	**129364.3**	**0**	**0**	**0**	**102500**	**0**	**0**	**11250**	**125000**	**£545,814**	
EXPENDITURE														
Employment Costs	0	0	0	20,400	0	0	0	20,400	0	0	0	20,400	£61,200	£78,700
Consultants/Contractors	0	0	0	2500	2500	2500	2500	2500	2500	2500	0	0	£17,500	
Product Costs	0	0	0	0	6850	0	0	0	0	6850	6850	6850	£27,400	£316,156
Service Costs	0	0	0	0	17200	17200	17200	17200	17200	17200	685	1370	£105,255	£120,305
R&D Costs	13333	13333	13333	25000	25000	25000	25000	25000	25000	25000	25000	25000	£264,999	
Rent Costs	450	450	450	800	800	800	800	800	800	800	3200	3200	£13,350	
Marketing	0	0	0	500	0	0	0	0	0	500	500	500	£2,000	
Travel	0	0	0	200	200	200	200	200	200	200	200	200	£1,800	
Corporate Insurance	0	0	0	1000	1000	1000	1000	1000	1000	1000	1000	1000	£9,000	
IT Costs	200	200	200	1089.9	1089.9	1089.9	1089.9	1089.9	1089.9	1089.9	1089	1089	£10,407	
Legal Fees	0	0	0	250	250	250	250	250	250	250	250	250	£2,250	
Total Cash Out	**13983**	**13983**	**13983**	**51,740**	**54,890**	**48,040**	**48,040**	**68,440**	**48,040**	**55,390**	**38,774**	**59,859**	**£515,161**	
Net Cash Flow	**163717**	**-13983**	**-13983**	**77,624**	**-54,890**	**-48,040**	**-48,040**	**34,060**	**-48,040**	**-55,390**	**-27,524**	**65,141**	**£30,653**	

Cost	Cost/day	Days / Month	Assumption
Workshop	45	10	Work for 30 days over the 3 months
VC Funding			
Innovate UK			
DASA			



Year 2 + Year 3 + ICE Framework — 24

Year 2



Month	Jan (£)	Feb (£)	Mar (£)	Apr (£)	May (£)	Jun (£)	Jul (£)	Aug (£)	Sep (£)	Oct (£)	Nov (£)	Dec (£)	Total (£)	
INCOME														
Grant Revenue	0	0	0	0	0	0	0	0	0	0	0	0	£0	£460,000
Investment	100,000	0	0	50,000	0	0	100,000	50,000	0	0	110,000	50,000	£460,000	
Subscription Revenue	2500	5000	7500	10000	12500	17500	20000	22500	25000	30000	32500	37500	£222,500	£522,500
Drone Revenue	20000	20000	20000	20000	20000	40000	20000	20000	20000	40000	20000	40000	£300,000	
Saas Sales	0	0	0	0	0	0	0	0	0	0	0	0	£0	
Total Income	**122500**	**25000**	**27500**	**80000**	**32500**	**57500**	**140000**	**92500**	**45000**	**70000**	**162500**	**127500**	**£982,500**	
EXPENDITURE														
Employment Costs	£13,750	£13,750	£13,750	£13,750	£13,750	£13,750	£13,750	£16,667	£16,667	£16,667	£16,667	£16,667	£179,583	£179,583
Consultants/Contractors	0	0	0	0	0	0	0	0	0	0	0	0	£0	
Product Costs	13700	13700	13700	13700	27400	13700	13700	13700	27400	13700	27400	27400	£219,200	£403,721
Service Costs	1370	2740	4110	5480	6850	9590	10960	12330	13700	16440	17810	20550	£121,930	£394,430
R&D Costs	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000	£60,000	
Rent Costs	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	£27,600	
Marketing	14,000	14,000	14,000	14,000	29,000	14,000	14,000	14,000	29,000	14,000	29,000	14,000	£213,000	
Travel	500	1000	1500	2000	2500	3500	4000	4500	5000	6000	6500	7500	£44,500	
Corporate Insurance	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	£12,000	
IT Costs	1889	2178	3267	4356	5445	7623	8712	9801	10890	13068	14157	16335	£96,921	
Legal Fees	250	250	250	250	250	250	250	250	250	250	250	250	£3,000	
Total Cash Out	**£52,959**	**£55,918**	**£58,877**	**£61,836**	**£93,495**	**£70,713**	**£73,672**	**£79,548**	**£111,207**	**£88,425**	**£120,084**	**£111,002**	**£977,734**	
Net Cash Flow	**£69,541**	**-£30,918**	**-£31,377**	**£18,164**	**-£60,995**	**-£13,213**	**£66,328**	**£12,952**	**-£66,207**	**-£18,425**	**£42,416**	**£16,498**	**£4,766**	

Convert to tab

Total Deployed Drones	10	20	30	40	50	70	80	90	100	120	130	150	170
Marketing Events					15,000				15,000		15,000		
Direct Marketing	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	
PR	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	

Salaries	Salary	Q1 Number	Total	Q3 Number	Total
Founders	30,000	4	120000	4	120000
Sales	40,000	0	0	0	0
Data Scientists	45,000	1	45000	1	45000
Engineers	50,000	0	0	0	0
Administritive	35000	0	0	1	35000
			13750		16666.66667

Year 3

Month	Jan (£)	Feb (£)	Mar (£)	Apr (£)	May (£)	Jun (£)	Jul (£)	Aug (£)	Sep (£)	Oct (£)	Nov (£)	Dec (£)	Total (£)	
INCOME														
Grant Revenue	0	0	0	0	0	0	0	0	0	0	0	0	£0	£3,000,000
Investment	0	0	0	0	0	0	0	0	0	0	0	3,000,000	£3,000,000	
Subscription Revenue	42500	45000	47500	50000	52500	57500	60000	62500	65000	70000	72500	77500	£702,500	£3,267,500
Drone Revenue	30000	30000	30000	30000	30000	60000	30000	30000	30000	60000	30000	60000	£450,000	
Saas Sales	15000	30000	60000	90000	120000	150000	180000	210000	240000	270000	300000	450000	£2,115,000	
Total Income	**87500**	**105000**	**137500**	**170000**	**202500**	**267500**	**270000**	**302500**	**335000**	**400000**	**402500**	**3587500**	**£6,267,500**	
EXPENDITURE														
Employment Costs	£34,167	£34,167	£34,167	£34,167	£34,167	£34,167	£34,167	£38,750	£38,750	£38,750	£38,750	£38,750	£432,917	£472,917
Consultants/Contractors	10,000	0	0	10,000	0	0	10,000	0	0	10,000	0	0	£40,000	
Product Costs	13700	13700	13700	13700	27400	13700	13700	13700	27400	13700	27400	13700	£205,500	£659,109
Service Costs	23290	24660	26030	27400	28770	31510	32880	34250	35620	38360	39730	42470	£384,970	£678,170
R&D Costs	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	£120,000	
Rent Costs	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	2300	£27,600	
Marketing	14,000	14,000	14,000	14,000	29,000	14,000	14,000	14,000	29,000	14,000	29,000	14,000	£213,000	
Travel	3400	3600	3800	4000	4200	4600	4800	5000	5200	5600	5800	6200	£56,200	
Corporate Insurance	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	£12,000	
IT Costs	18513	19602	20691	21780	22869	25047	26136	27225	28314	30492	31581	33759	£306,009	
Legal Fees	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	£12,000	
Total Cash Out	**£131,370**	**£124,029**	**£126,688**	**£139,347**	**£160,706**	**£137,324**	**£149,983**	**£147,225**	**£178,584**	**£165,202**	**£186,561**	**£163,179**	**£1,810,196**	
Net Cash Flow	**-£43,870**	**-£19,029**	**£10,812**	**£30,653**	**£41,794**	**£130,176**	**£120,017**	**£155,275**	**£156,416**	**£234,798**	**£215,939**	**£3,424,321**	**£4,457,304.33**	

Total Deployed Drones	170	180	190	200	210	230	240	250	260	280	290	310	320
Marketing Events					15,000				15,000		15,000		
Direct Marketing	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	
PR	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	
Total Saas Deployed	5	10	20	30	40	50	60	70	80	90	100	150	

Salaries	Salary	Q1 Number	Total	Q3 Number	Total
Founders	50,000	4	200000	4	200000
Sales	50,000	1	50000	1	50000
Data Scientists	55,000	1	55000	2	110000
Engineers	60,000	1	60000	1	60000
Administritive	45,000	1	45000	1	45000
			34166.66667		38750

ICE Framework

Sales Channel	Impact (I)	Confidence (C)	Ease (E)	Justification	Score (I × C × E)
Trade Shows	9	9	7	High-impact channel for generating leads and awareness. Proven success in the S&R and defense industries. Requires significant effort and cost.	567
Referrals (Word of Mouth)	8	10	9	Cost-effective with high conversion rates due to trust and credibility. Minimal effort once initial customers are satisfied.	720
Pilot Programs	10	9	6	Essential for credibility and data collection. Requires significant upfront investment but drives long-term success.	540
Partnerships with S&R Teams	8	8	6	Builds credibility and expands reach through trusted organizations. Moderate effort for setup and execution.	384
Conferences & Industry Events	8	8	7	Ideal for live demos and direct engagement with decision-makers. Requires moderate resources but aligns with trade shows.	448
Direct Outreach	6	7	7	Effective for targeting niche, high-value prospects. Moderate confidence due to reliance on personalized efforts.	294
PR Around Our Drone	7	8	8	Amplifies the success of primary channels and builds credibility. Moderate confidence as press results vary.	448
Community Building	5	6	7	Long-term strategy but slower to drive direct leads. Limited short-term applicability for S&R or defense sales.	210
Speaking Engagements	6	8	6	Good for thought leadership but less scalable than trade shows. Best paired with other efforts like demos and PR.	288
Unconventional PR	5	6	7	May attract attention but is less applicable for B2B S&R and defense-focused industries.	210
Affiliate Marketing	4	4	9	High ease but low relevance for a hardware- and SaaS-focused B2B market.	144
SEO	5	7	6	Useful for inbound interest but less impactful in a B2B market where direct engagement is more effective.	210
Social Media Marketing	6	7	7	Good for raising awareness and engagement. Less effective in driving direct sales in niche B2B markets.	294
Engineering as Marketing	5	6	7	Demonstrates innovation but less impactful for immediate sales. Better as a long-term trust-building strategy.	210
Demos	9	9	7	Highly effective for engaging decision-makers. Best paired with events and pilots to showcase product capabilities.	567
Cross-Government Relations	7	8	5	Useful for long-term defense data sales. Requires significant effort to establish but builds credibility for scaling.	280
Public Relations (General PR)	7	7	8	Supports credibility and awareness but requires pairing with primary channels for maximum effect.	392
Business Development	7	8	7	Effective for partnerships, scaling, and long-term relationships. Useful for securing large contracts.	392
Target Market Blogs	5	6	7	Slow to generate leads but useful for building credibility in niche markets over time.	210
Offline Events	6	7	6	Useful but overlaps heavily with trade shows and conferences. Less scalable than digital or targeted strategies.	252
Newspapers	5	6	8	Limited relevance to B2B markets unless targeting defense decision-makers. Works better as part of PR campaigns.	240
Investment Funds	6	8	7	Supports R&D and scaling but not directly tied to sales. Critical for long-term growth.	232

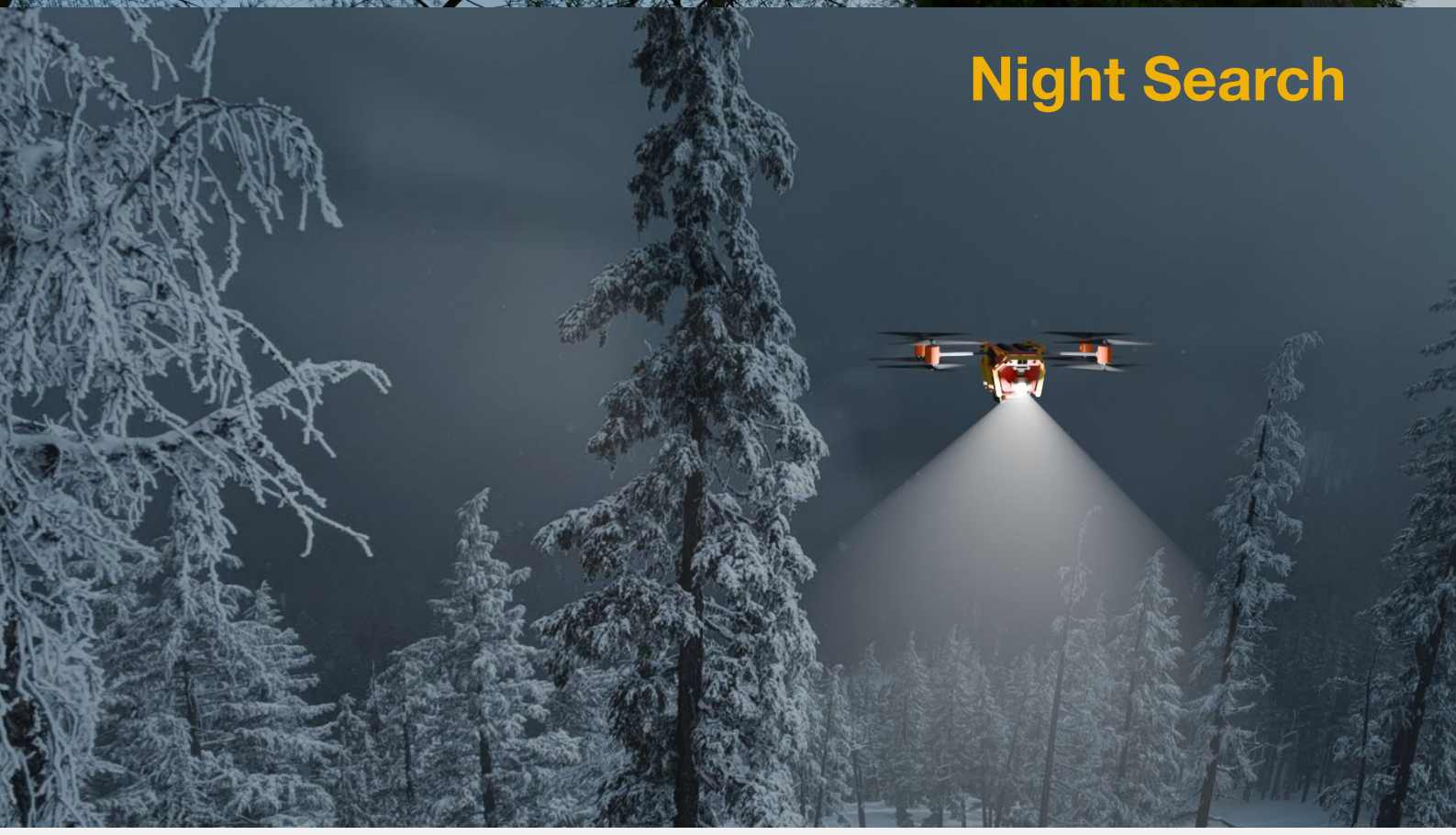
Valley Search

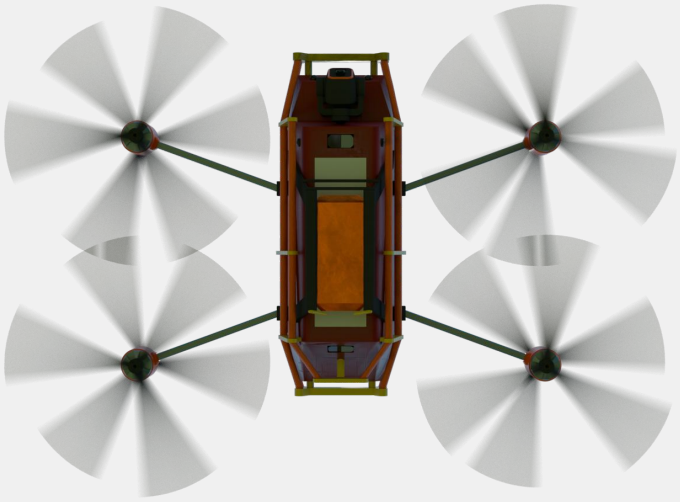
Close Up

Lost Hiker

Avalanche Rescue

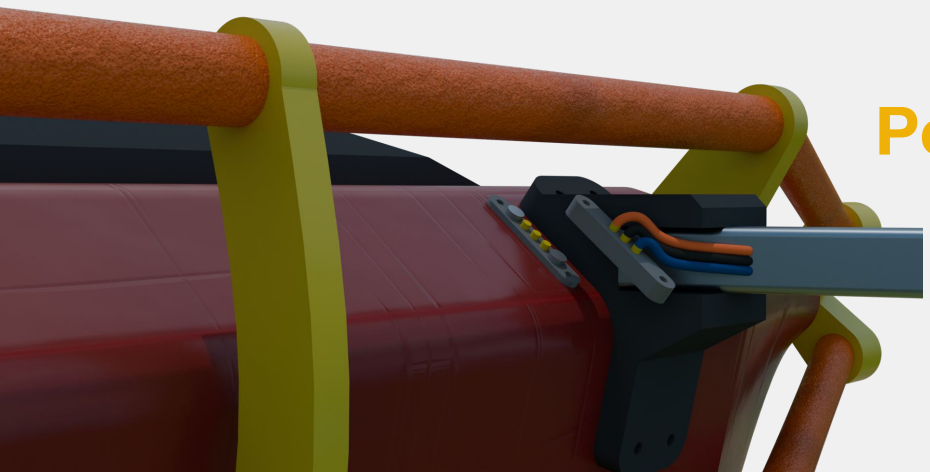
Falling Box


Night Search



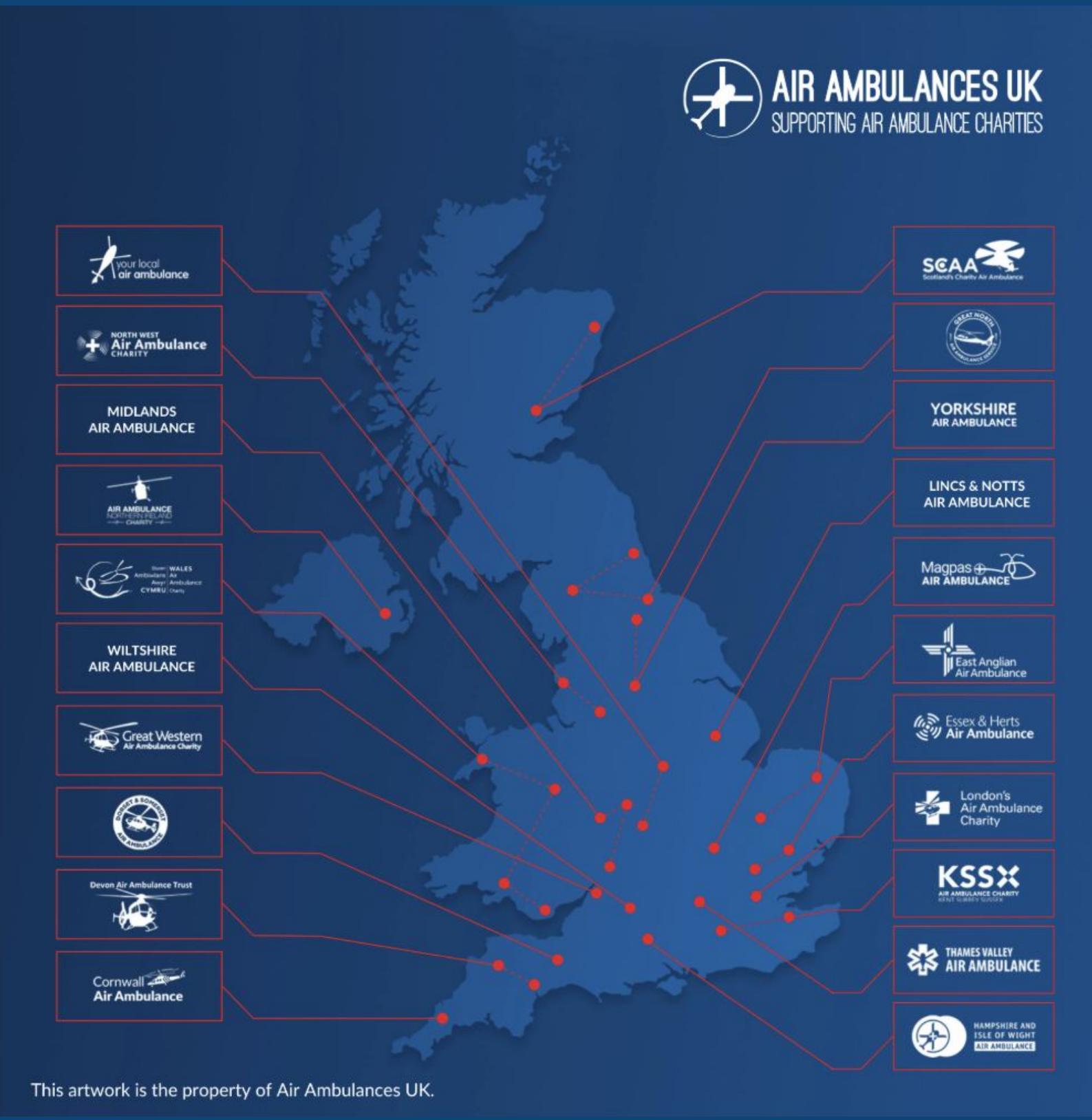
Active Motors

Stationary

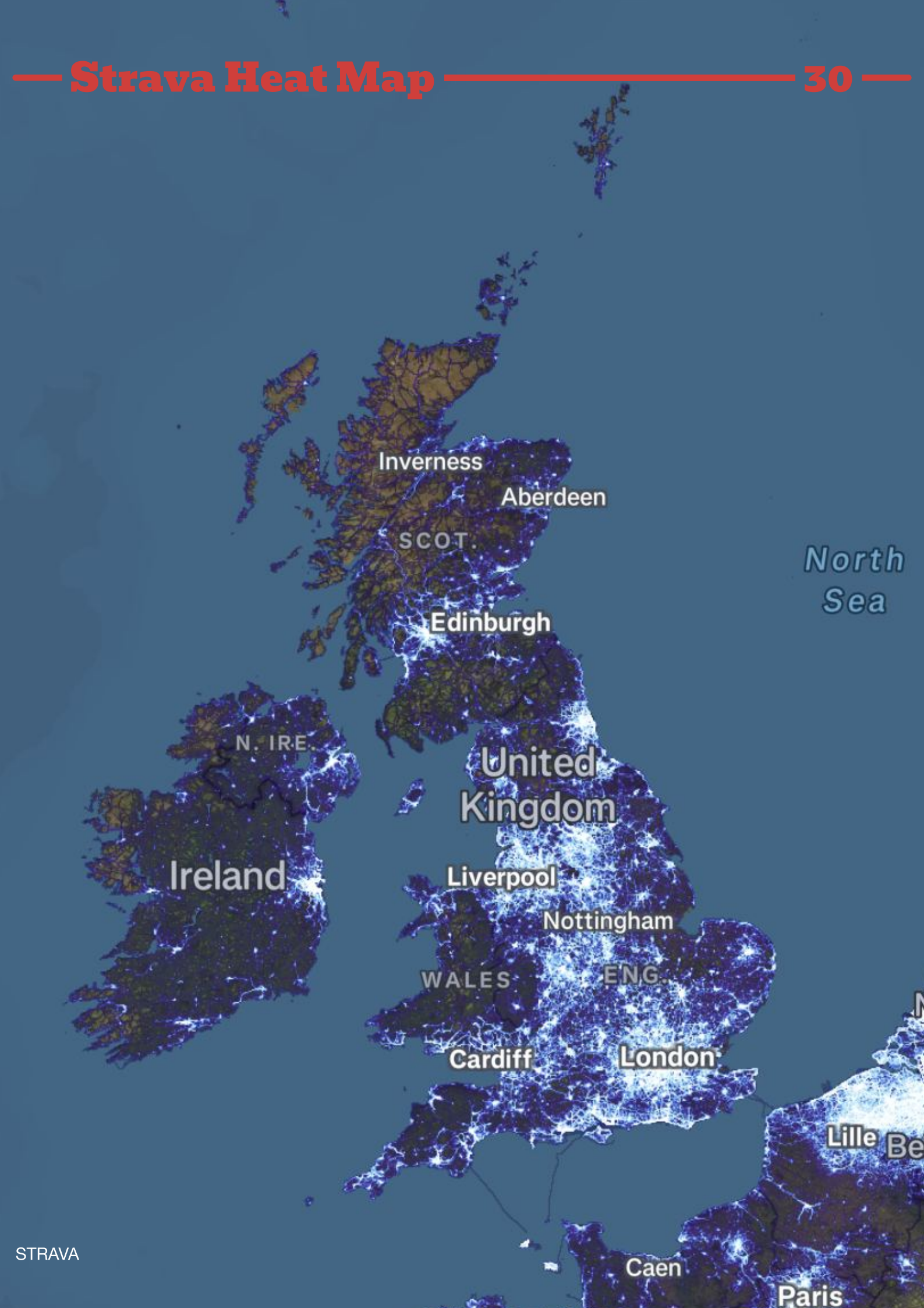


Pogo Pin Connector

Air Ambulance Location - UK 29



Air Ambulances UK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North Sea
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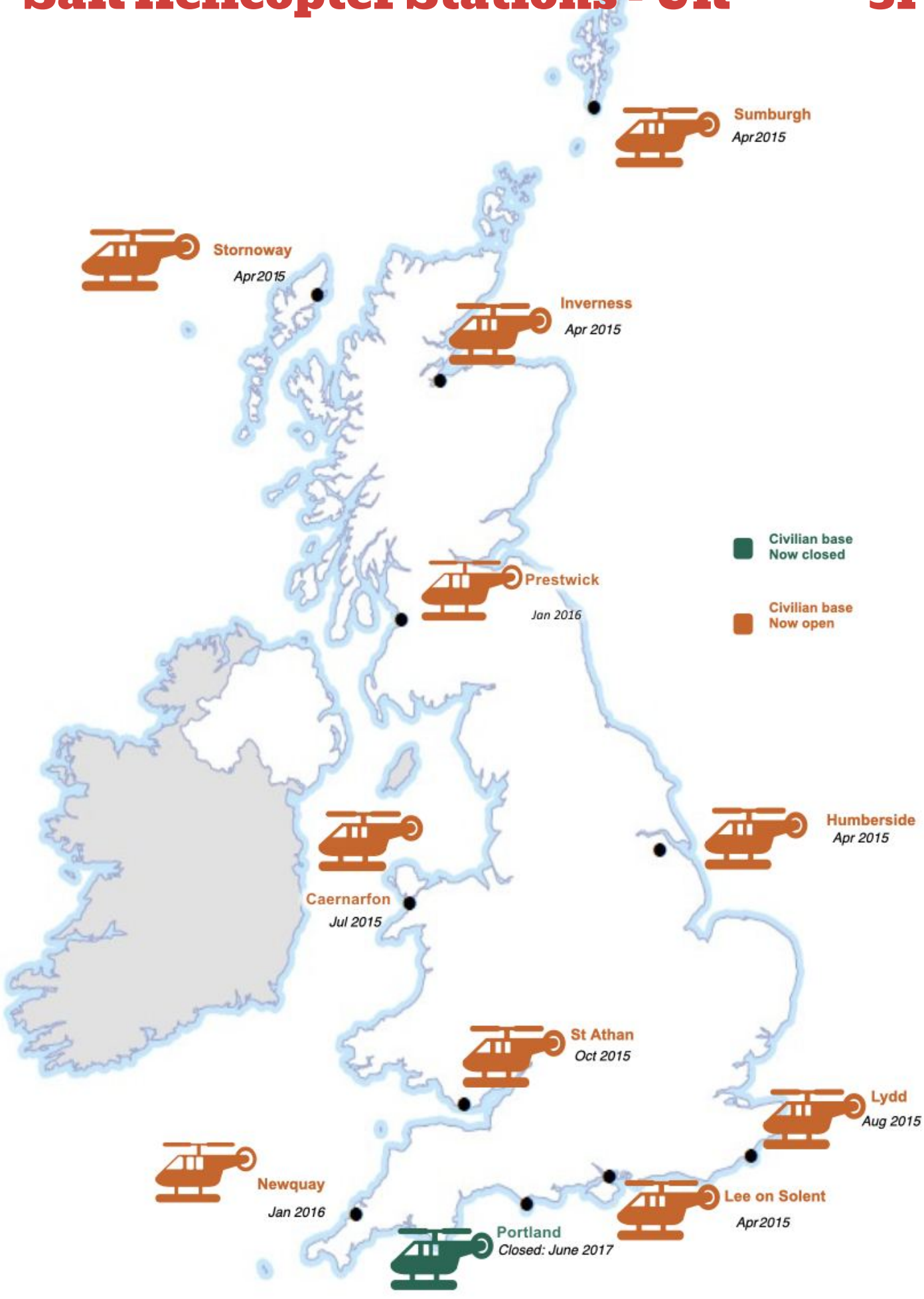
London
Lille
Caen
Paris

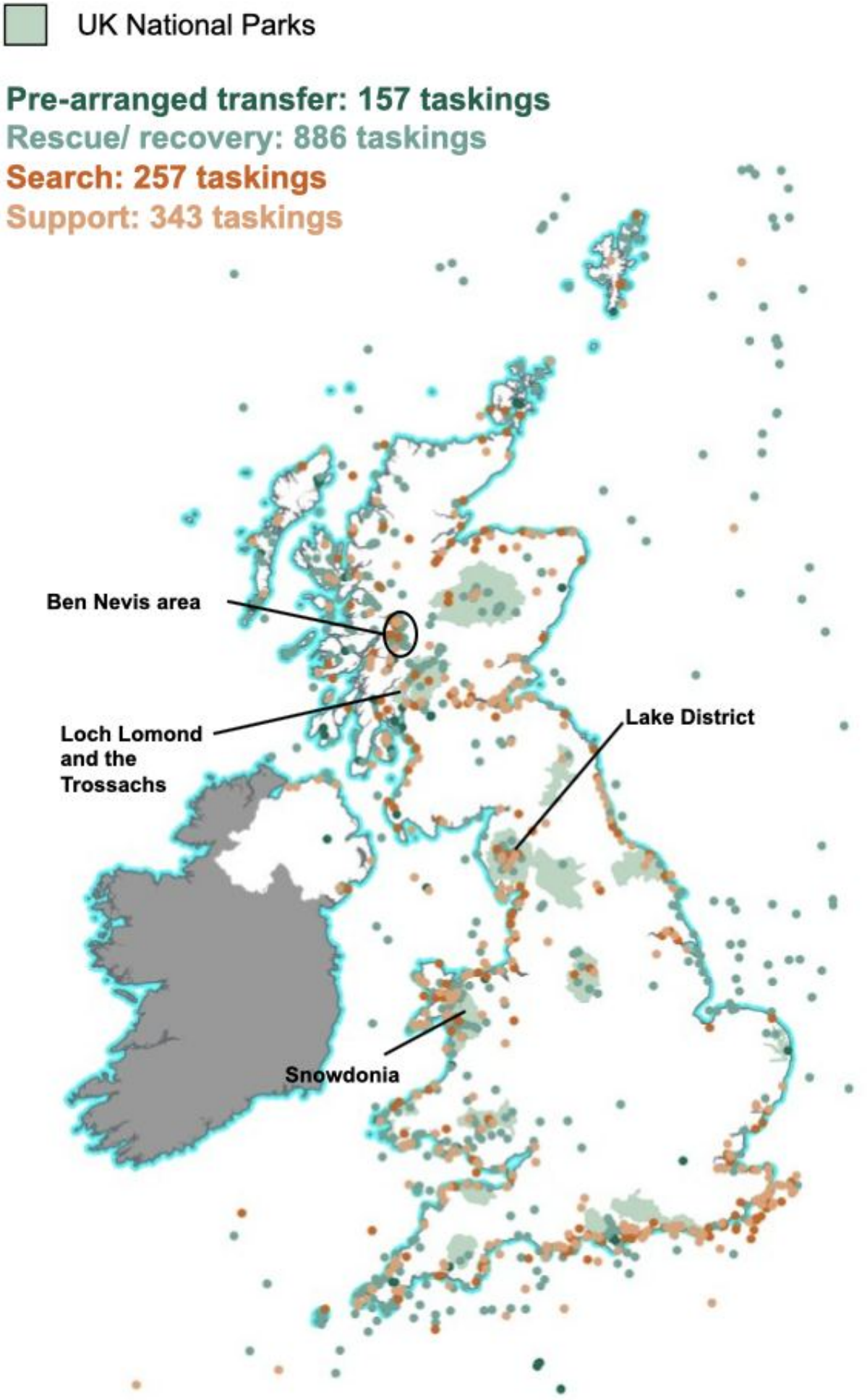
STRAVA

— SaR Helicopter Stations - UK —— 31 —



.GOV *(UK)

UK SaR Taskings



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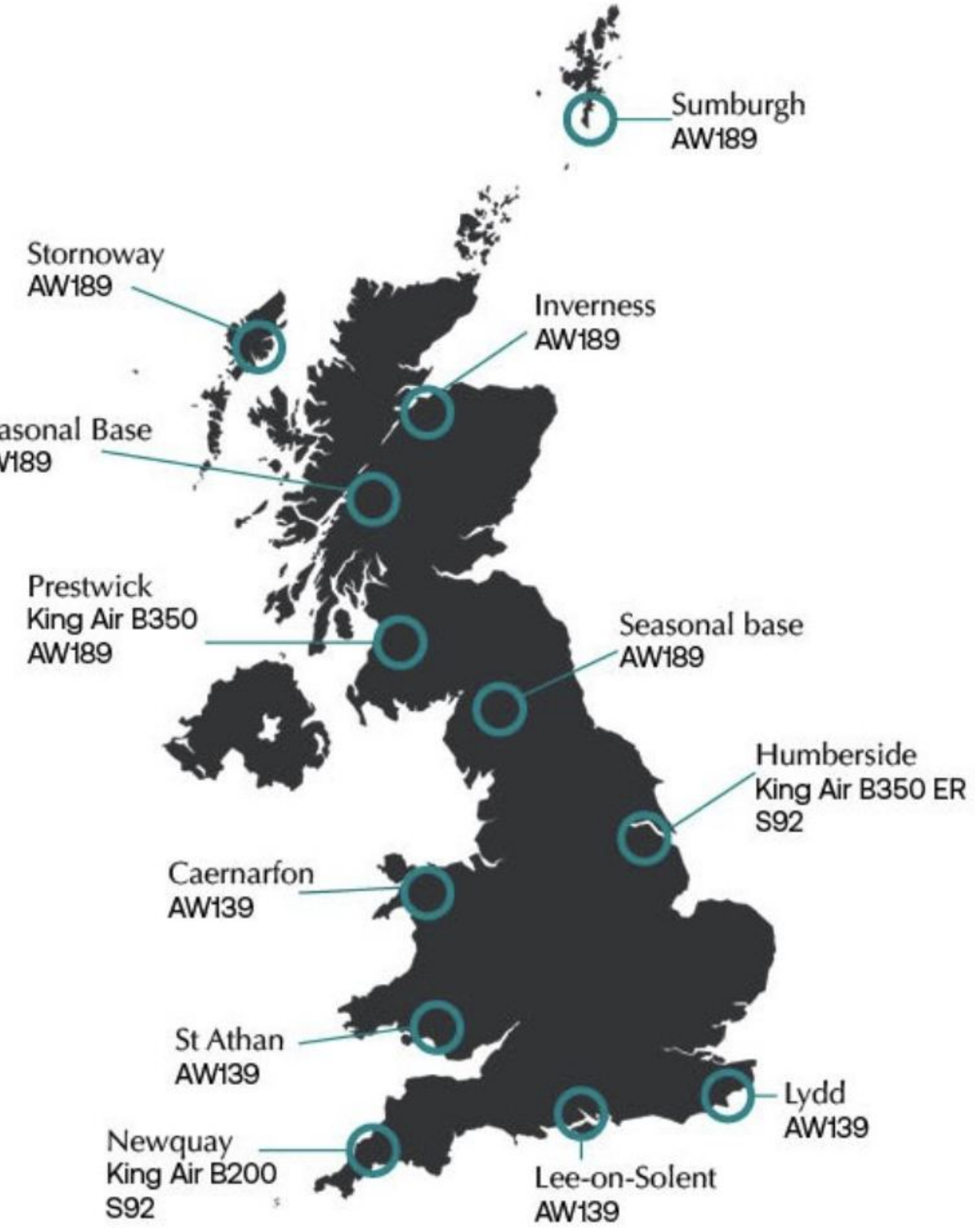
— UK Defence Helicopter Stations — 33 —



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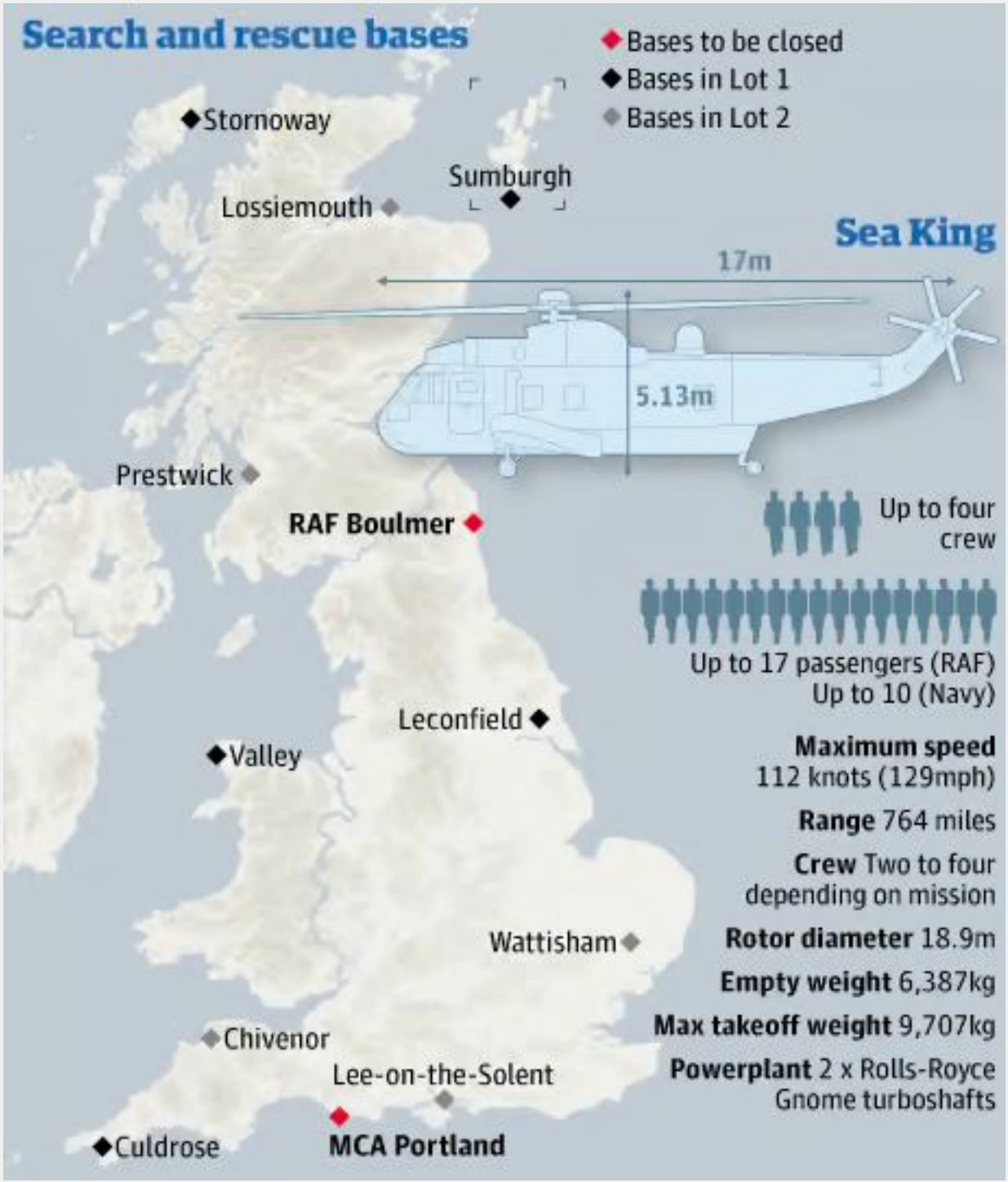
— Pre-Privatisation SaR Heli Map – 34 —



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Page 3
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[3.3] 'ArcGIS Dashboards' (2024) ArcGIS. Available at: https://www.arcgis.com/apps/dashboards/c251b6e1cd7947a68d4d76d231196e16 (Accessed: 14 December 2024).
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